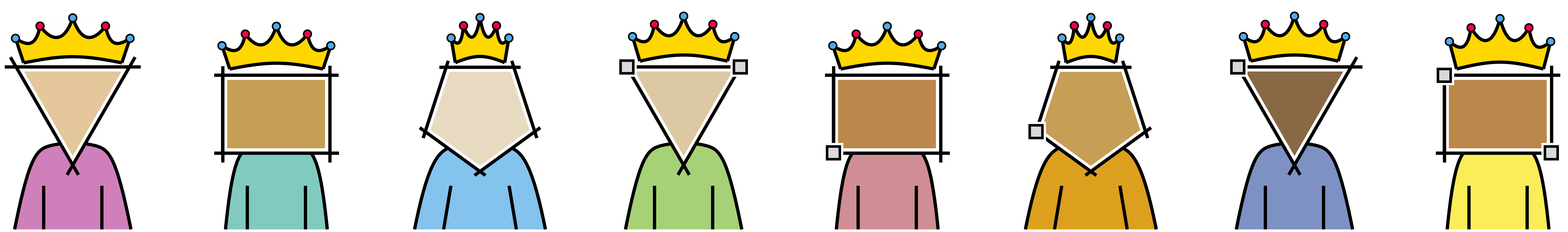
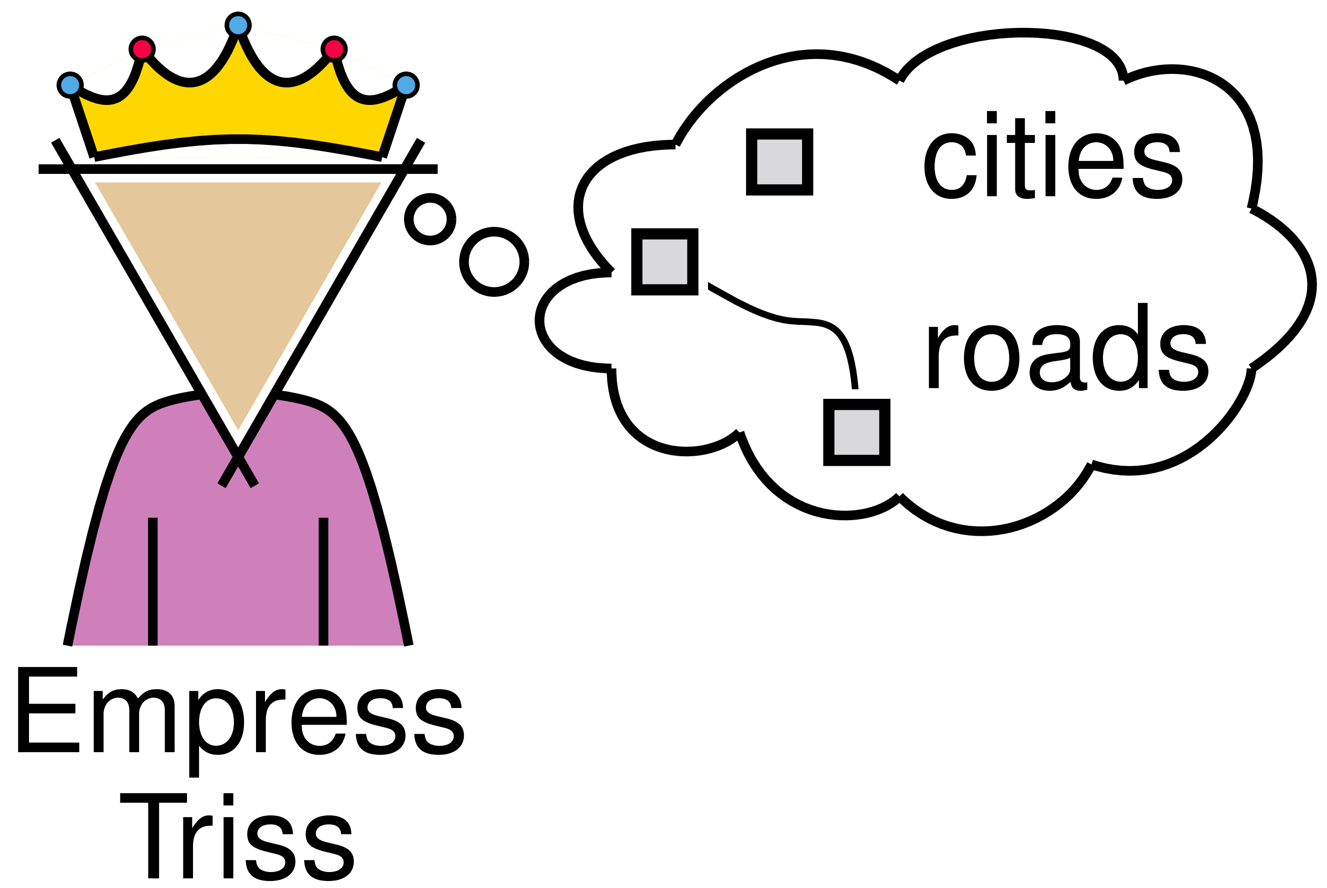
Edge densities of drawings with one forbidden cell

Benedikt Hahn Torsten Ueckerdt Birgit Vogtenhuber



Planning an empire



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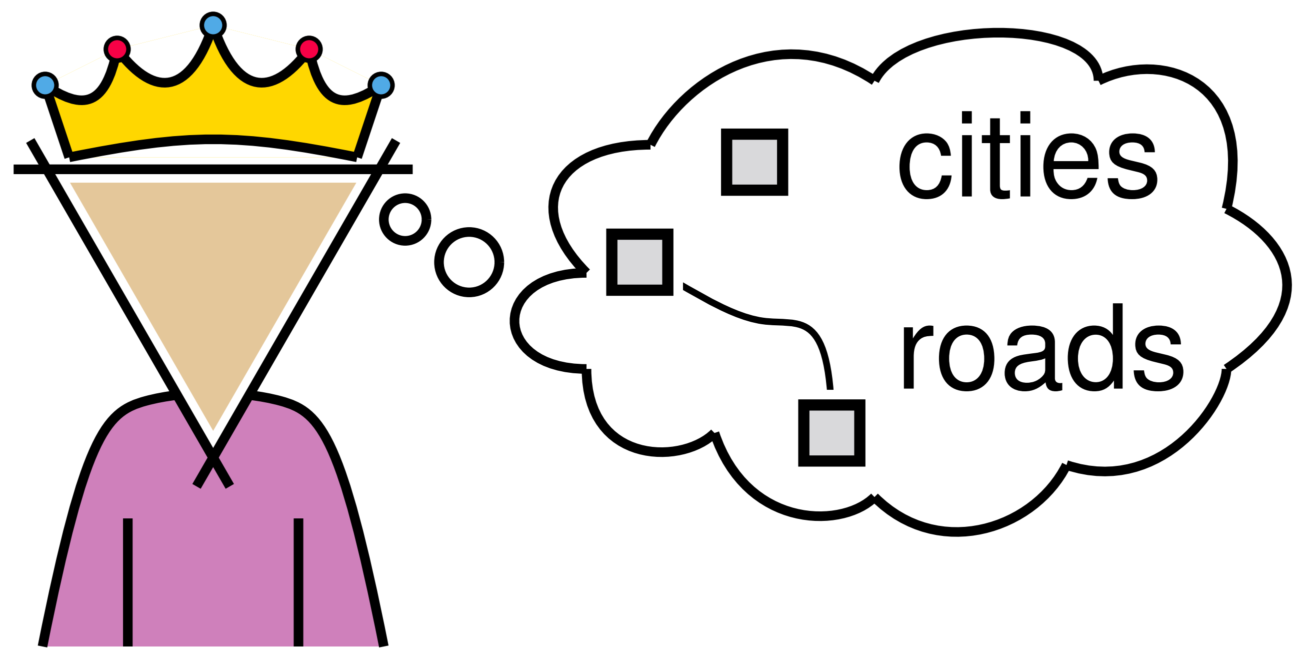
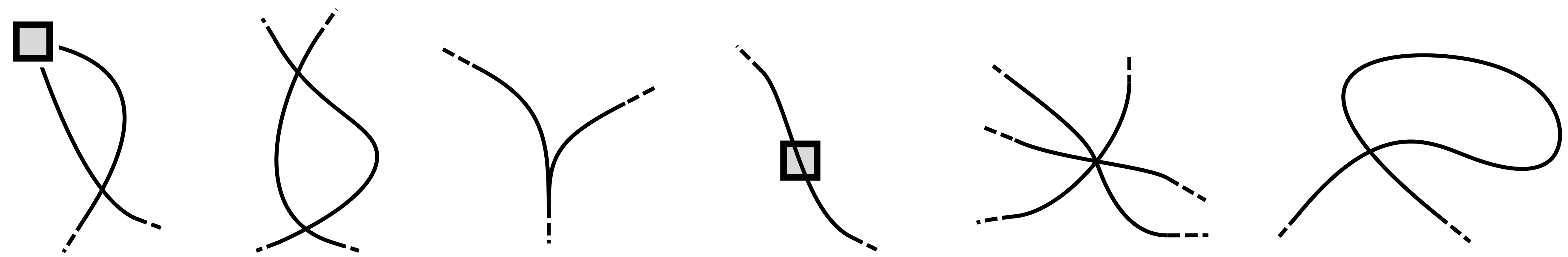


Empress
Triss



violates Simplicity | violates Purity | violates Integrity | violates Clarity | violates Efficiency

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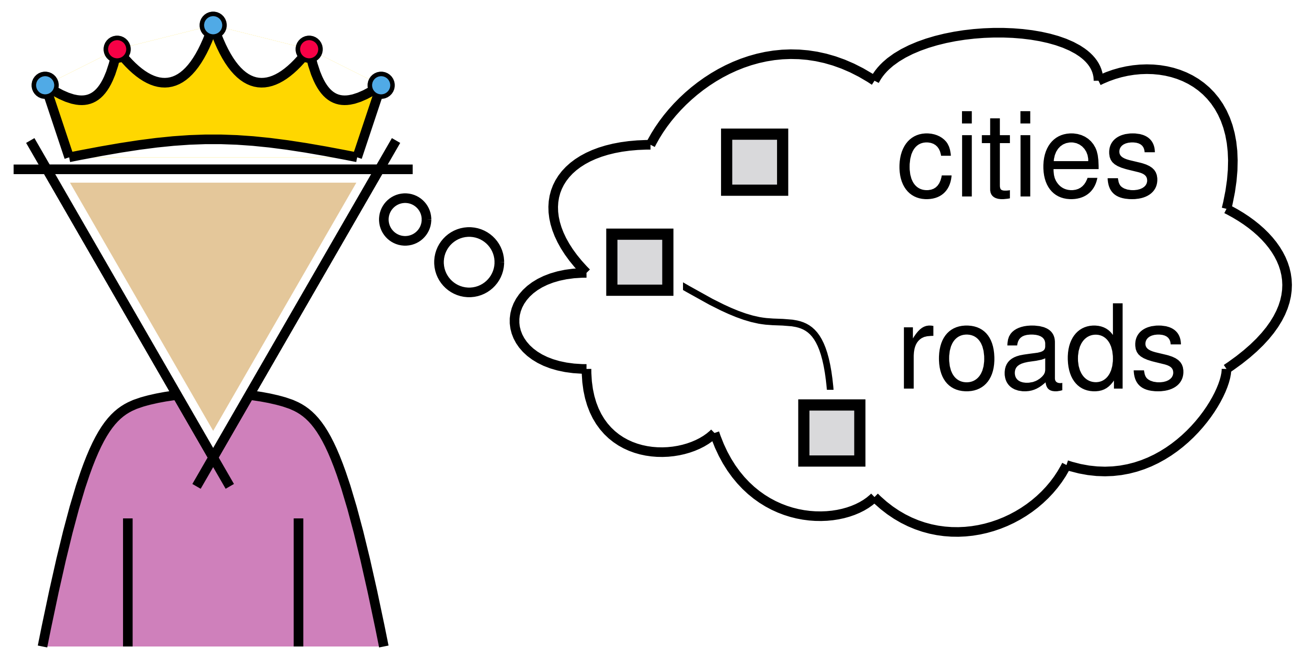

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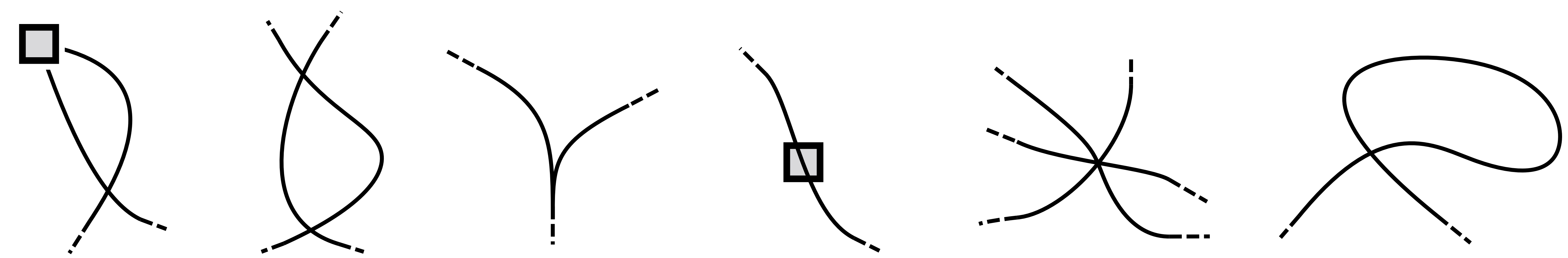

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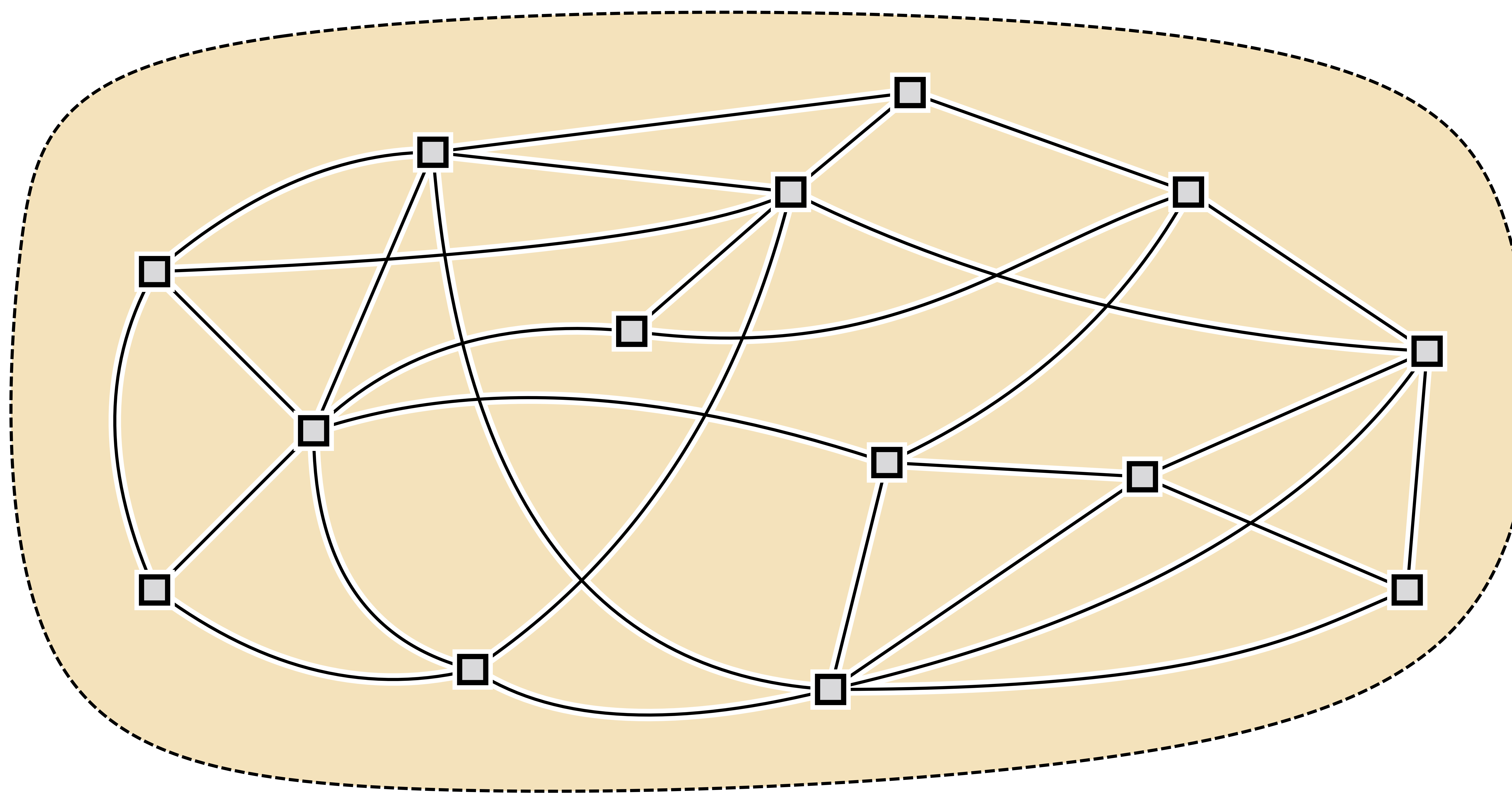

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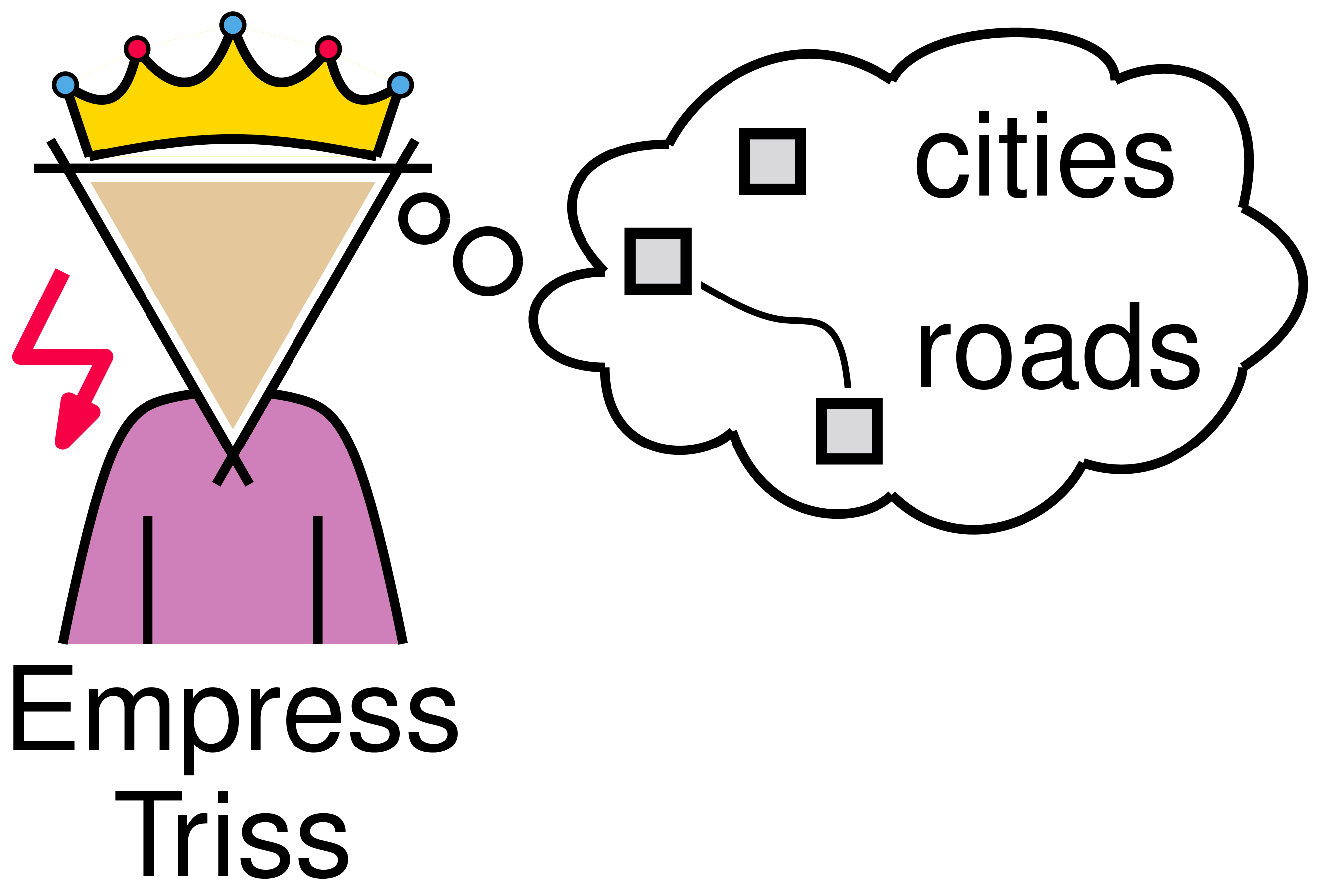

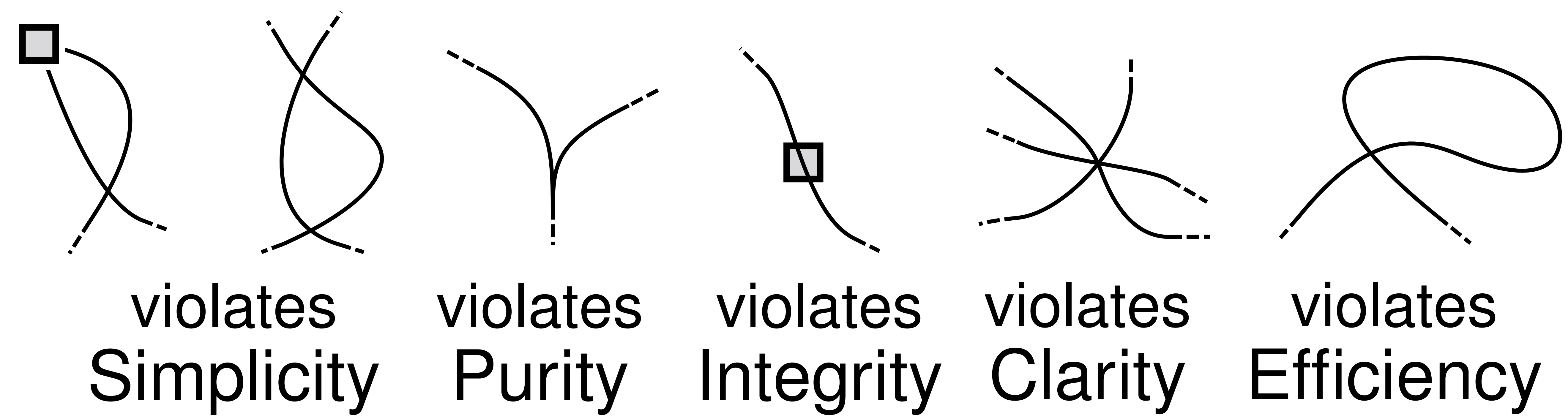

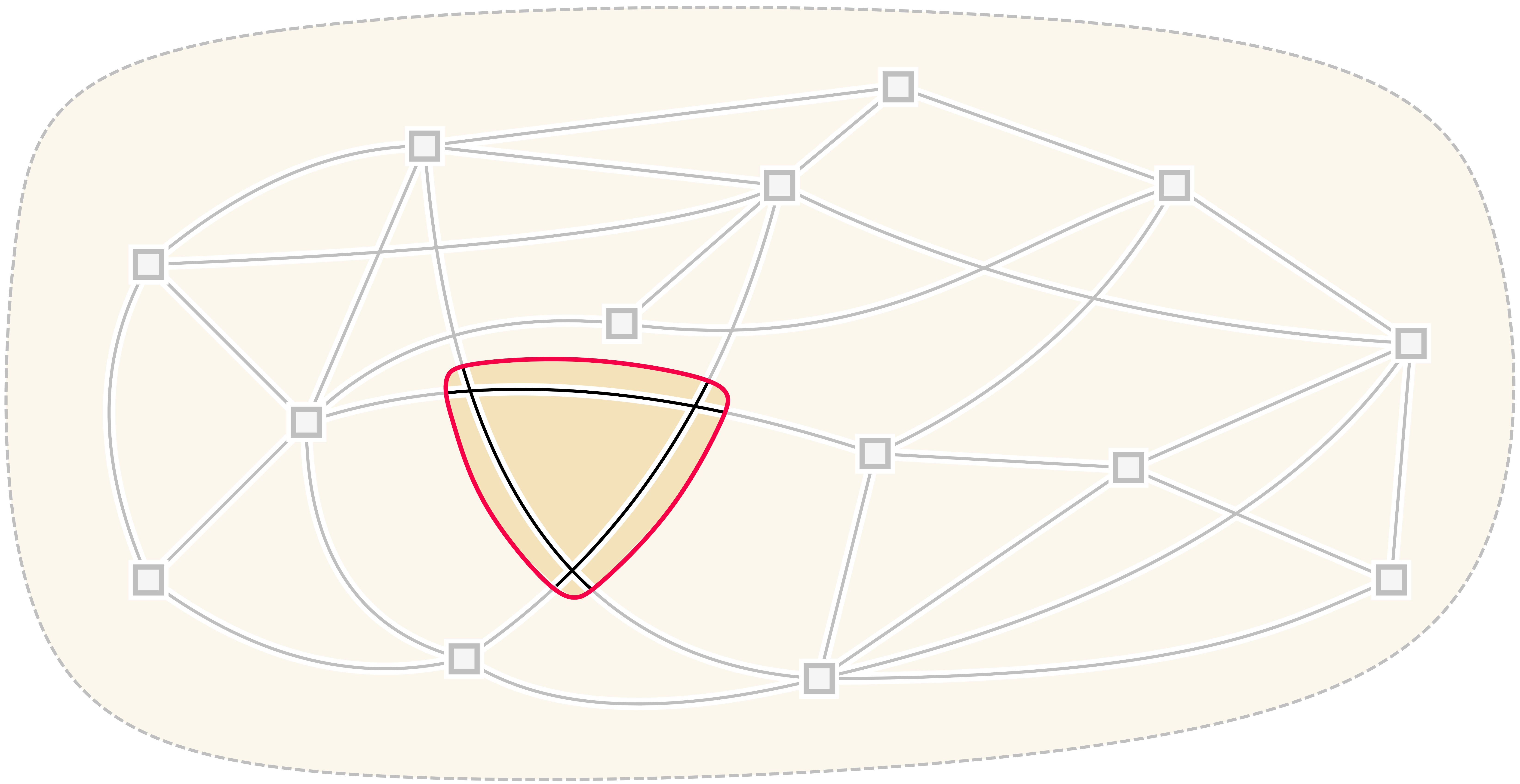

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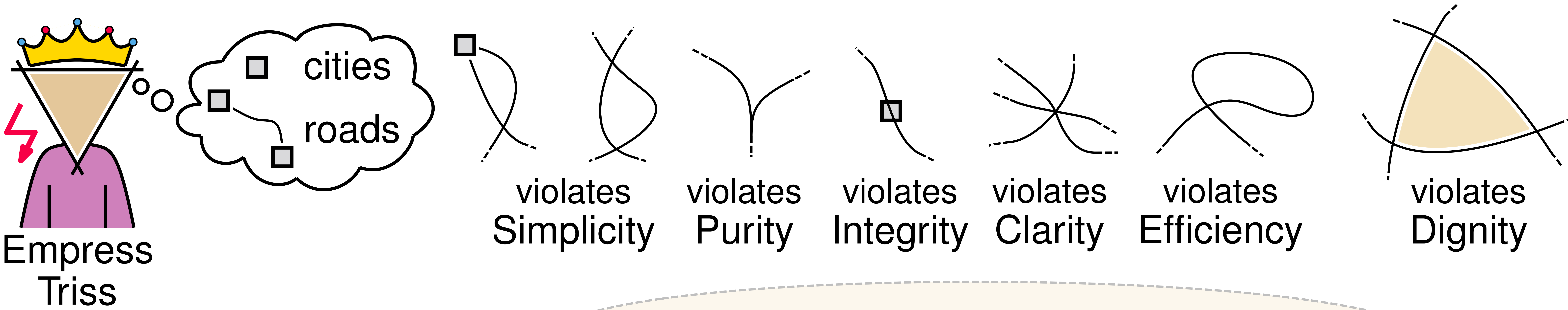

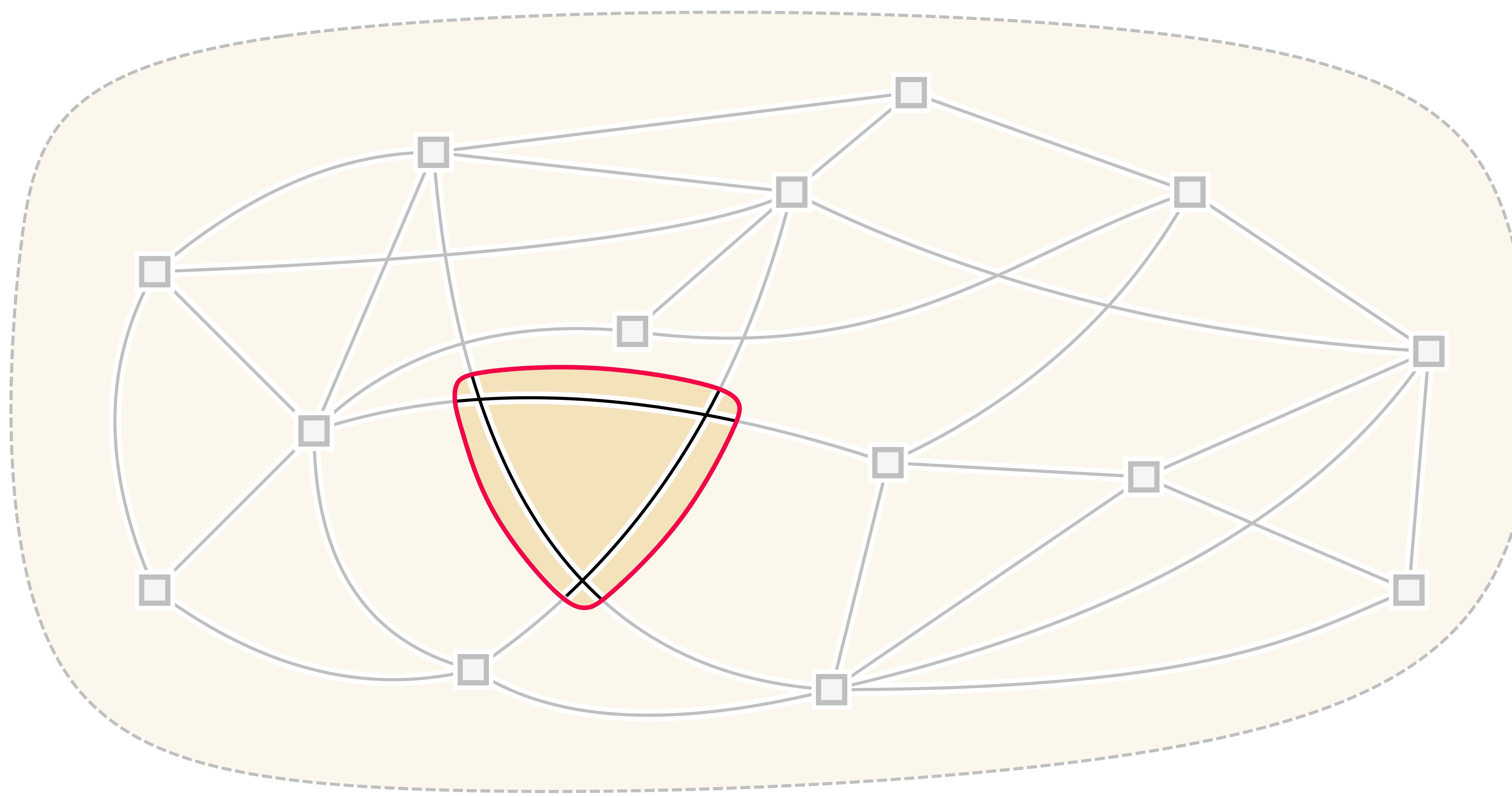

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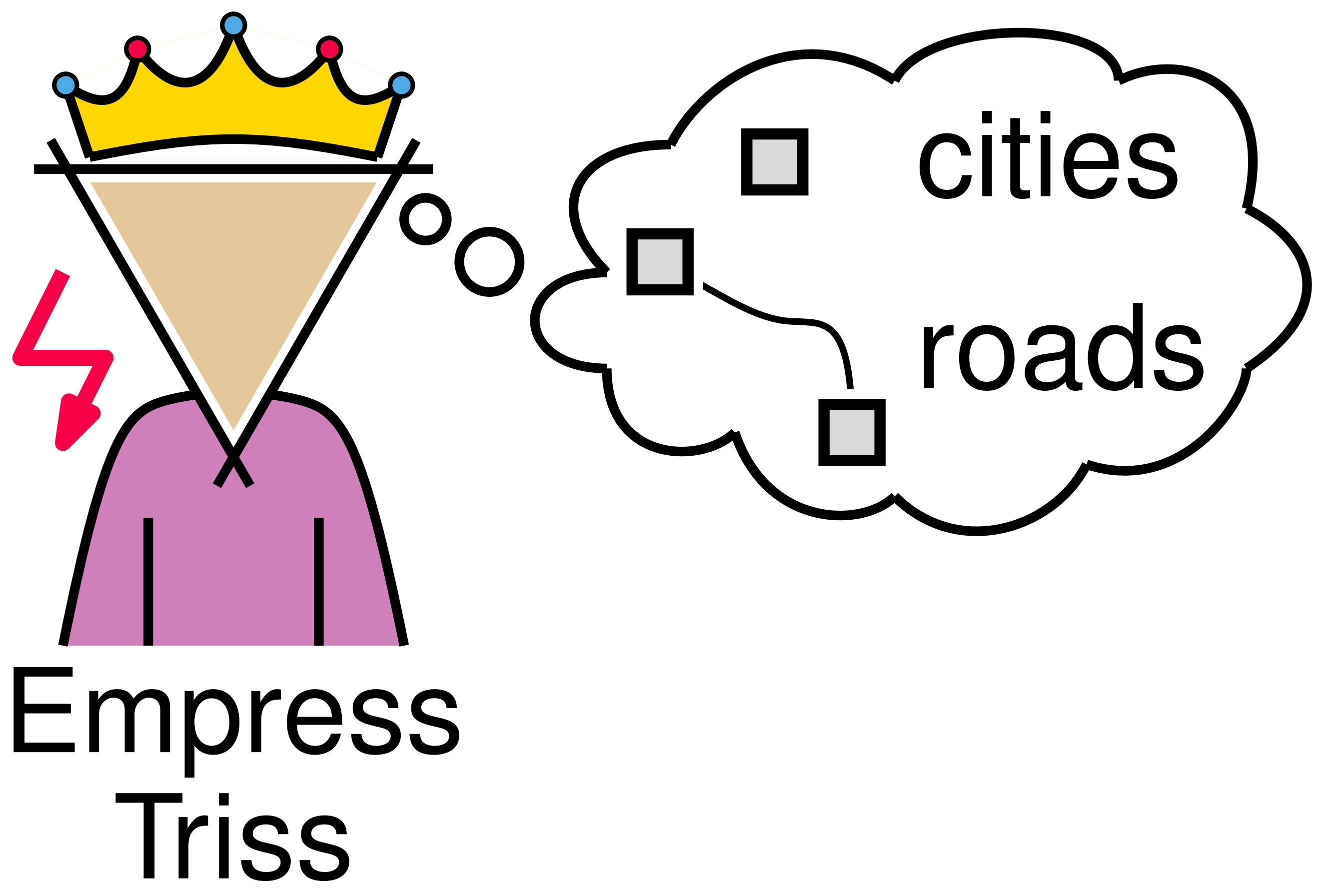

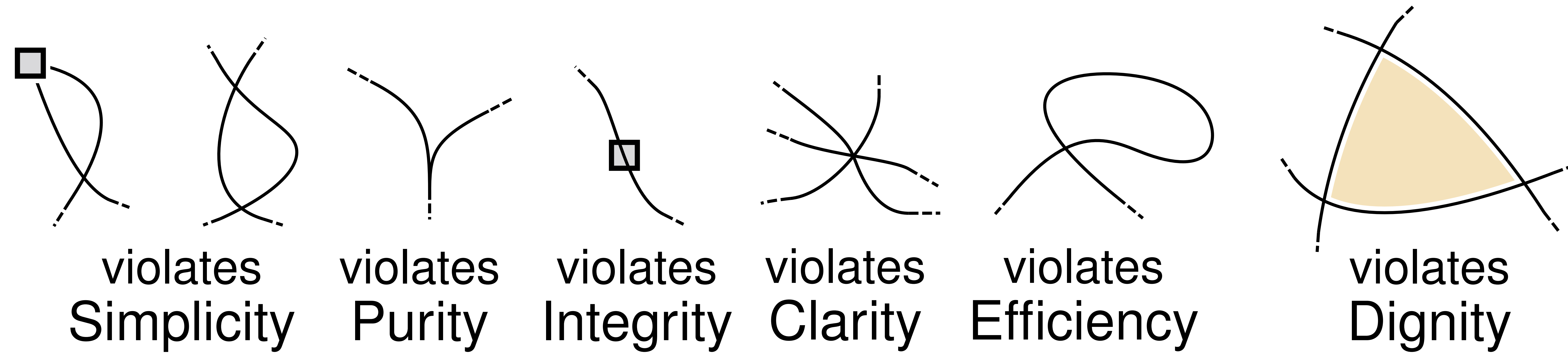

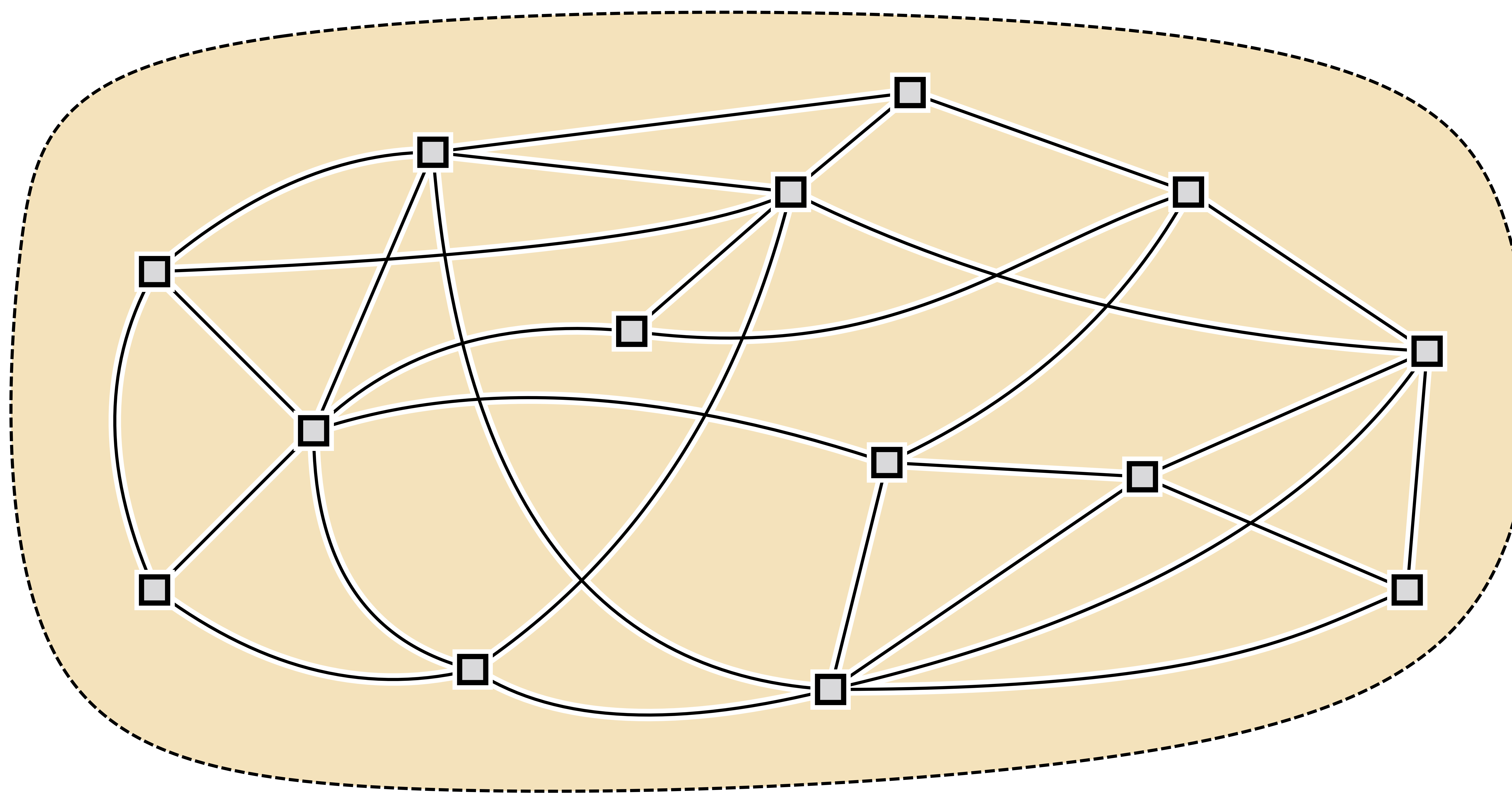

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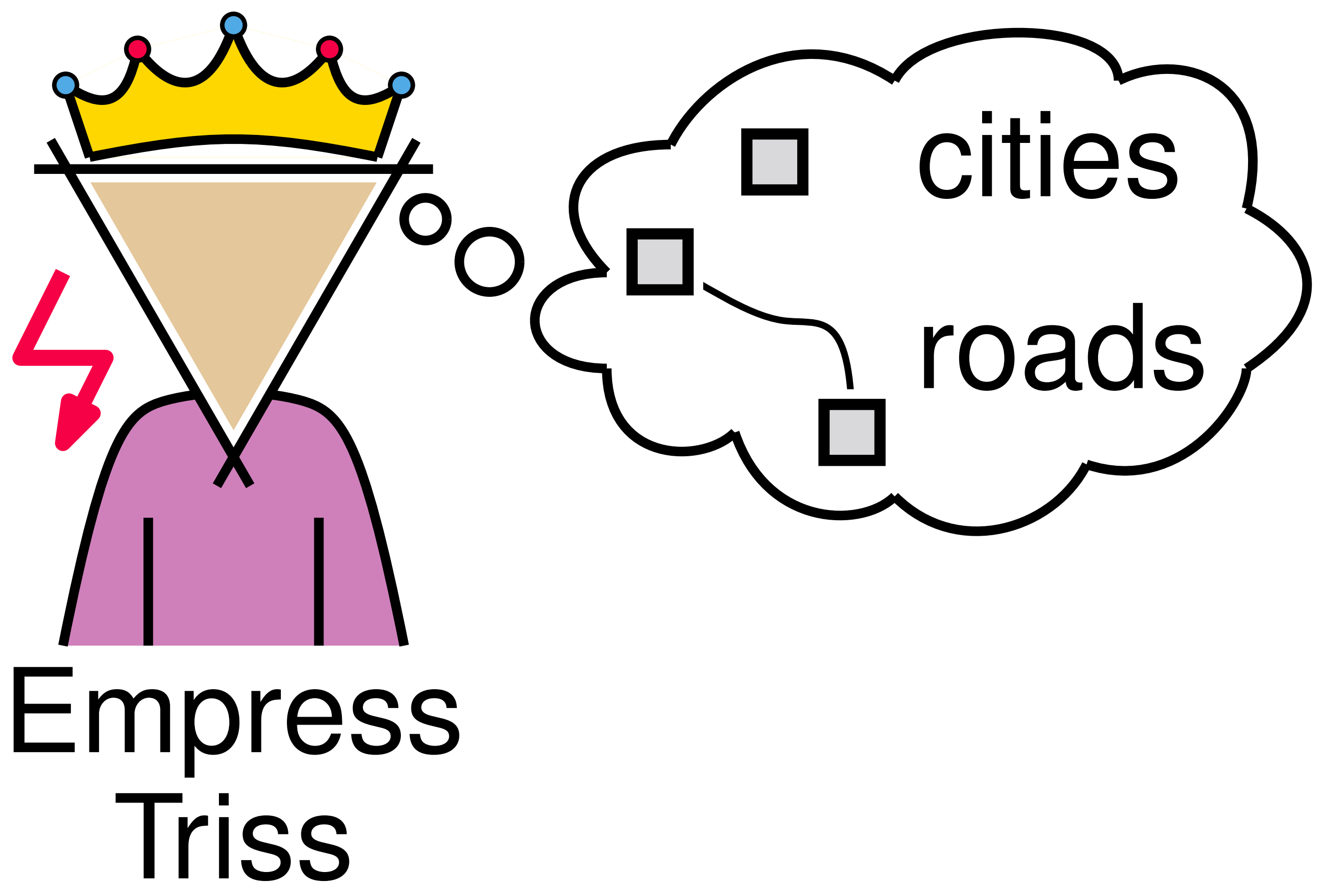

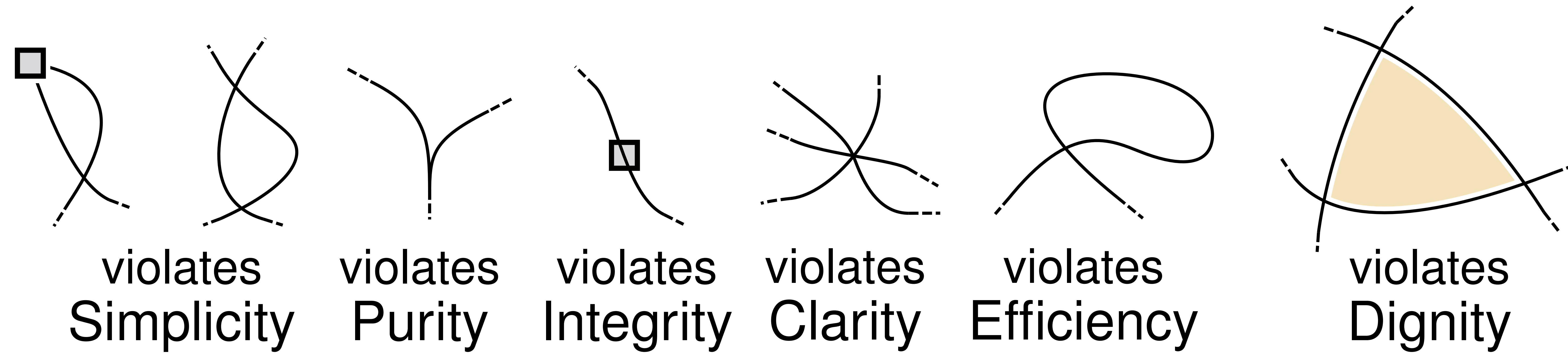

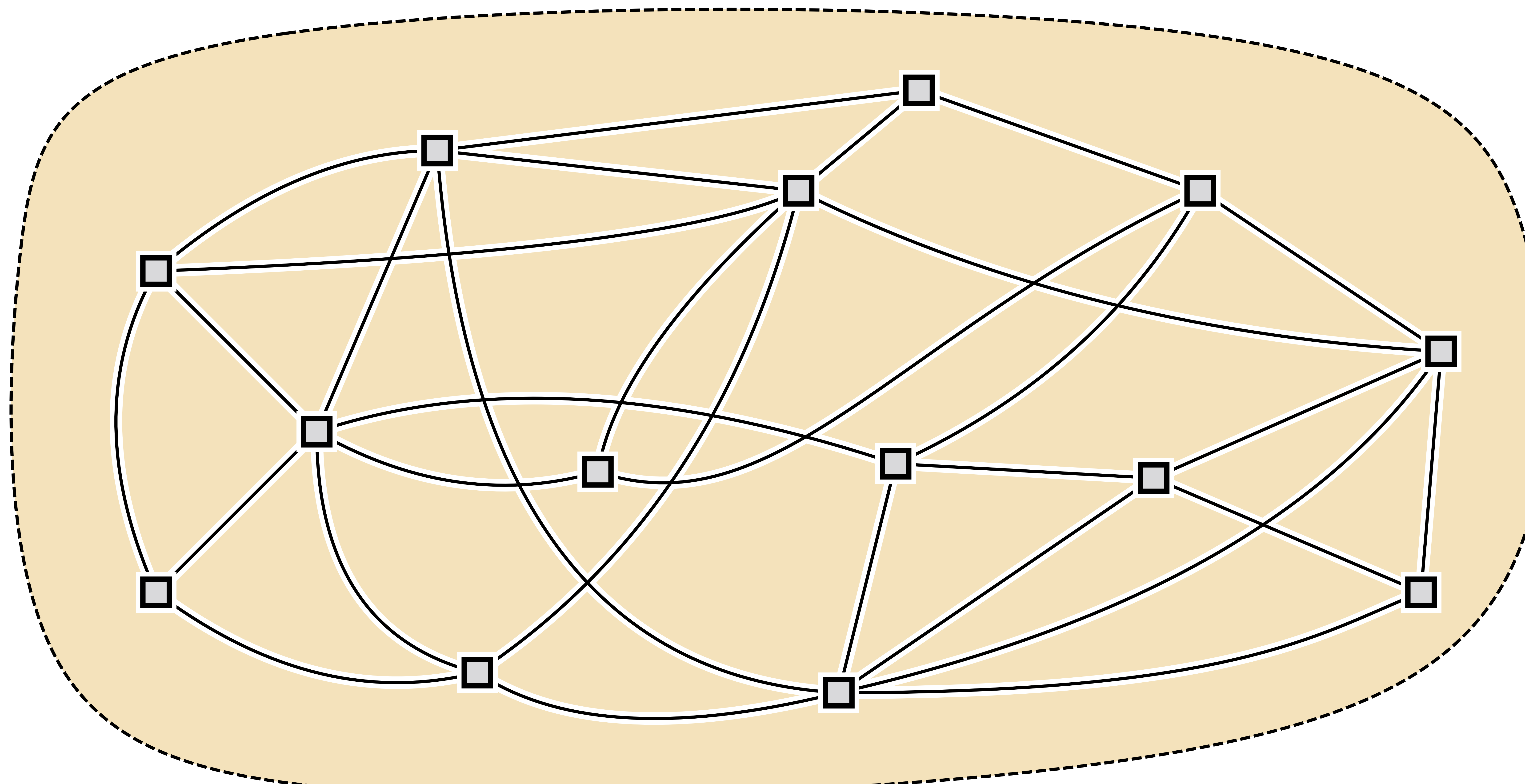

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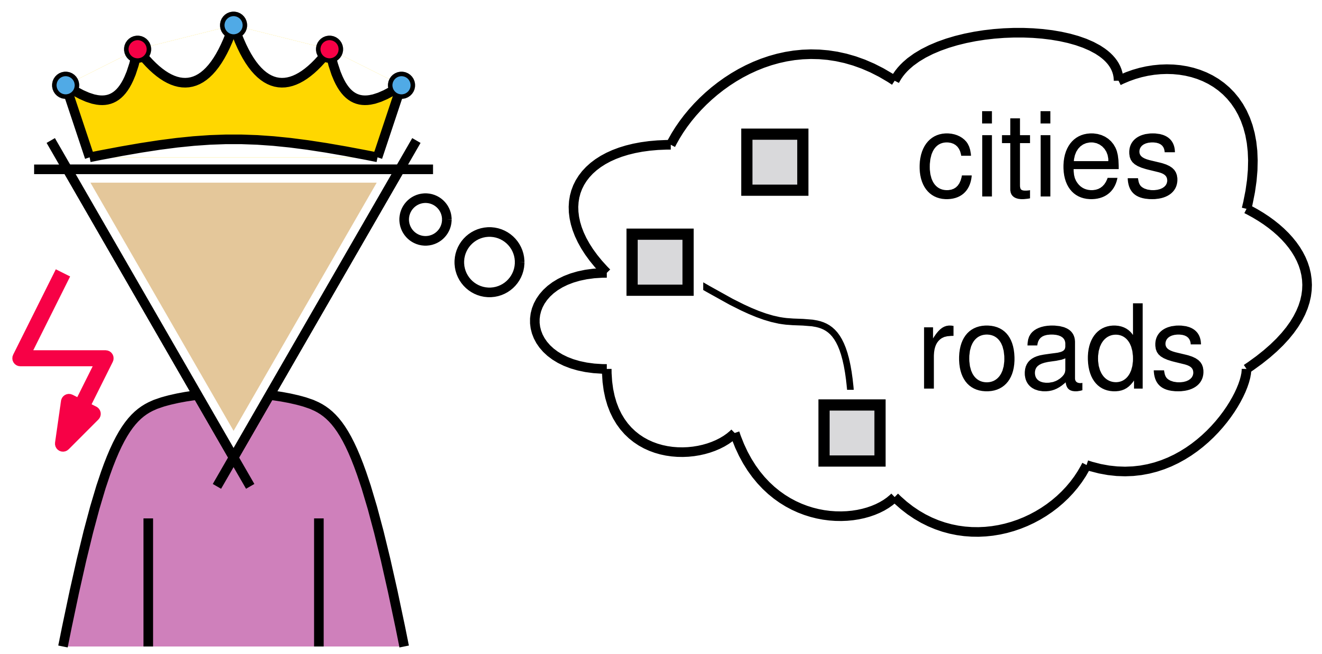
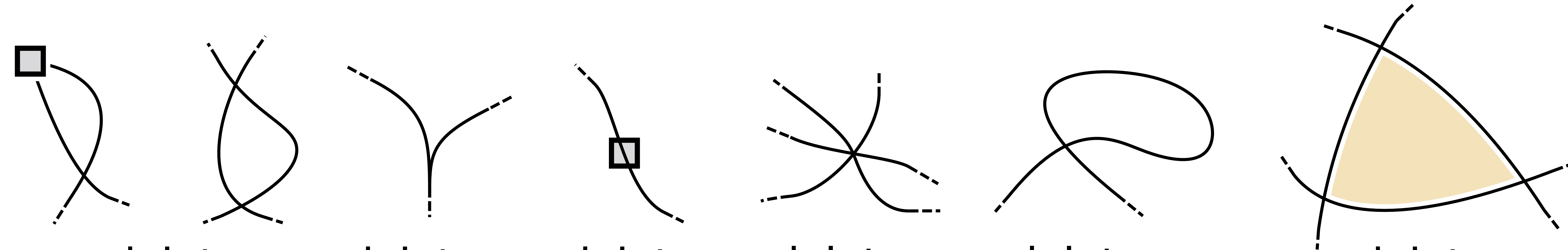
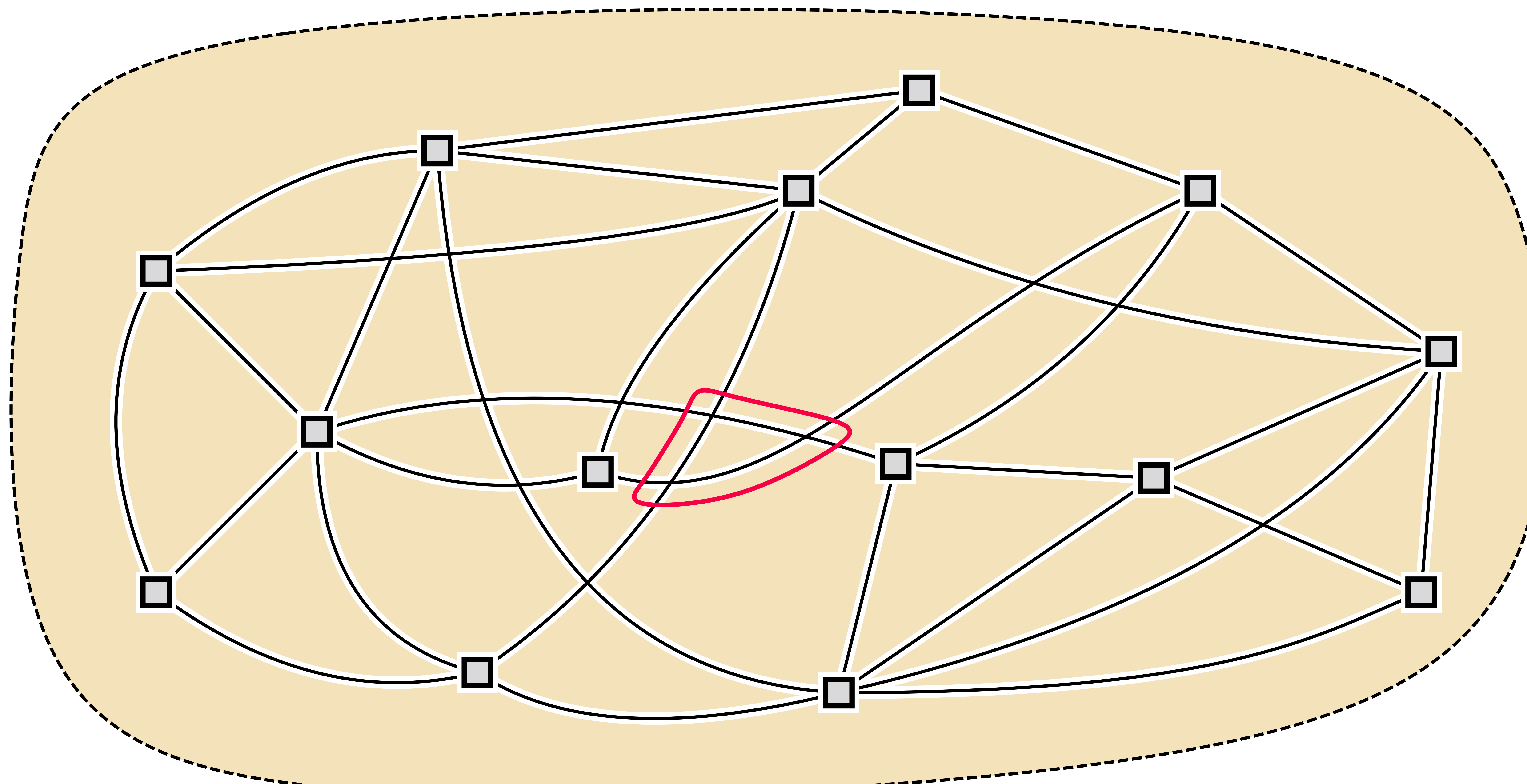


Empress Triss



violates Simplicity | violates Purity | violates Integrity | violates Clarity | violates Efficiency | violates Dignity



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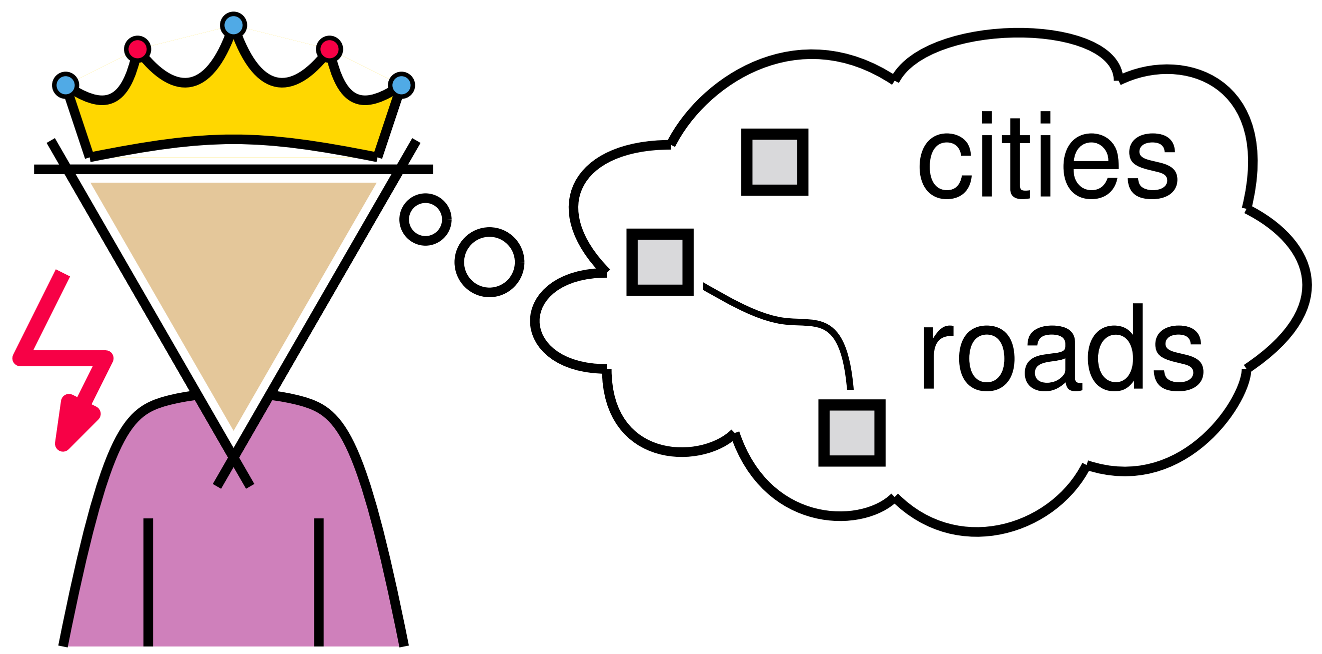

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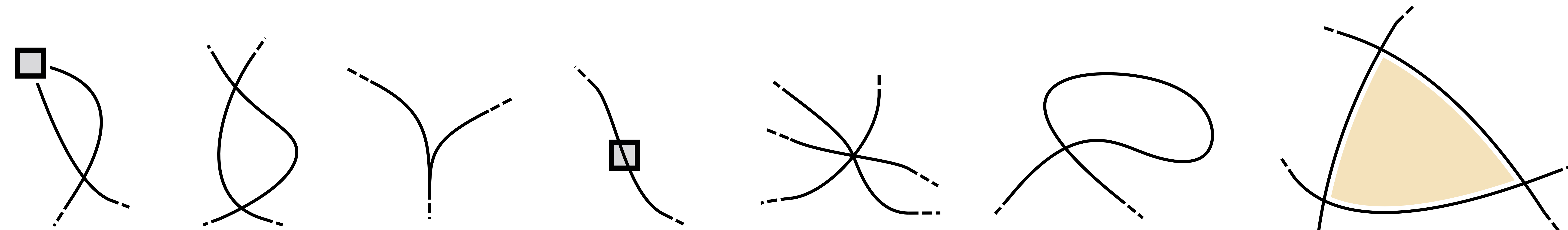

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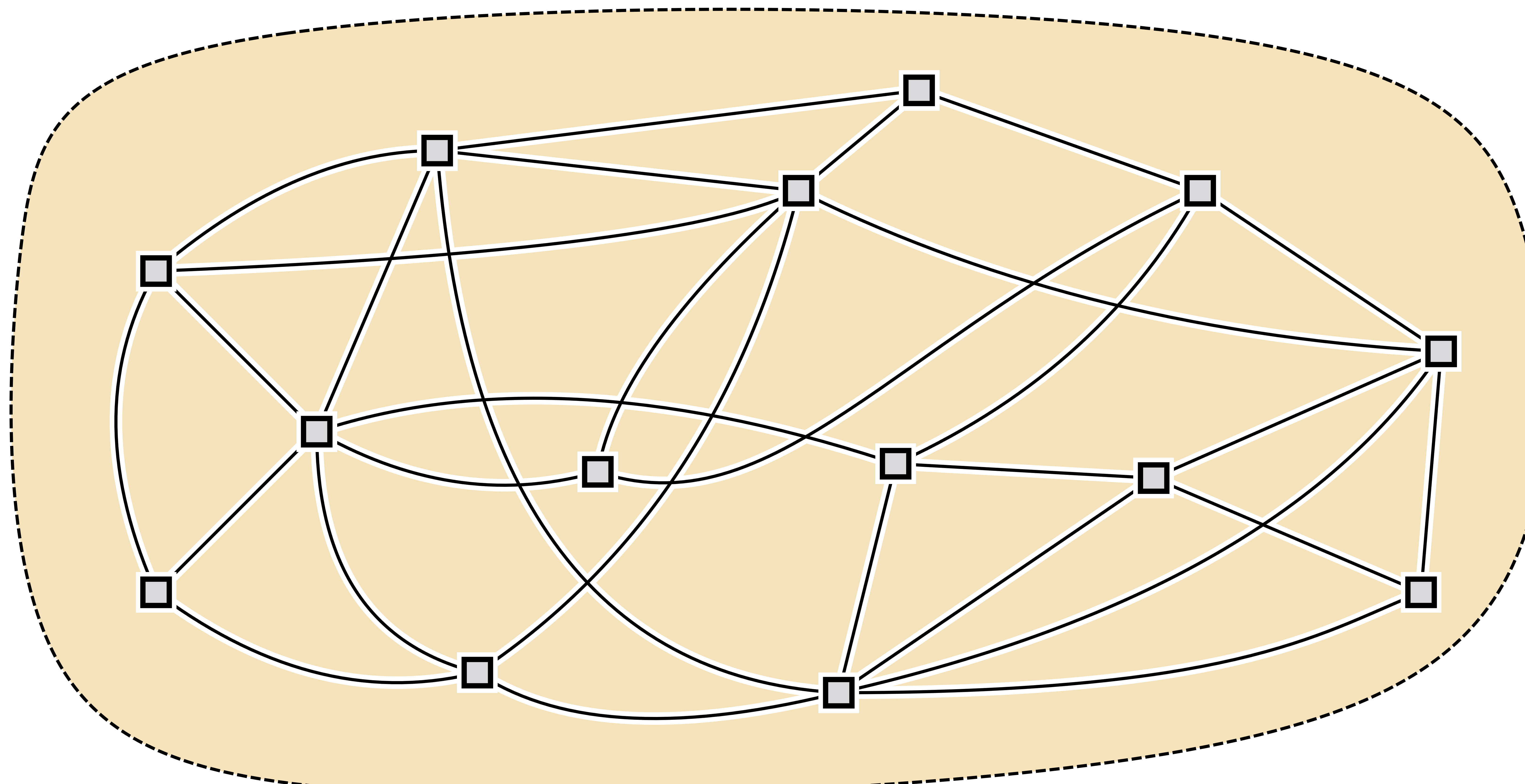

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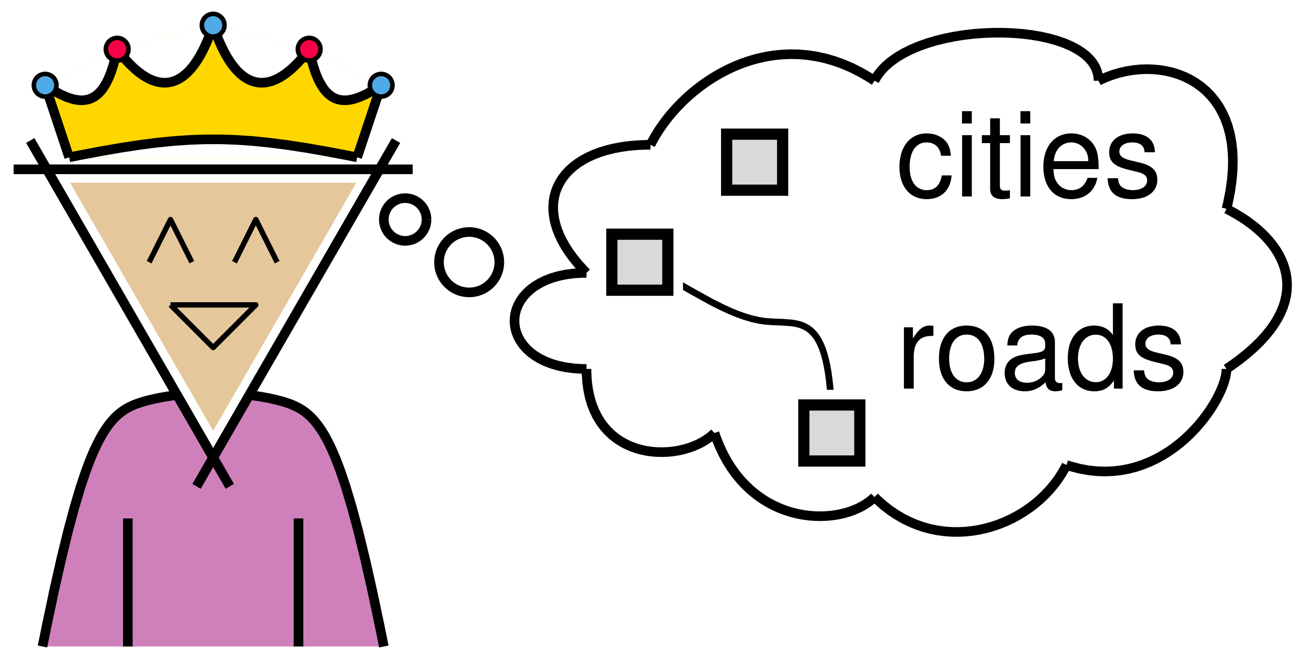

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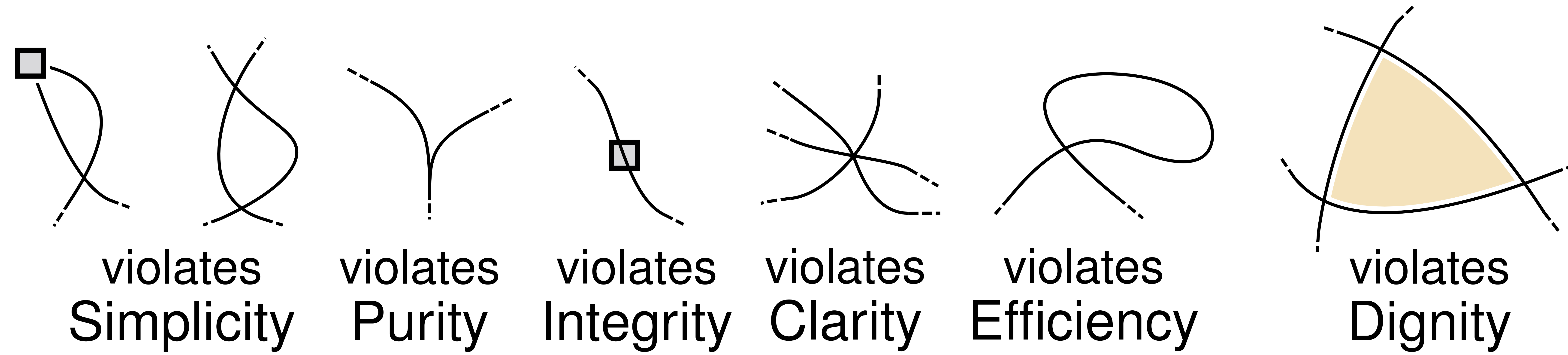

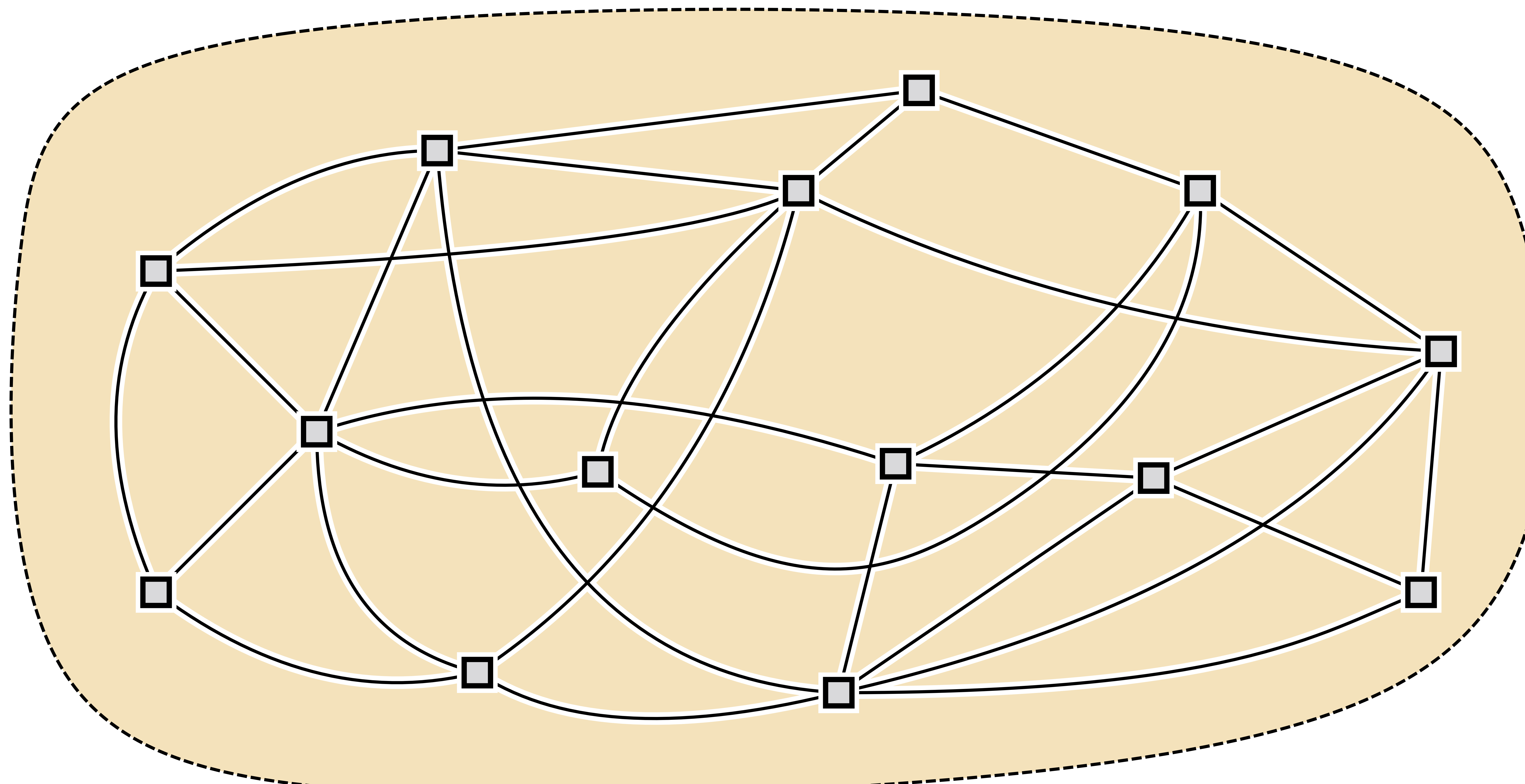

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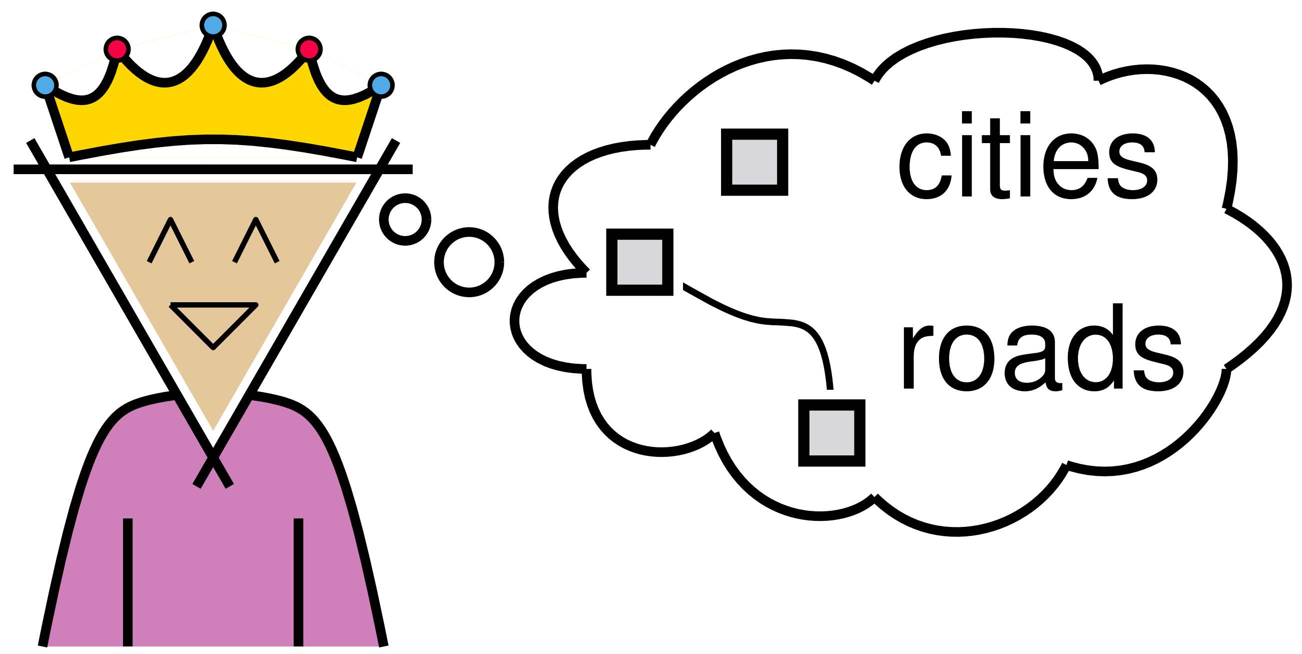

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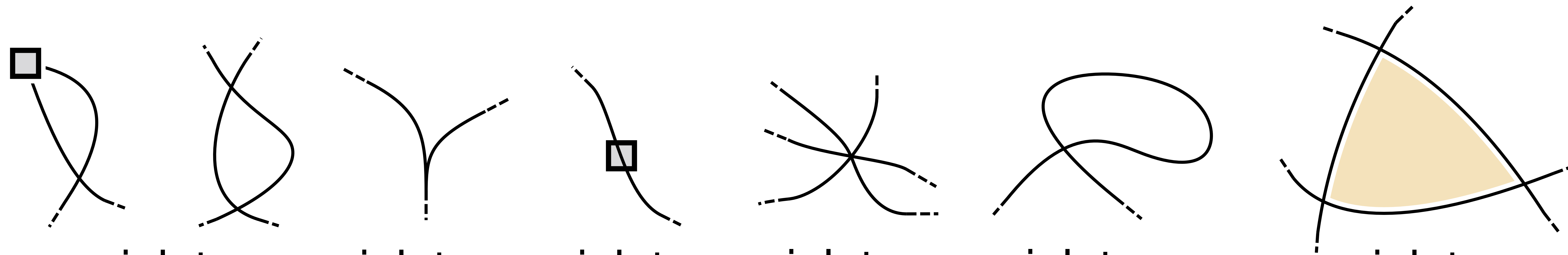

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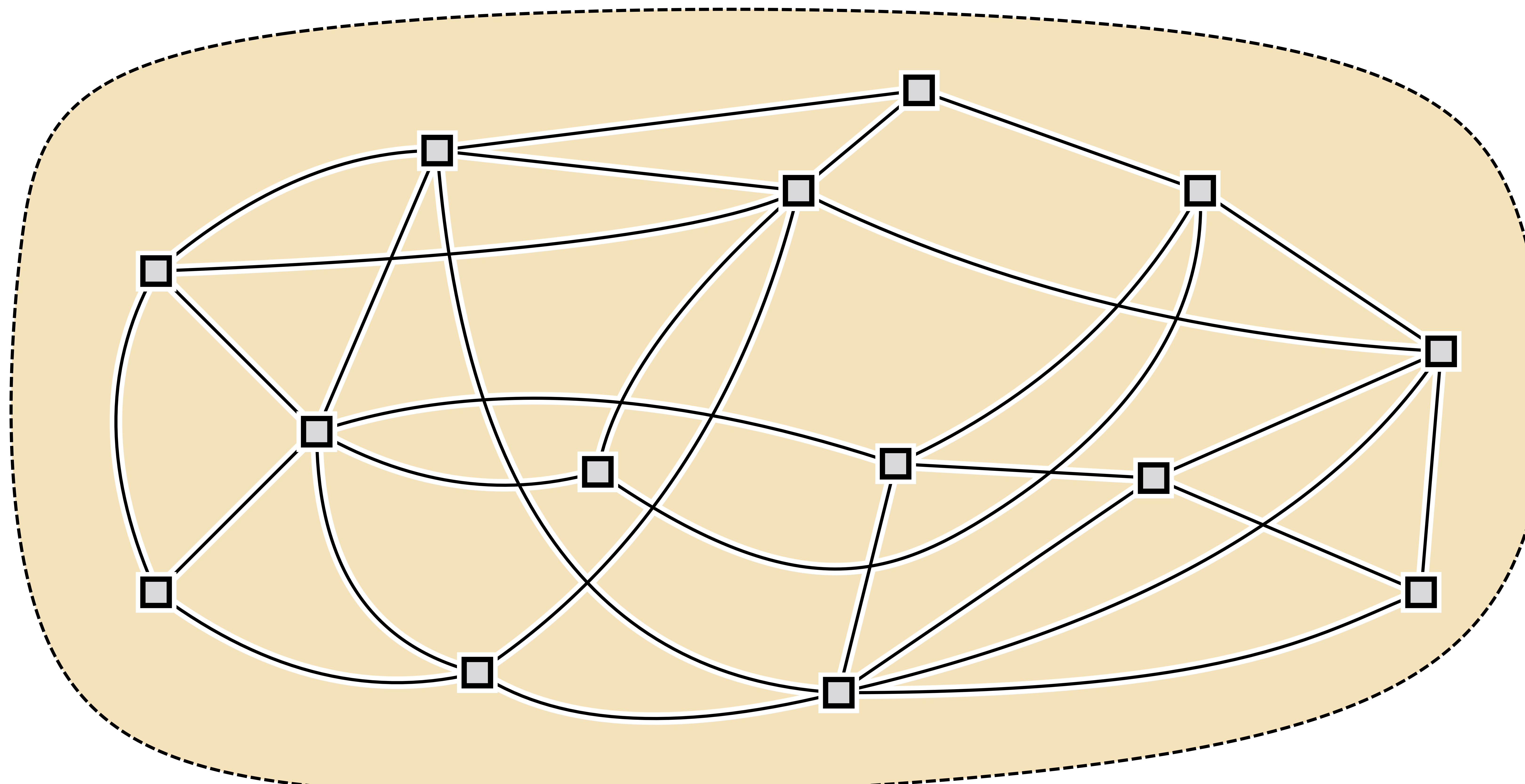
- Now, the empress demands a direct road between every pair of her 15 cities



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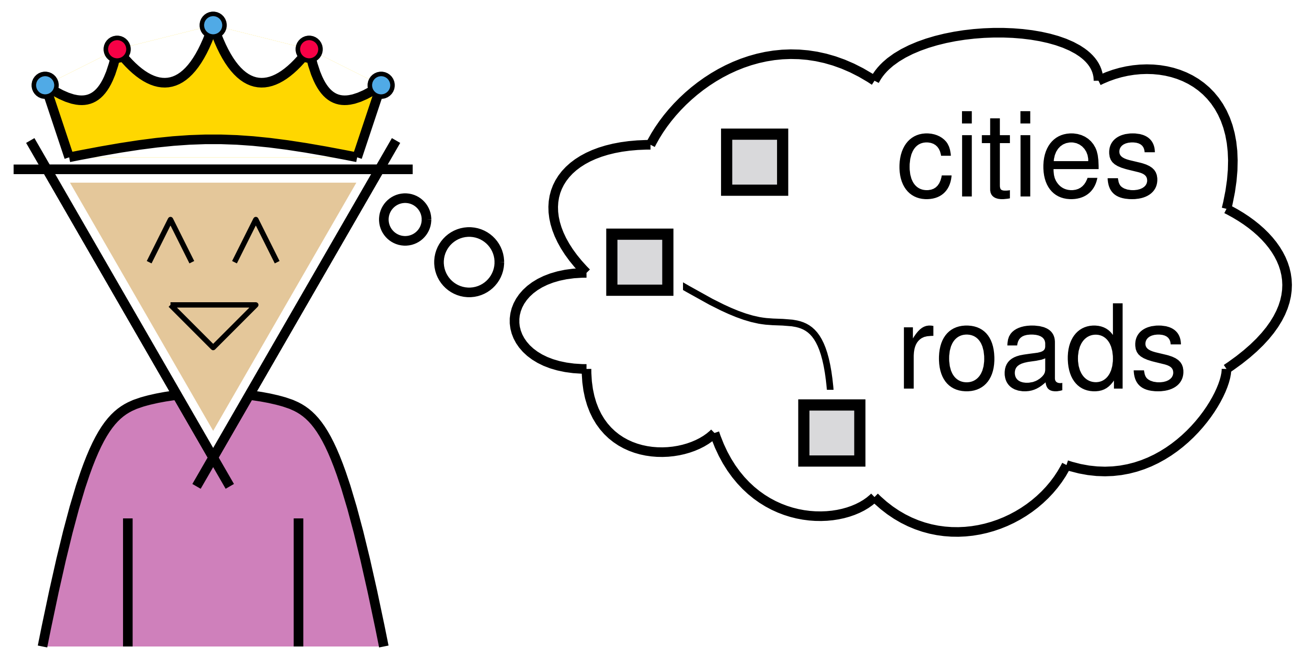

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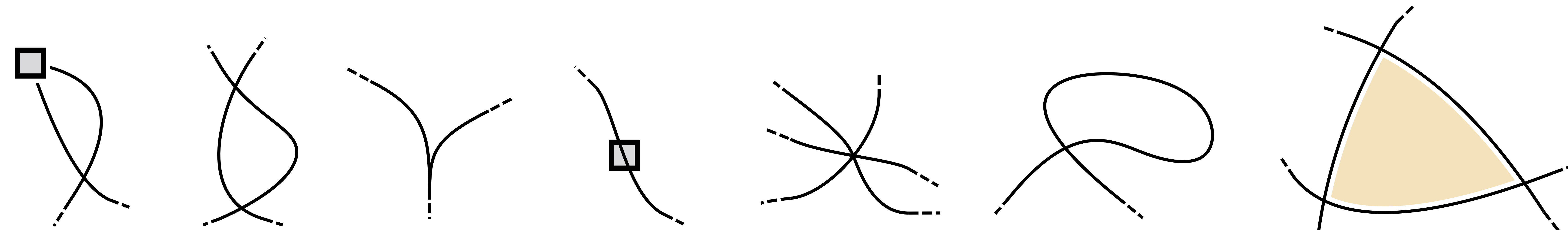

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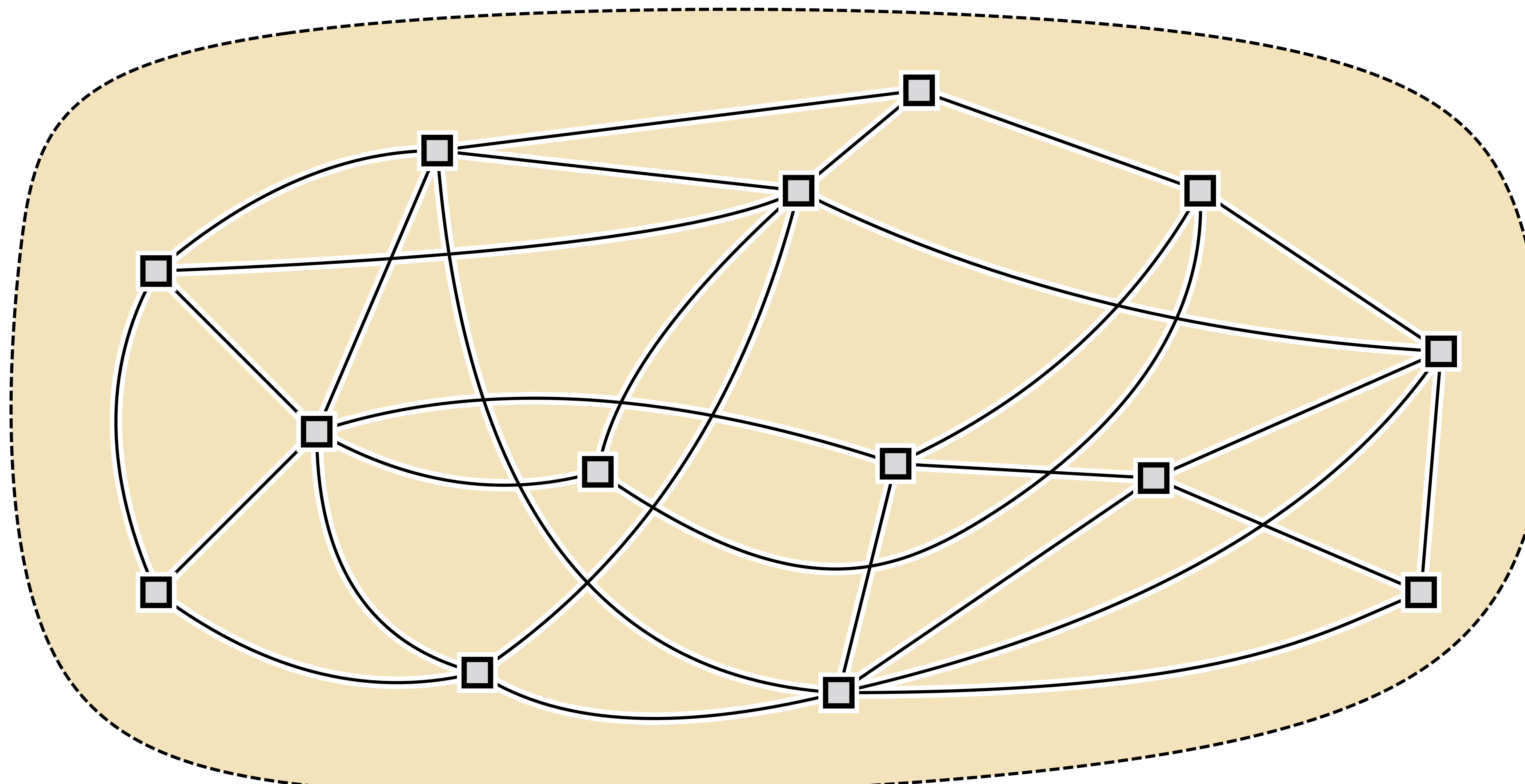
- [Ackerman, Tardos; 2007] Under SPICED constraints, an n-city empire contains at most $8n - 20$ roads.



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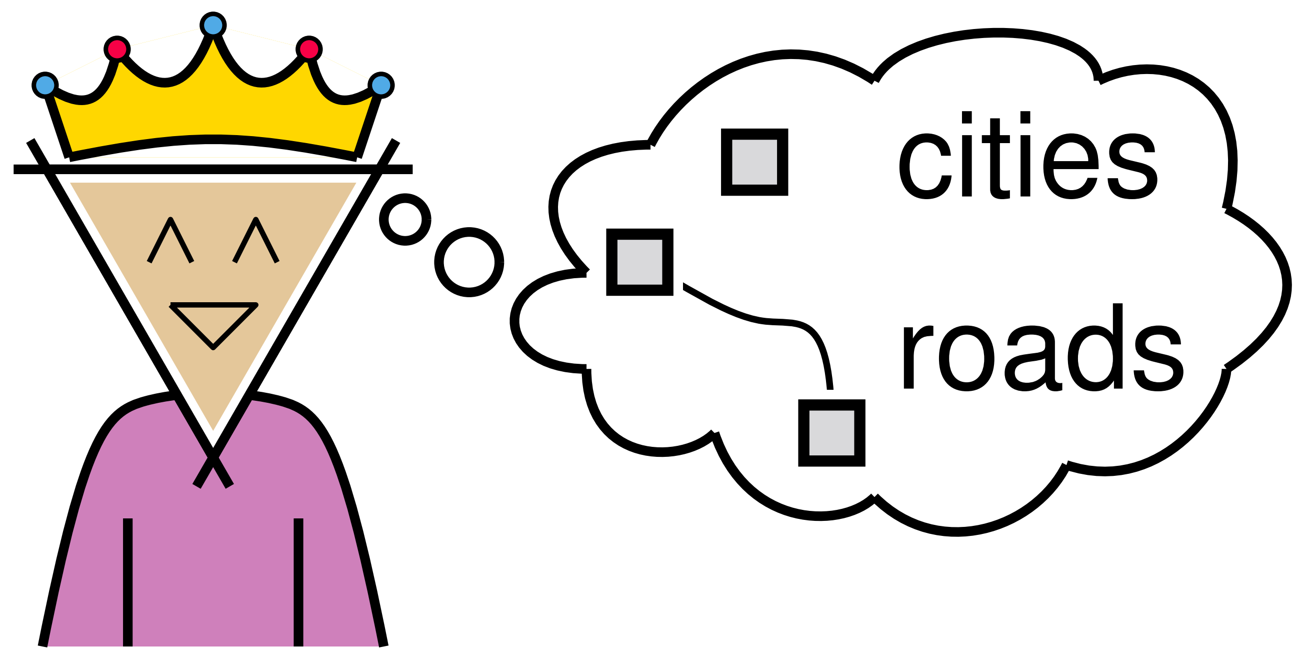

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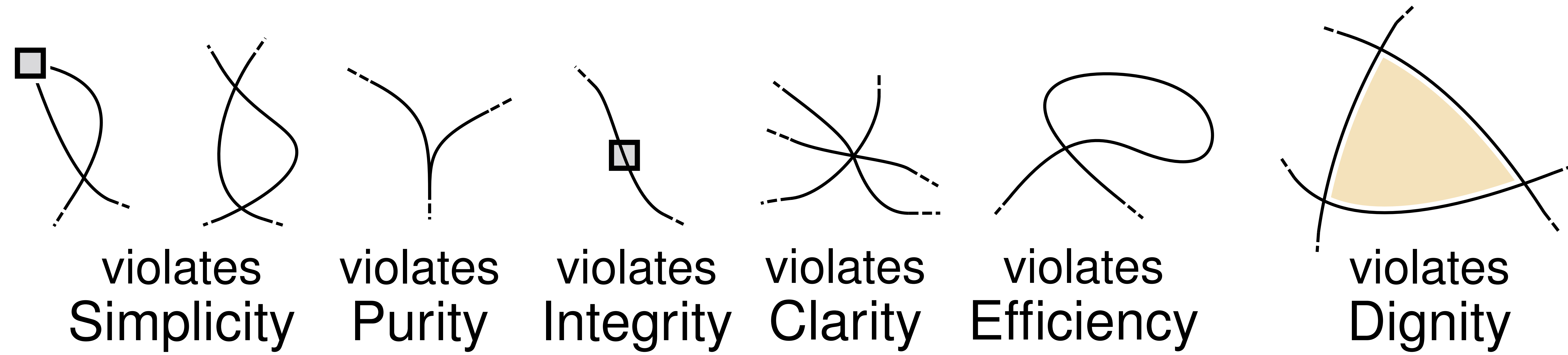

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  $$\binom{15}{2} > 8 \cdot 15 - 20$$

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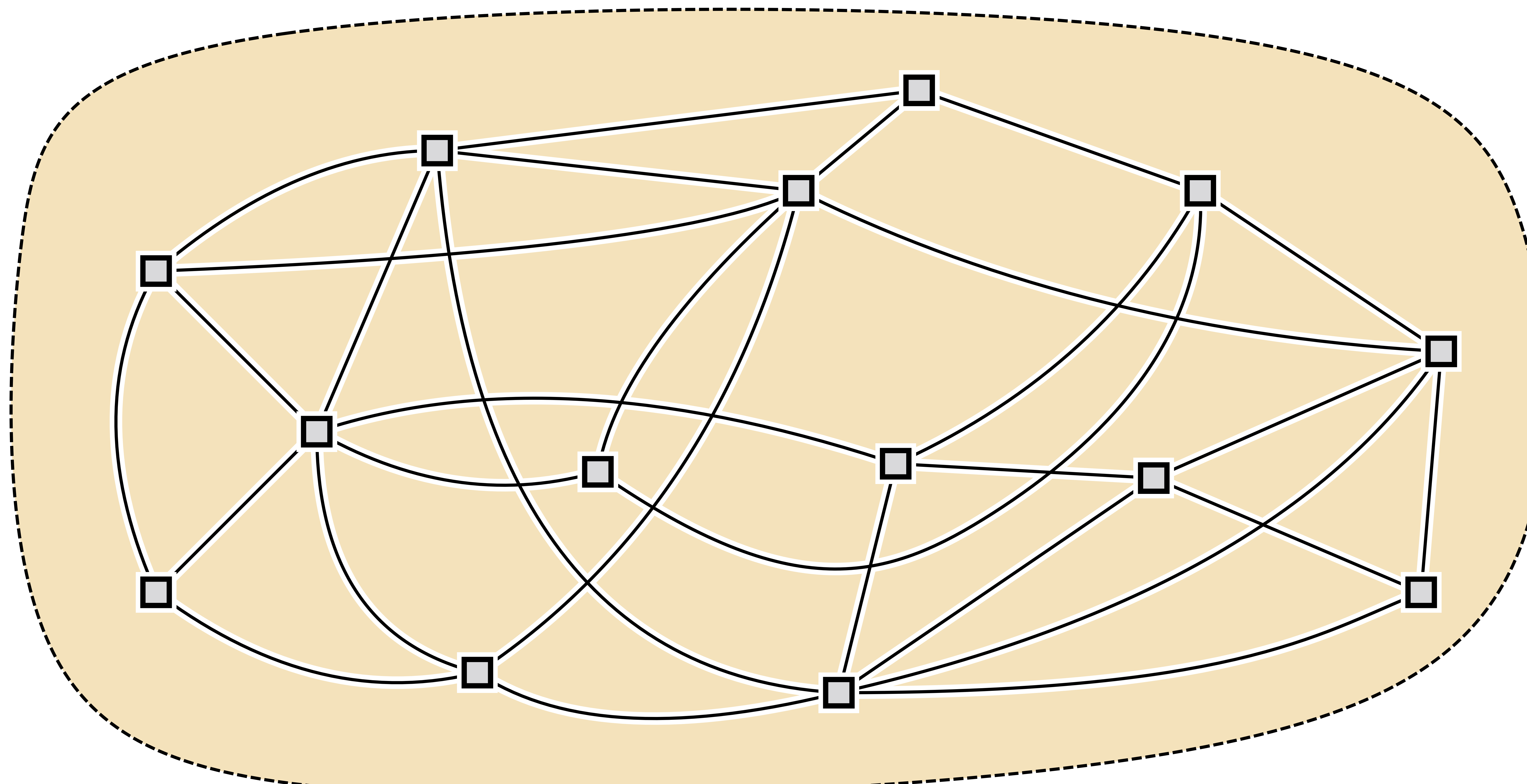

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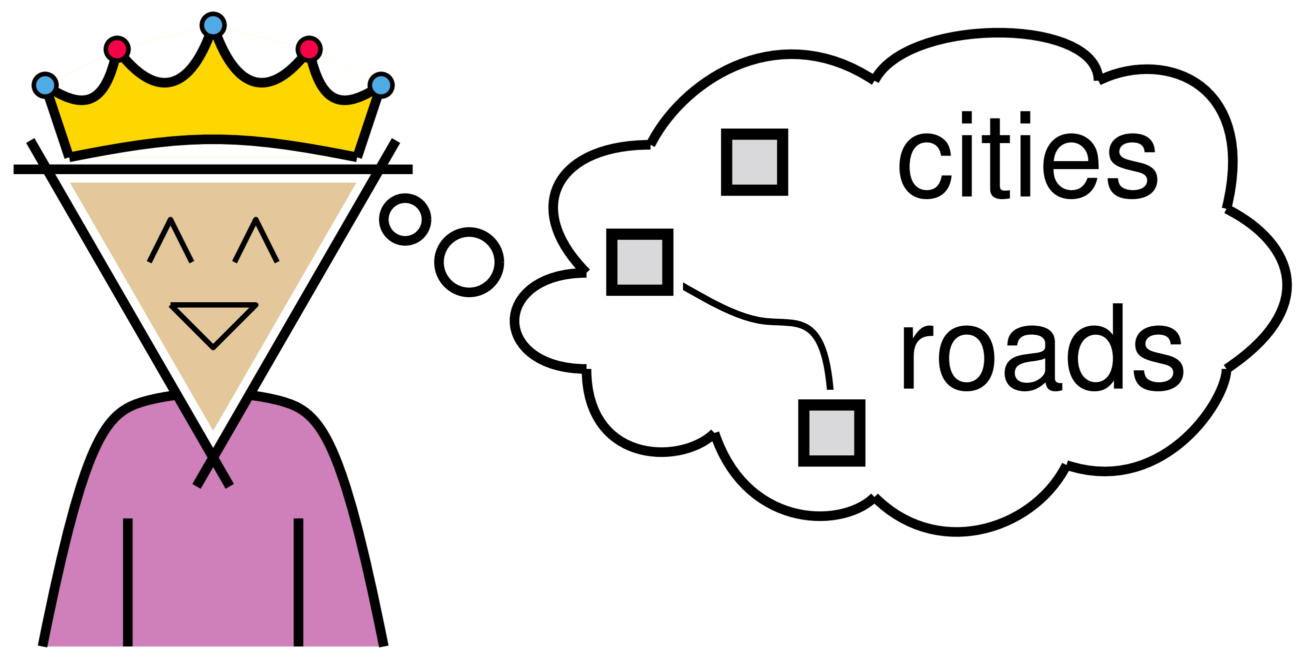

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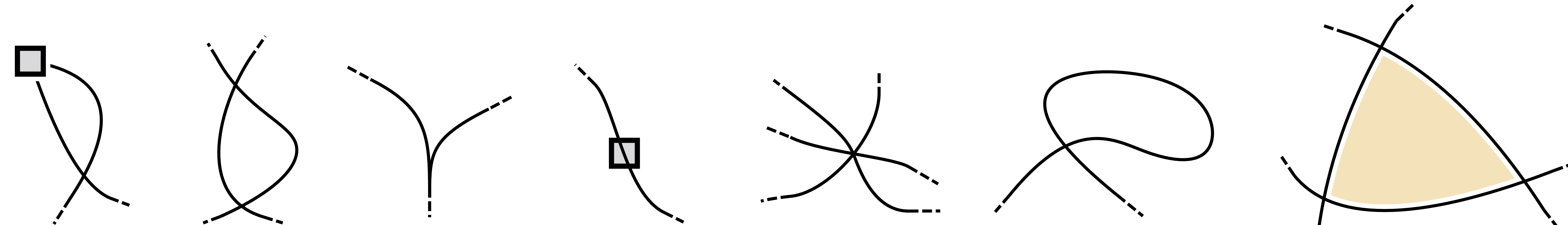

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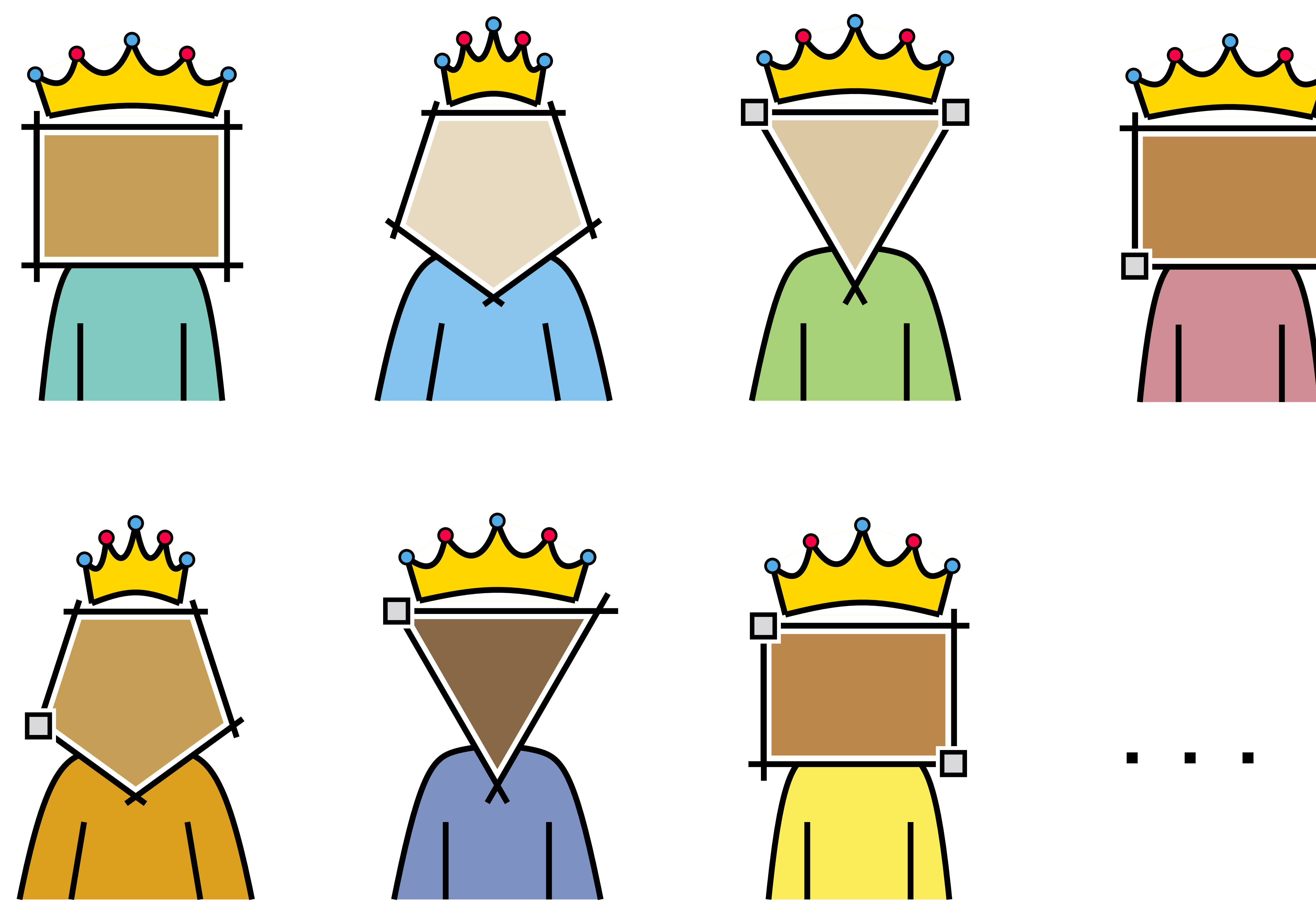
  $$\binom{15}{2} > 8 \cdot 15 - 20$$
- [Ackerman, Tardos; 2007] Under SPICED constraints, an n-city empire contains at most $8n - 20$ roads.
- **Question:** For a given emperor/empress, what is the maximum number of roads in an n-city empire fulfilling SPICED?



Most rulers allow complete empires

- **Question:**
  For a given cell type c, what is the maximum number of edges in an n-vertex c-free simple drawing?

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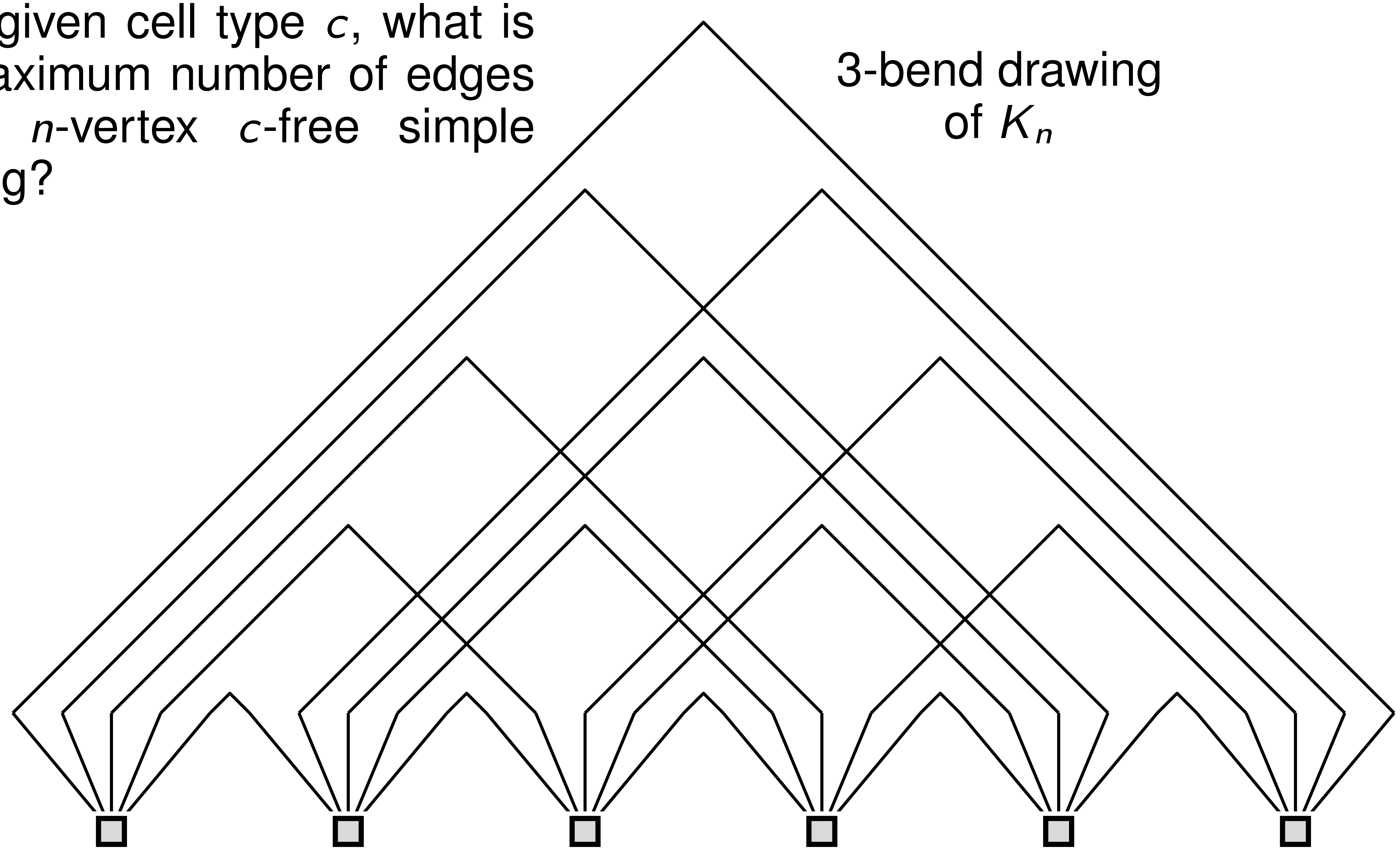

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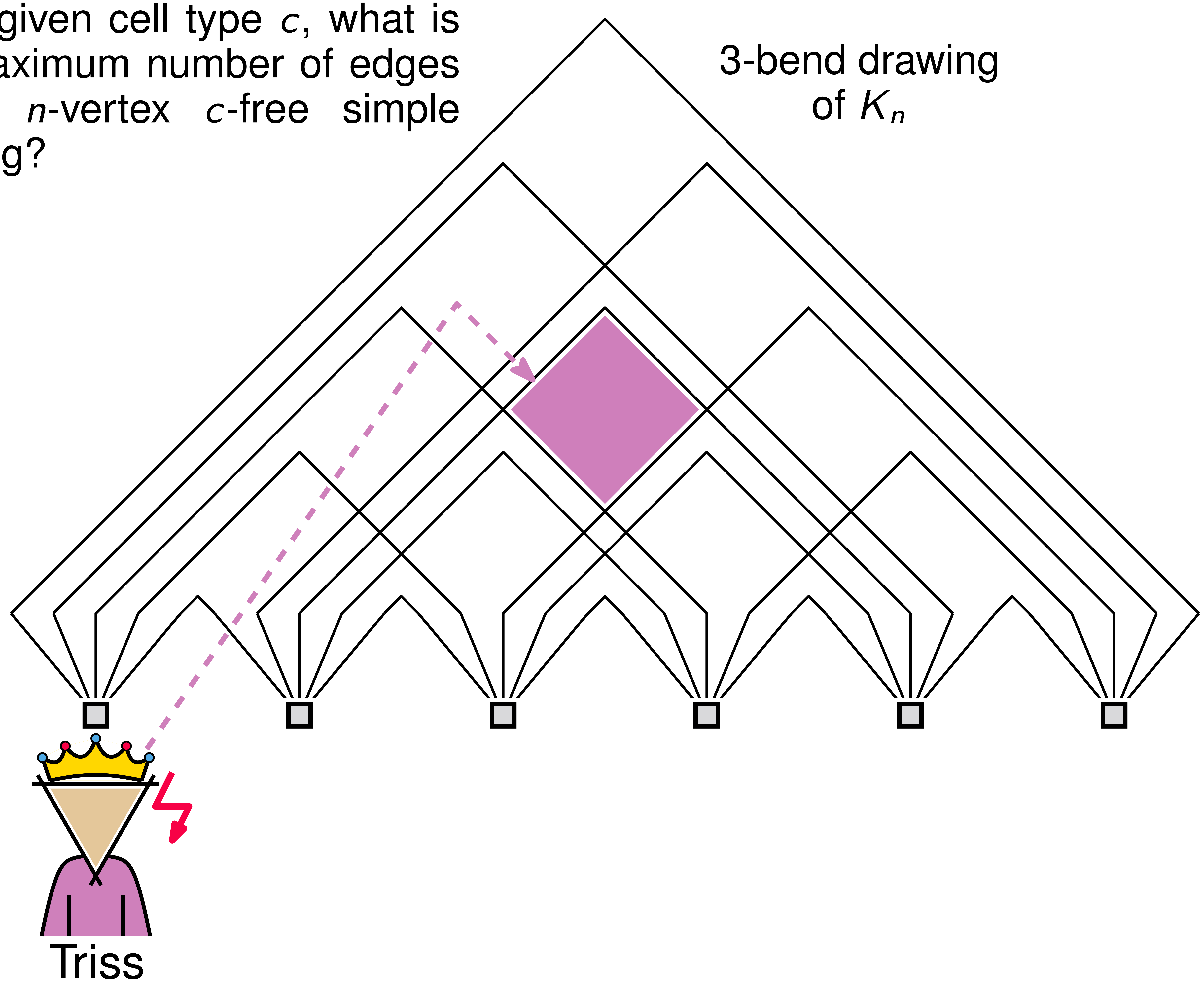

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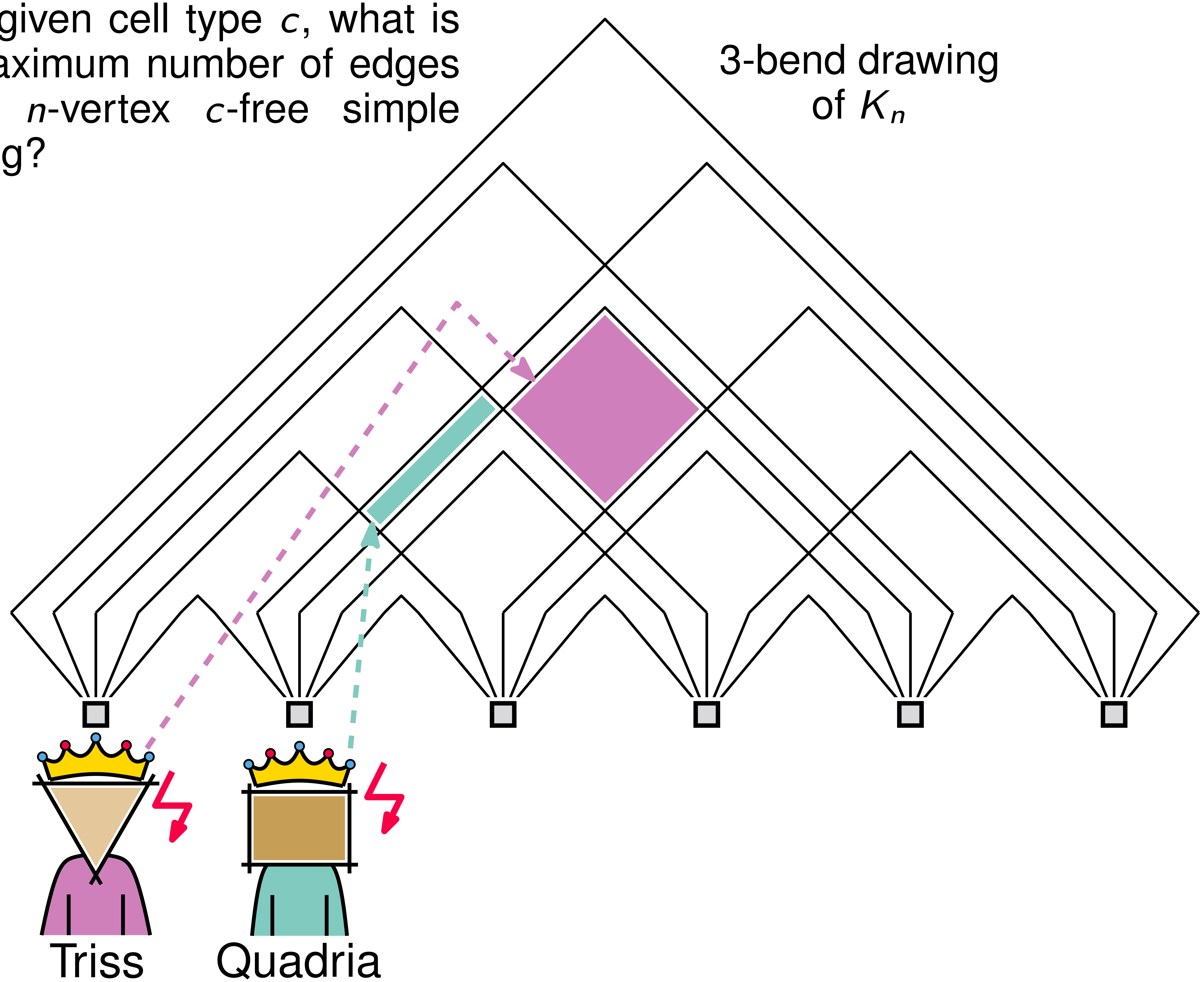

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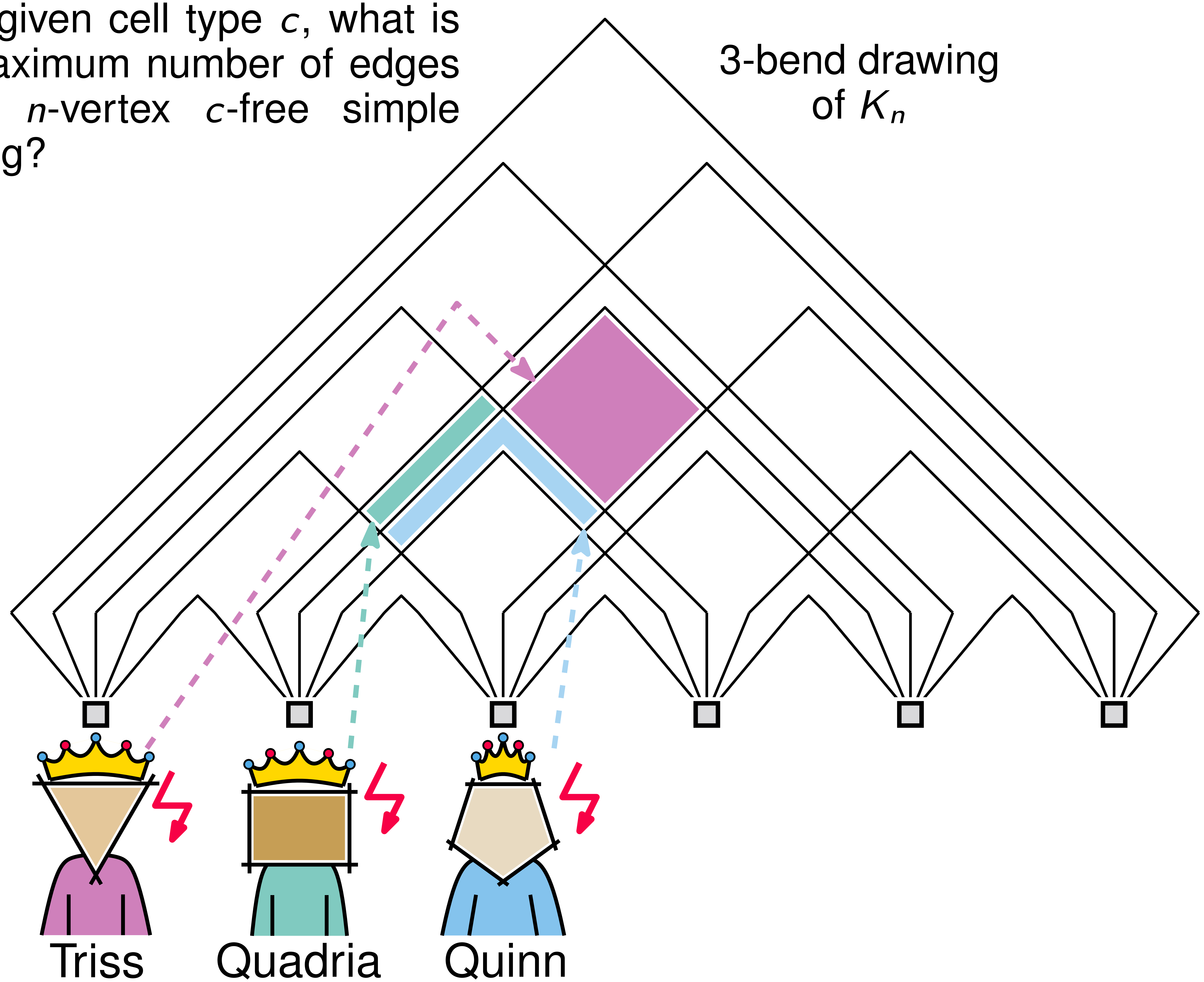

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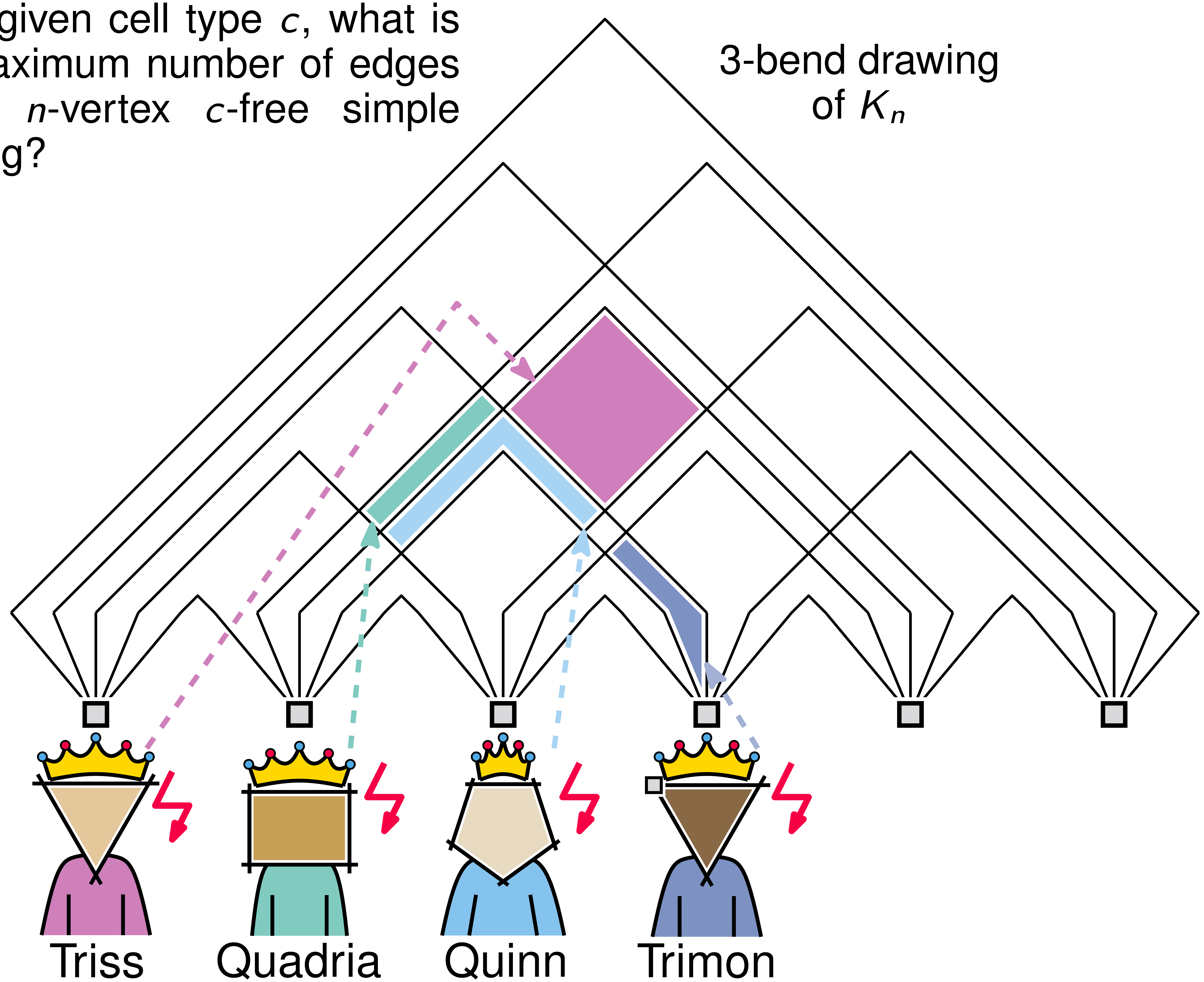

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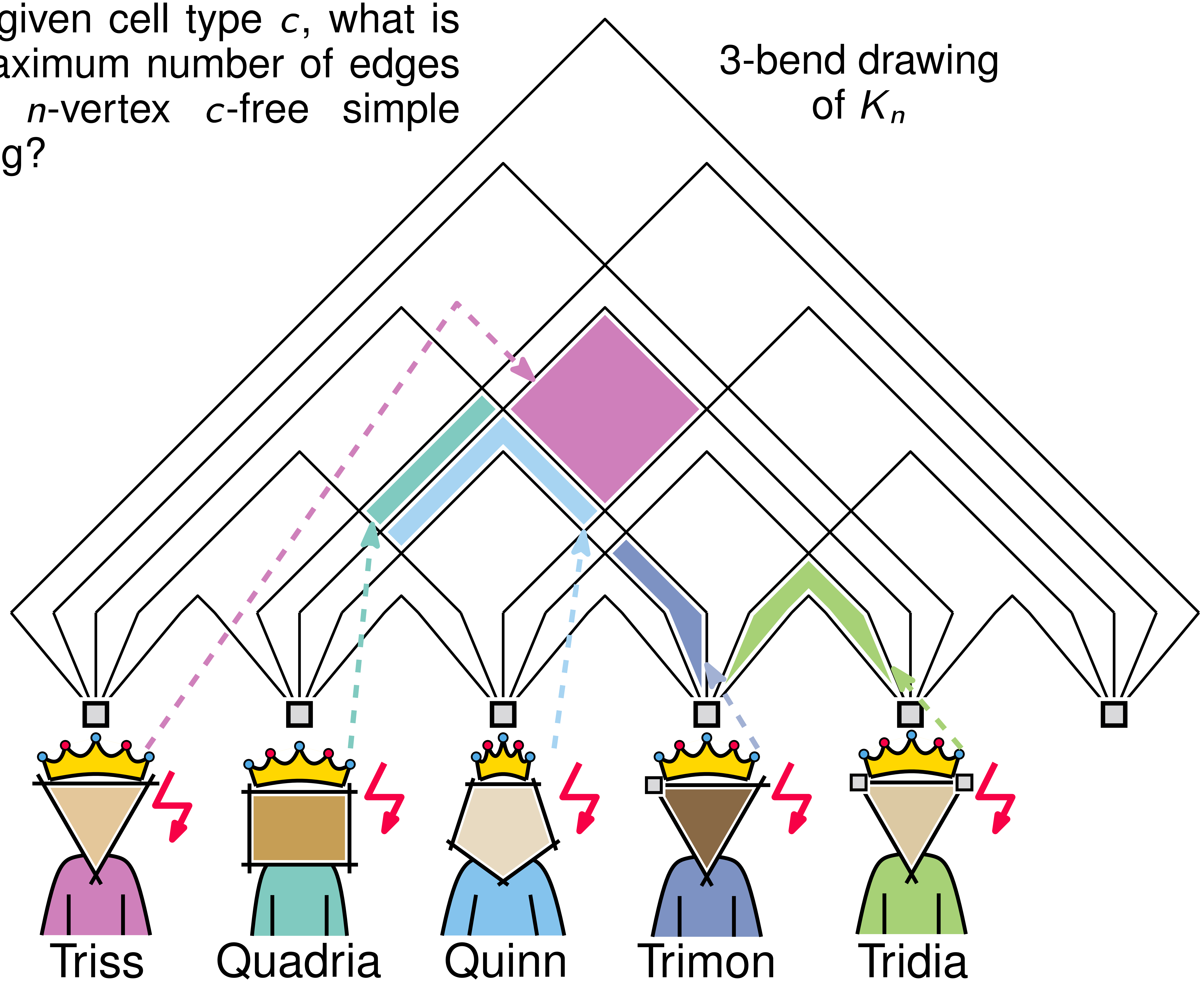

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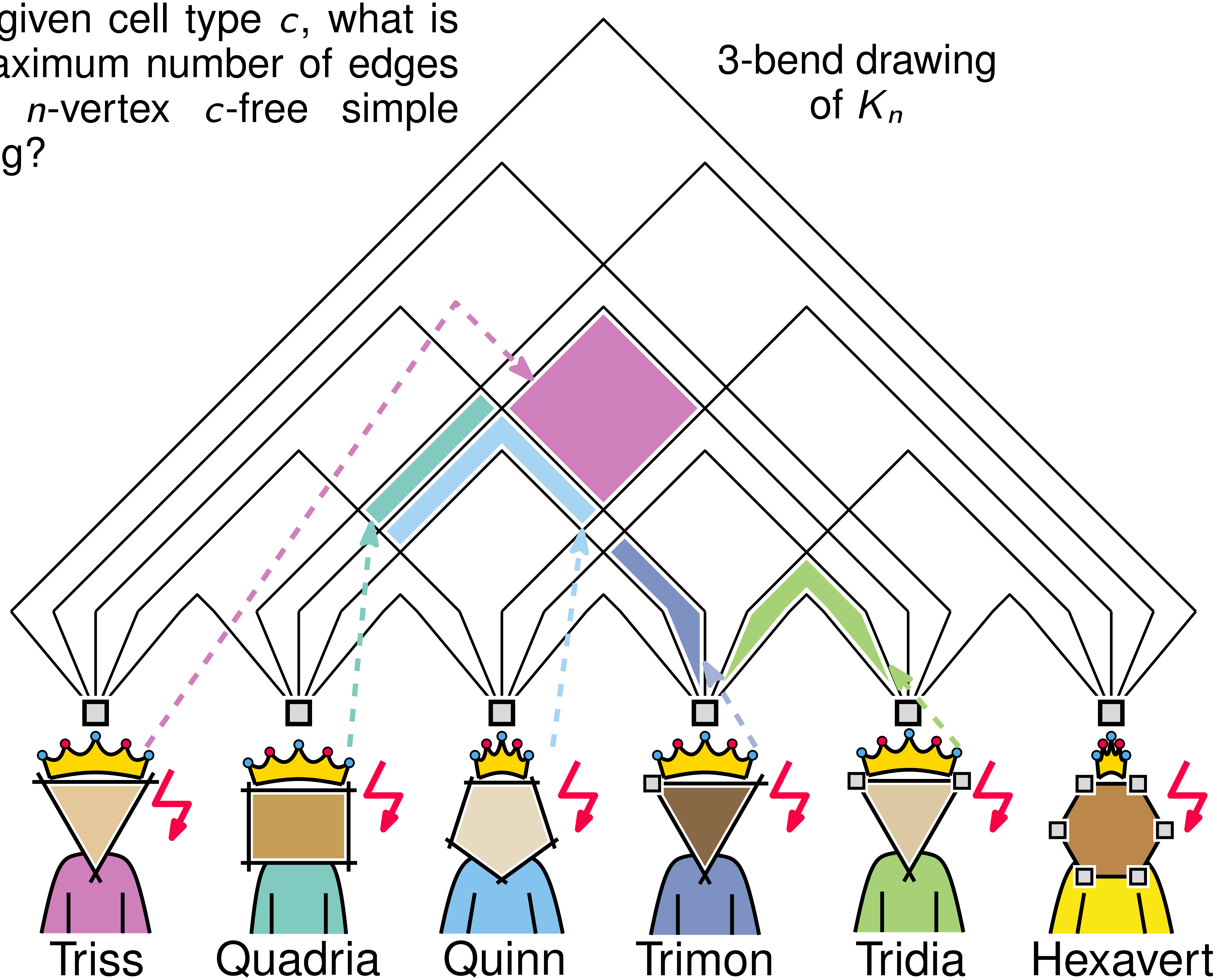

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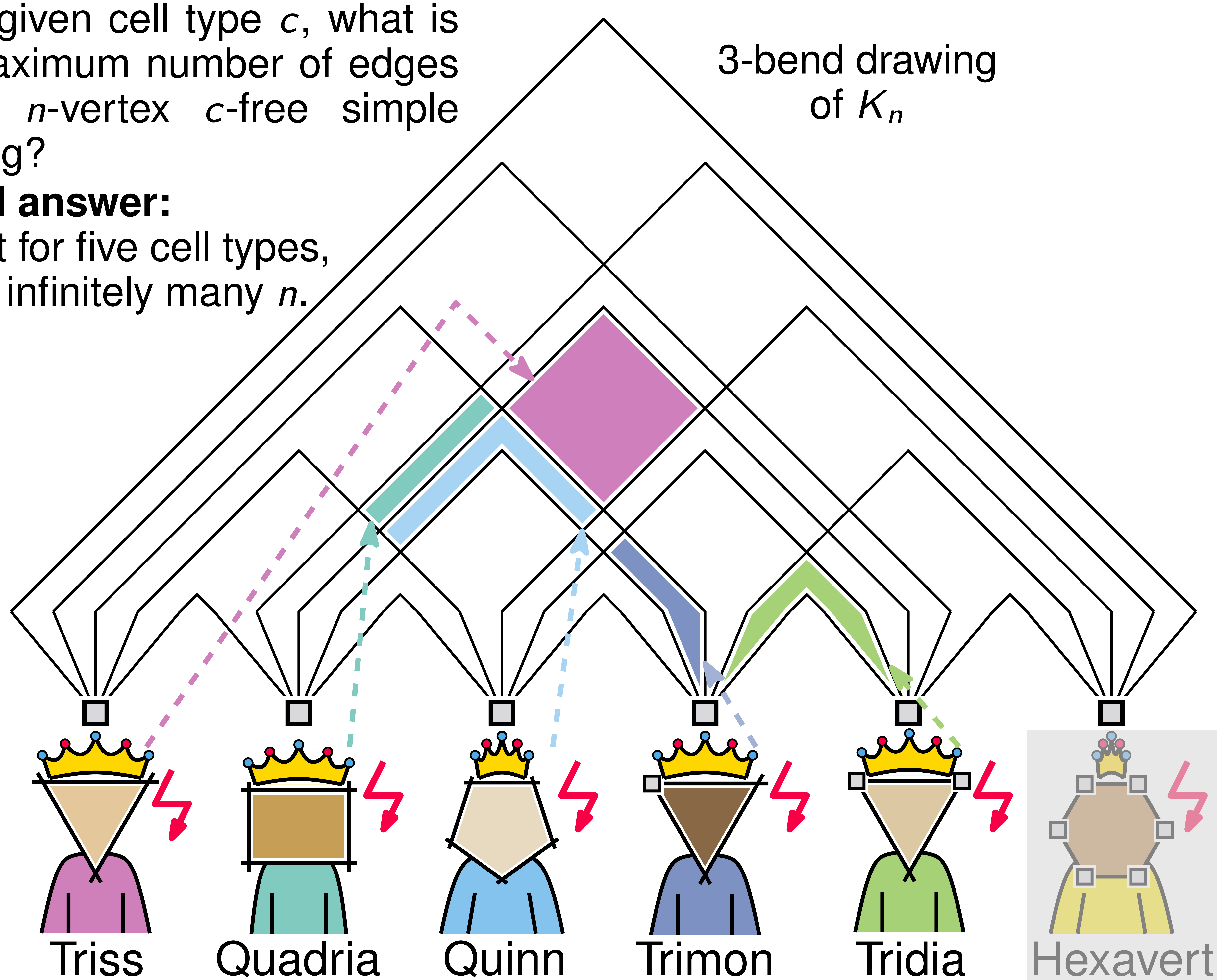
- **Question:**
  For a given cell type c, what is the maximum number of edges in an n-vertex c-free simple drawing?
- **Partial answer:**
  Except for five cell types, $\binom{n}{2}$ for infinitely many n.



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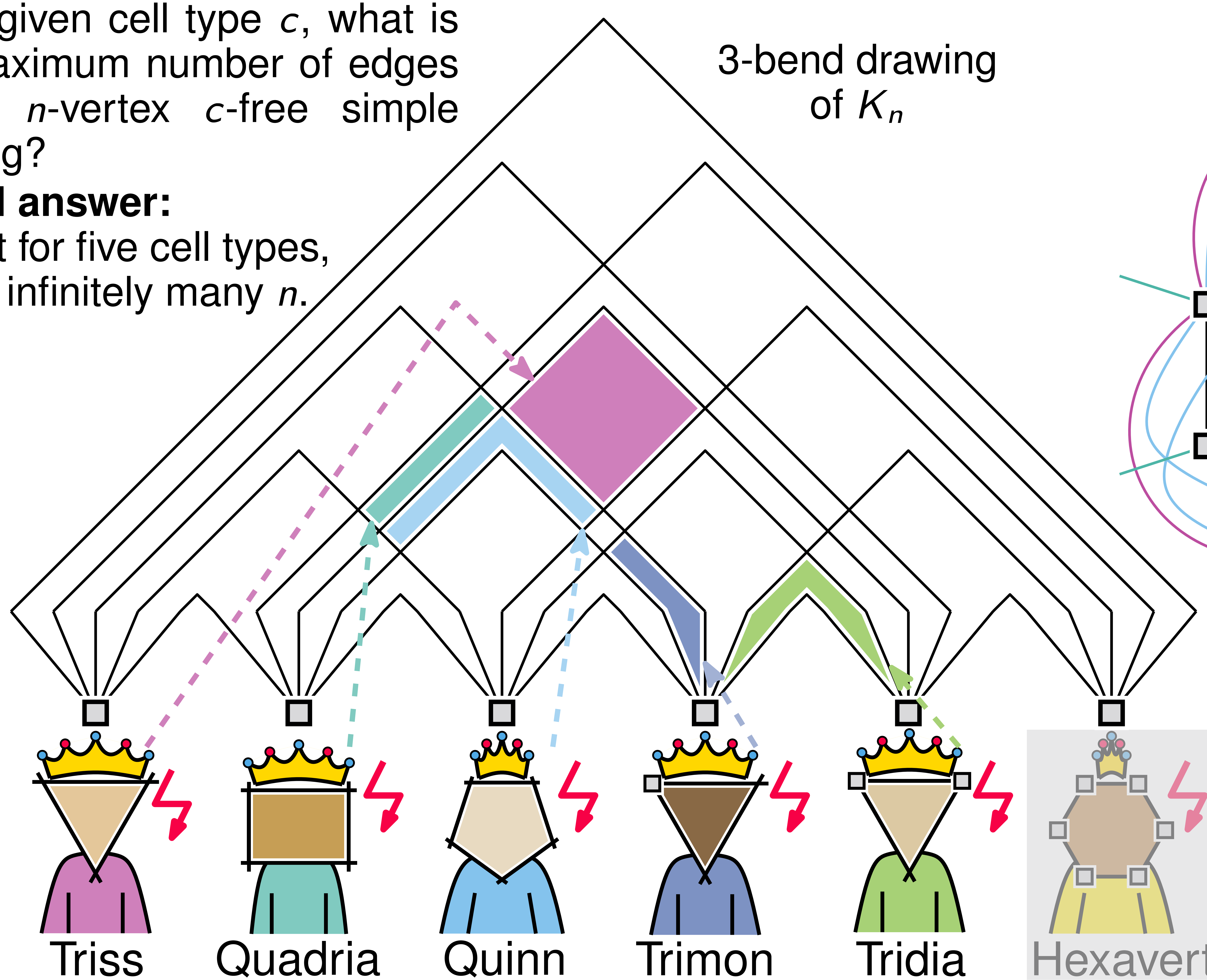

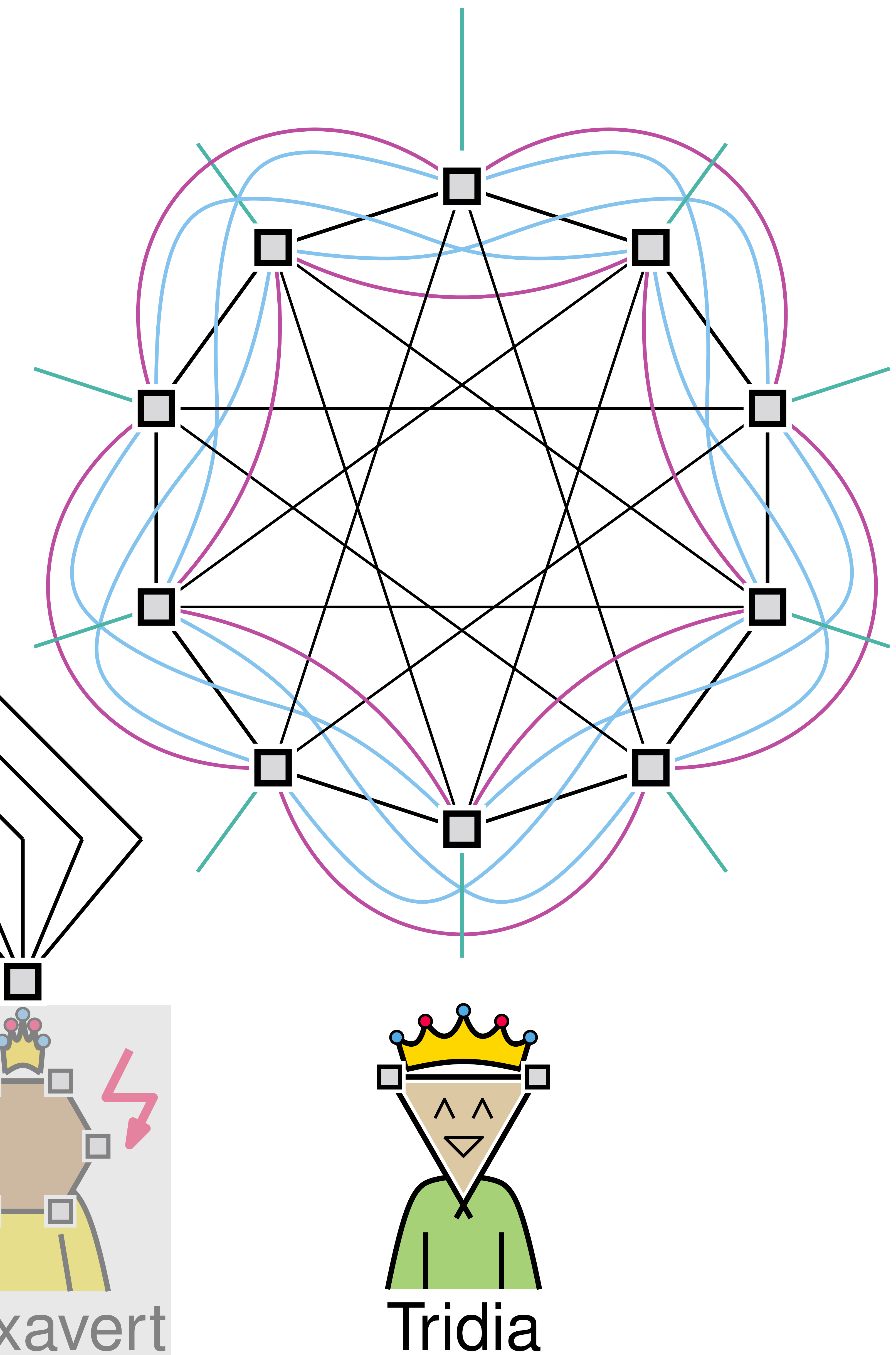

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- **Question:**
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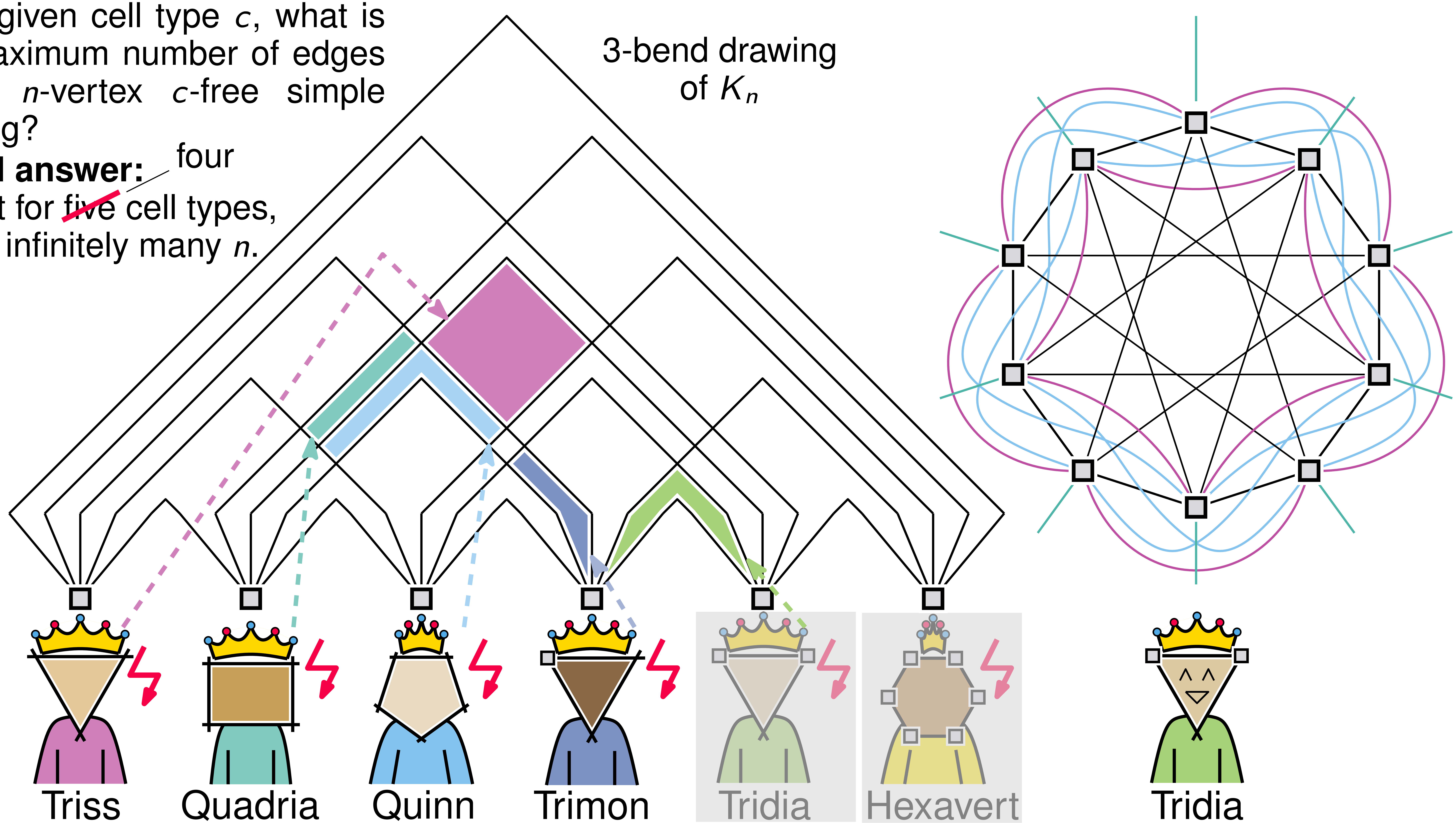
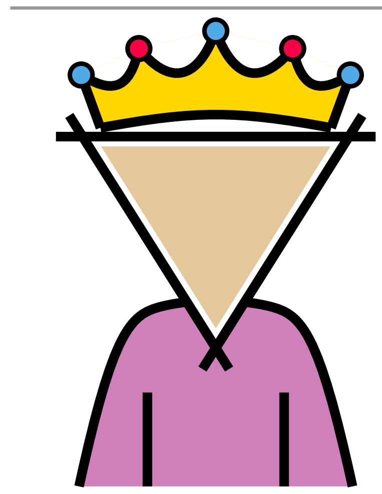
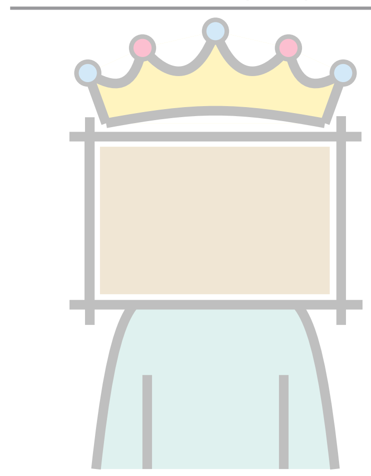
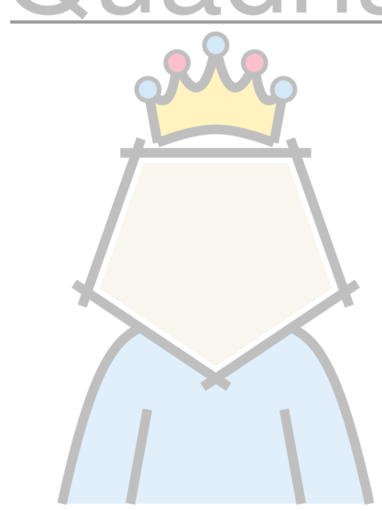
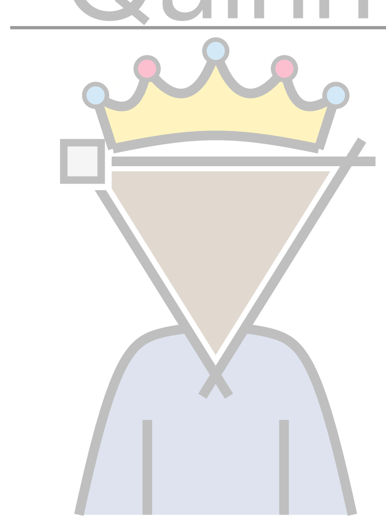
- **Partial answer:**
  Except for ~~five~~ four cell types, $\binom{n}{2}$ for infinitely many n.



Some rulers don't

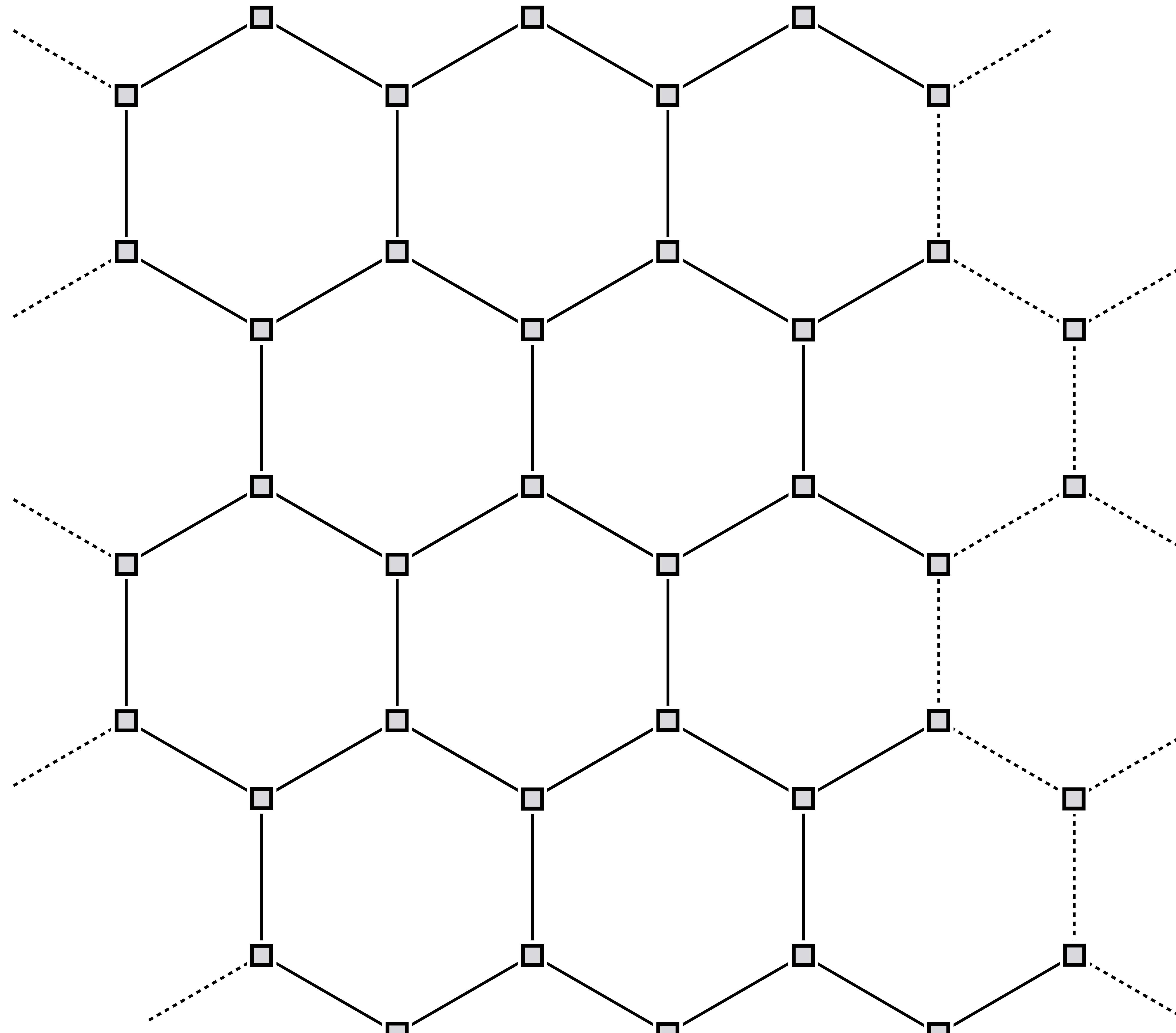
Forbidden cell	Lower Bound	Upper Bound
 Triss		$8n - 20$ [Ackerman, Tardos; '07]
 Quadria		
 Quinn		
 Trimon		

Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
Triss	$7n - 30$ based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07]
Quadria		
Quinn		
Trimon		

Hexagons on a cylinder

$\frac{3}{2}n - O(1)$



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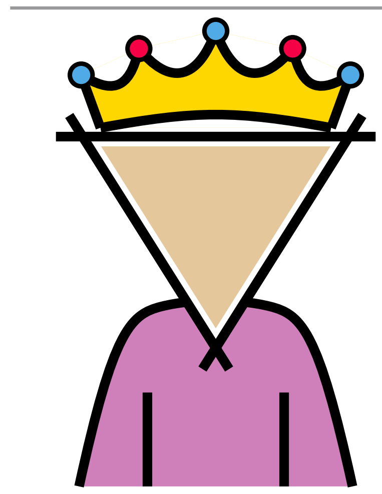
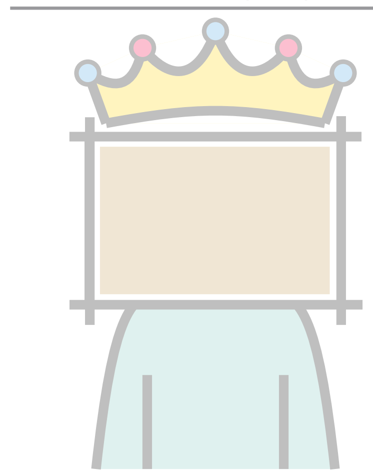
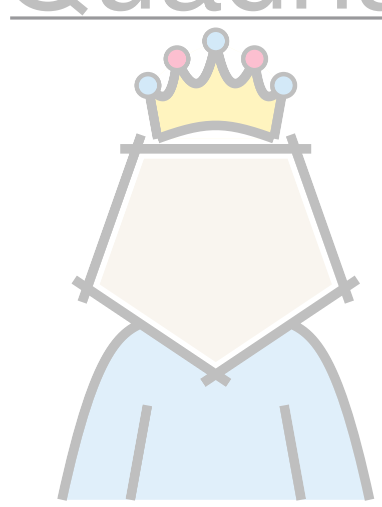
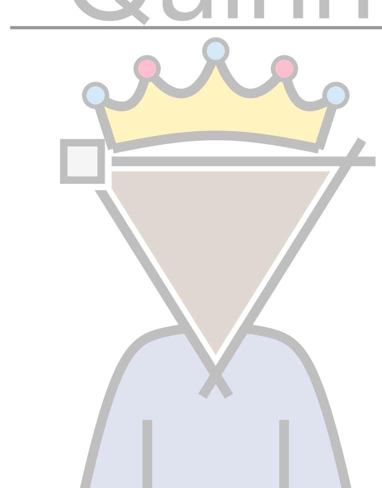
 Triss	**$7n - 30$** based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07]
 Quadria		
 Quinn		
 Trimon		

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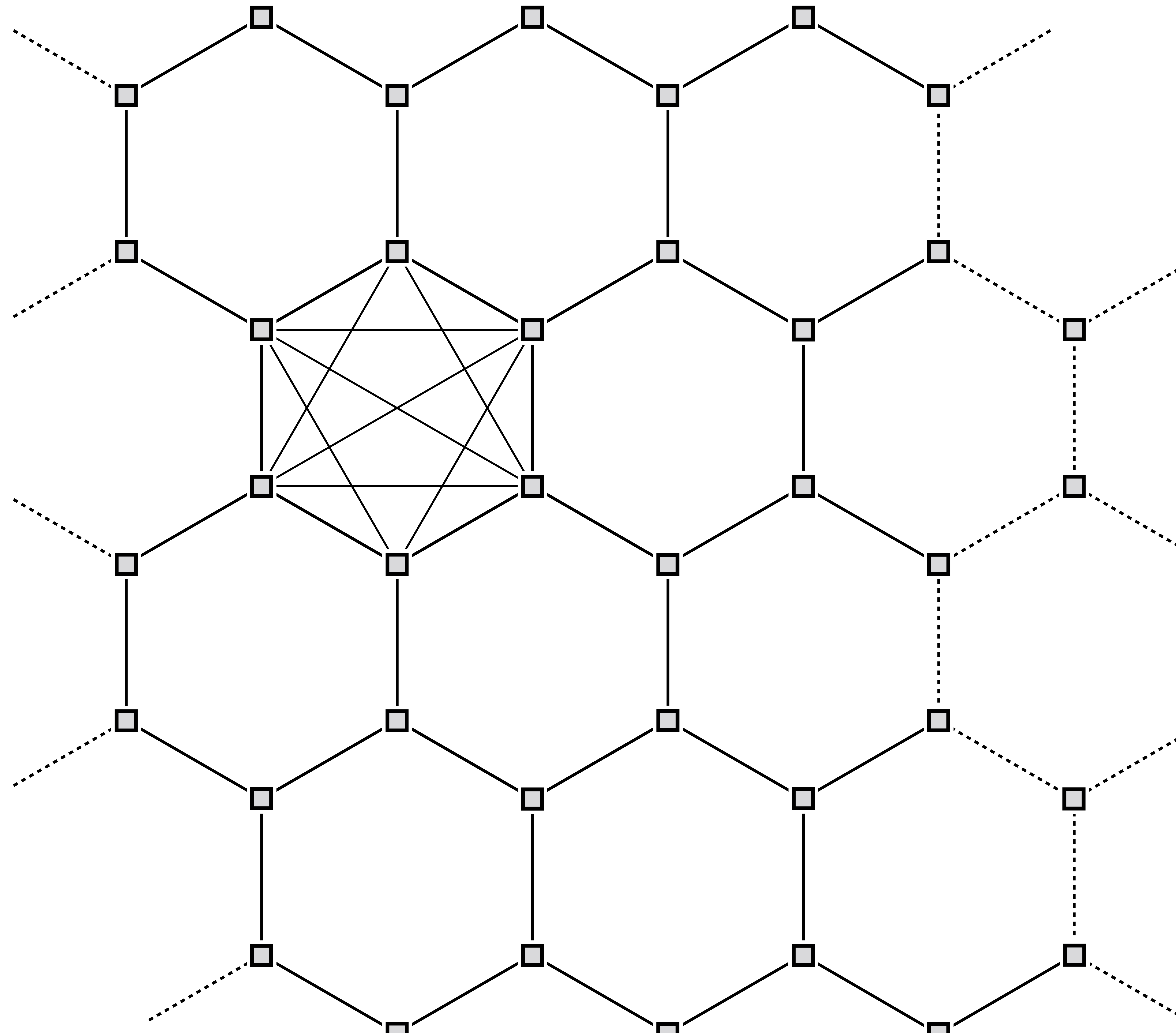

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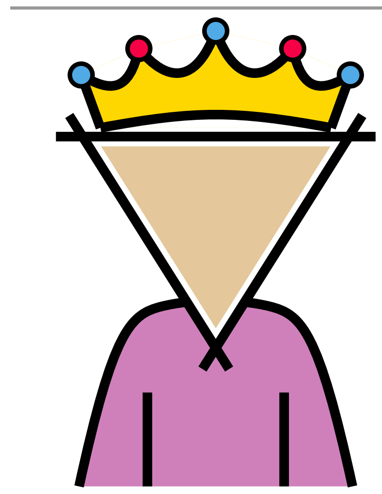
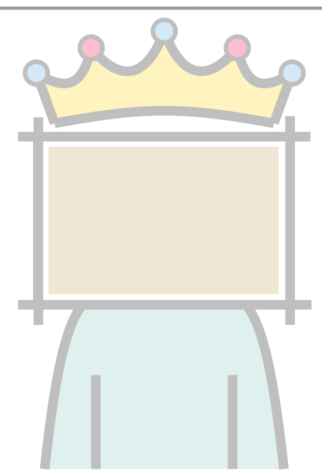
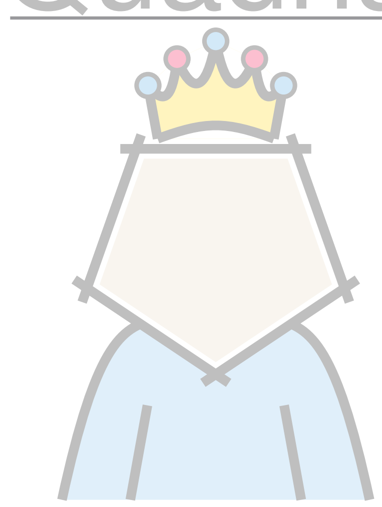
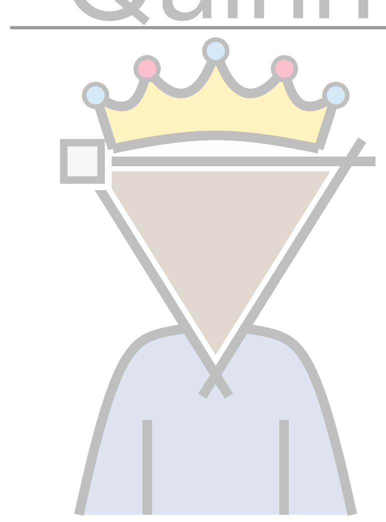
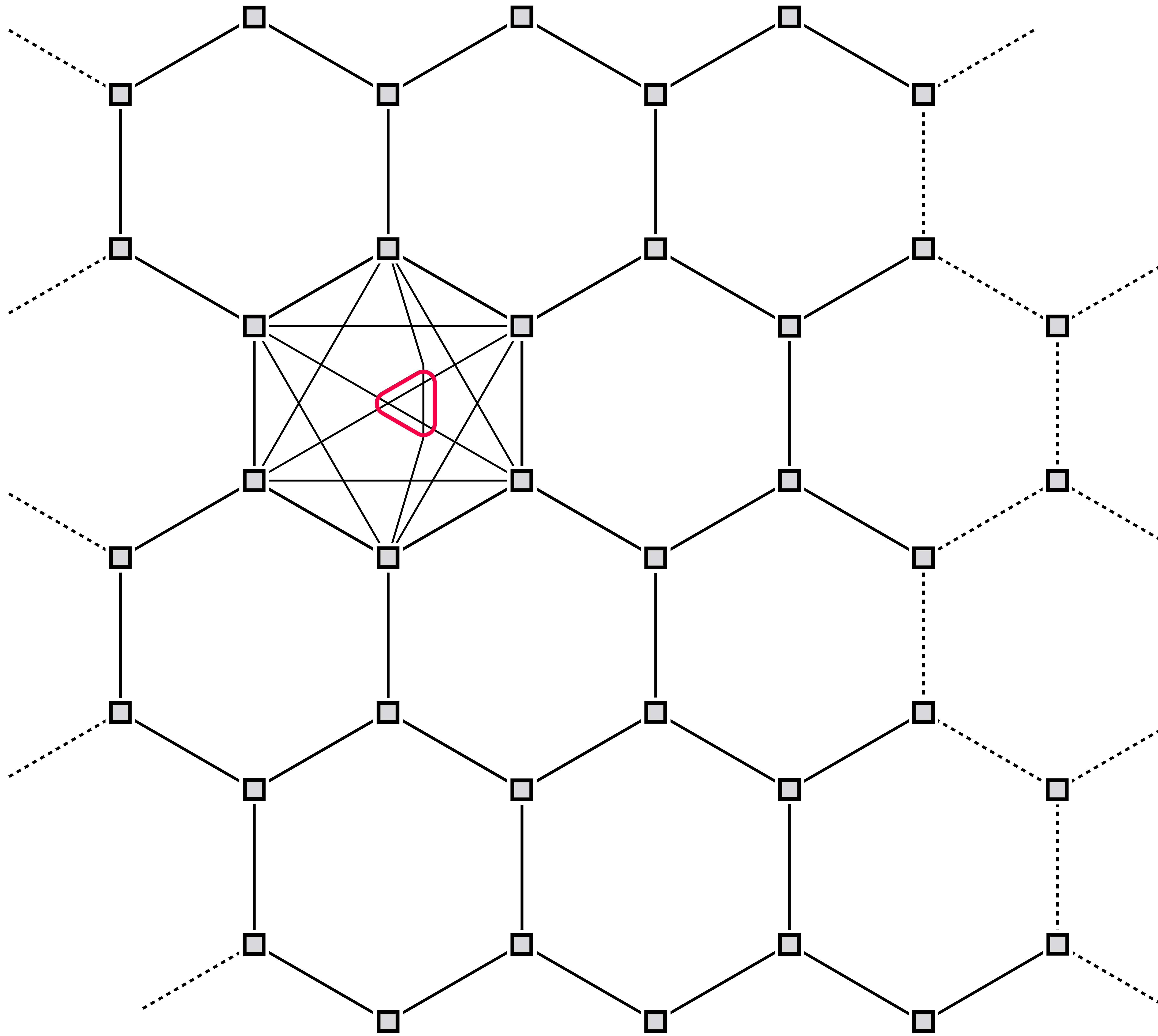
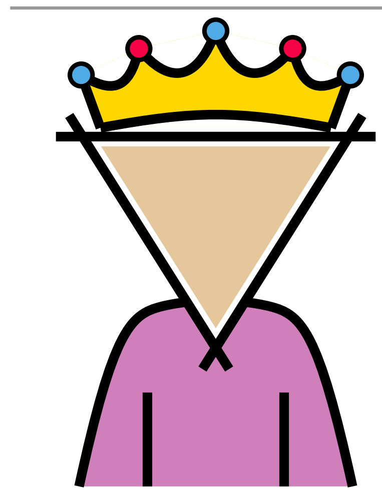
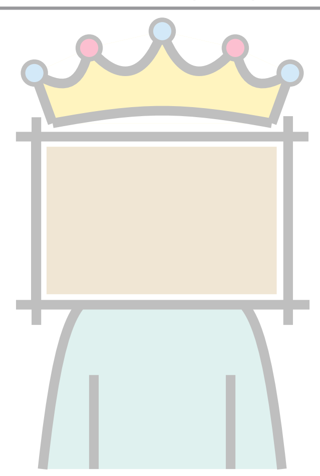
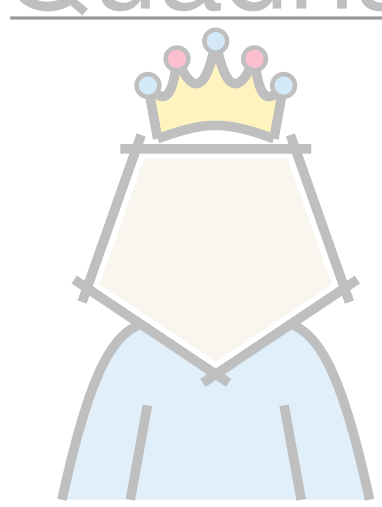
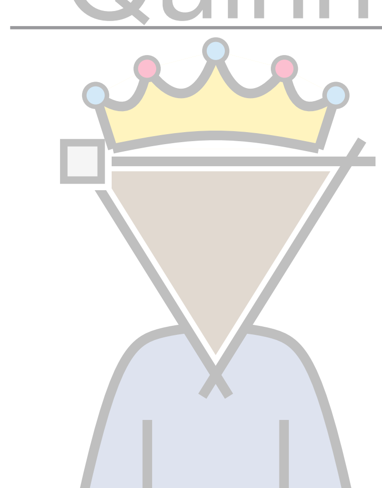
 Triss	**$7n - 30$** based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07]
 Quadria		
 Quinn		
 Trimon		

Hexagons on a cylinder

$\frac{3}{2}n - O(1)$

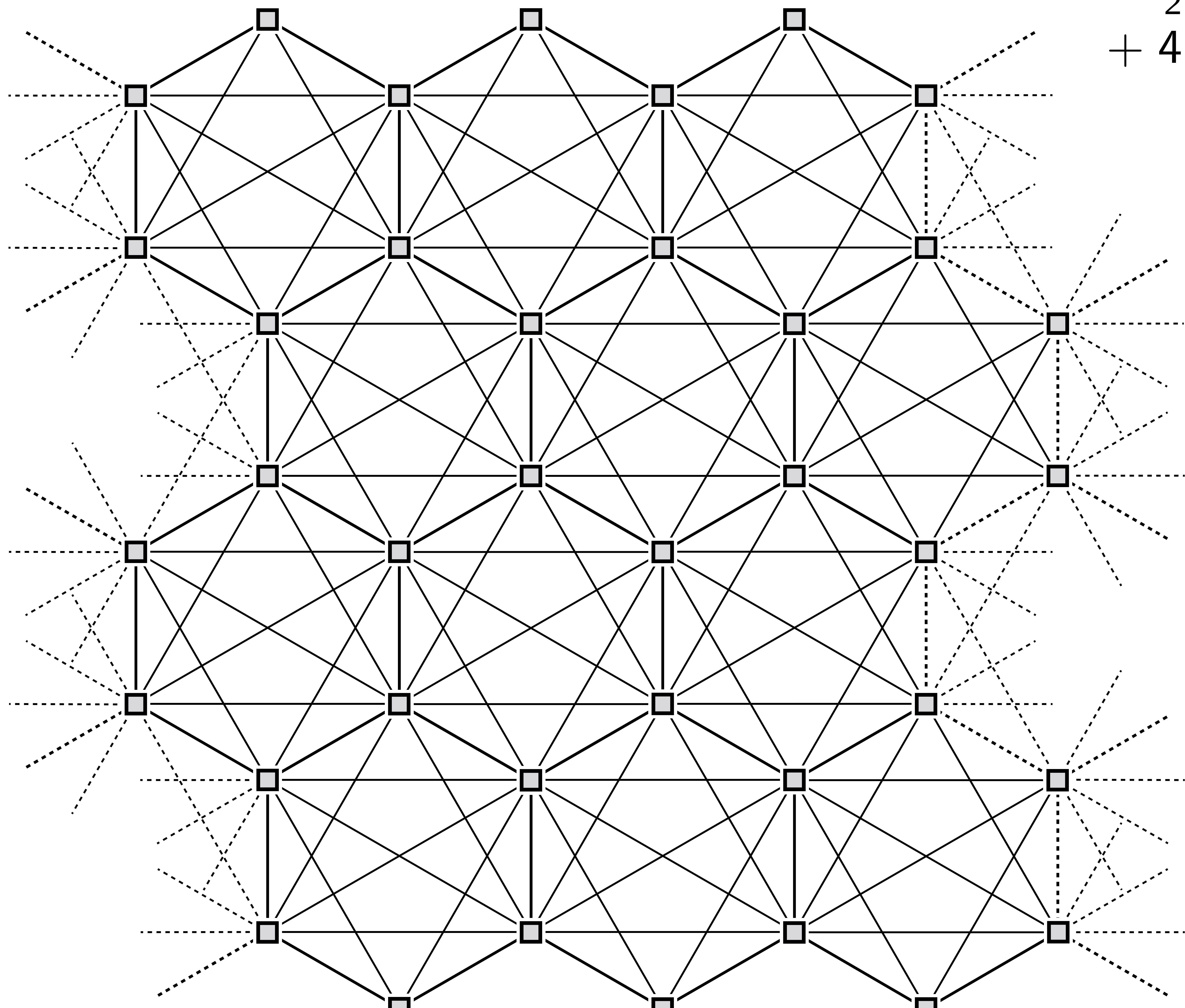


Some rulers don't

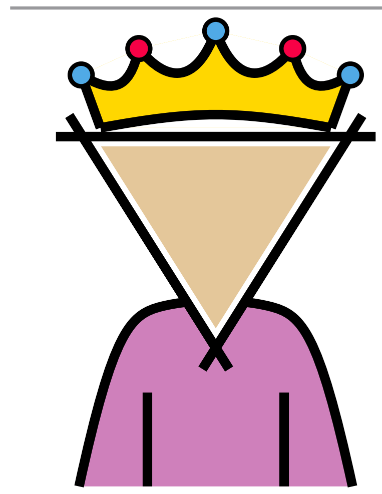
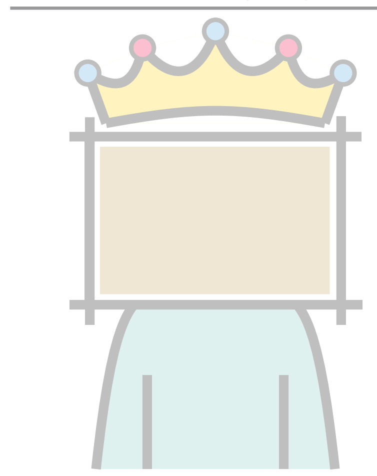
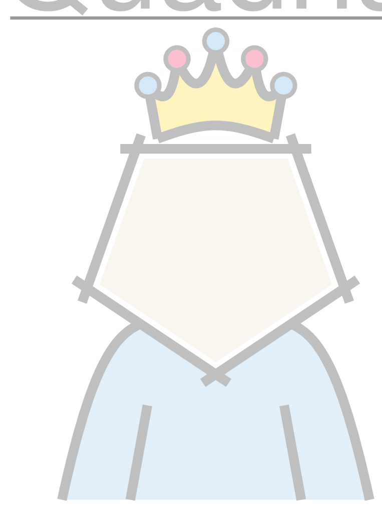
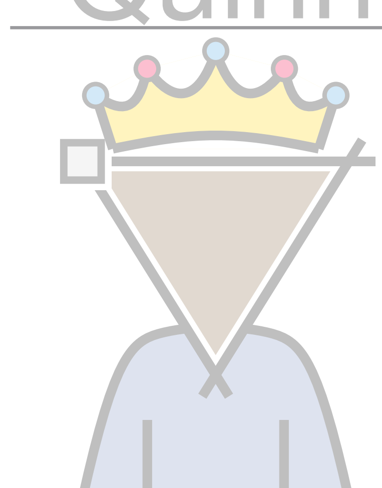
Forbidden cell	Lower Bound	Upper Bound
 Triss	**$7n - 30$** based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07]
 Quadria		
 Quinn		
 Trimon		

Hexagons on a cylinder

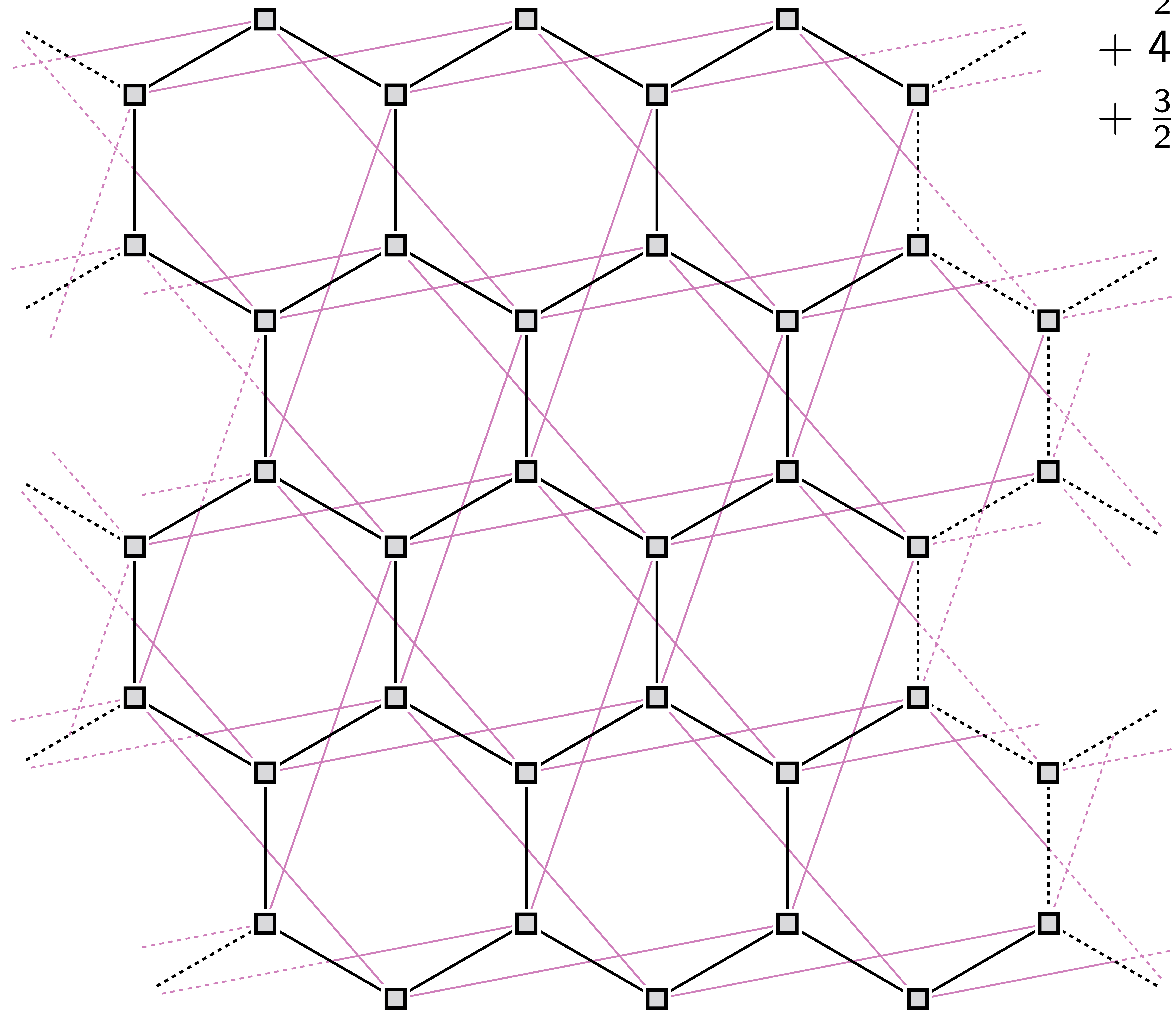
$$\frac{3}{2}n - O(1)$$
$$+ 4n - O(1)$$



Some rulers don't

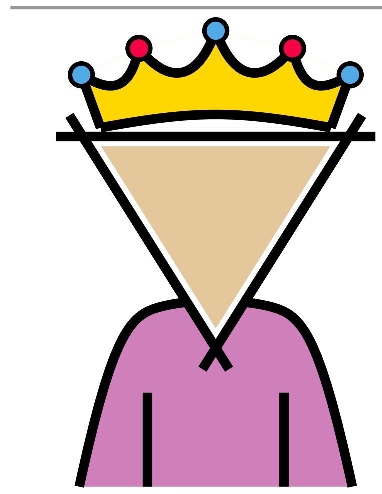
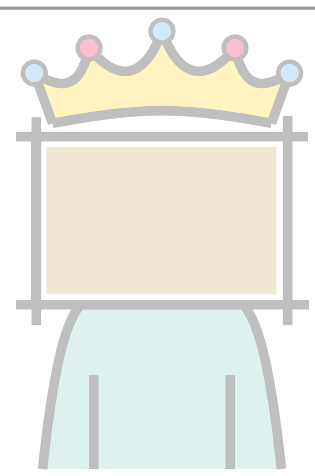
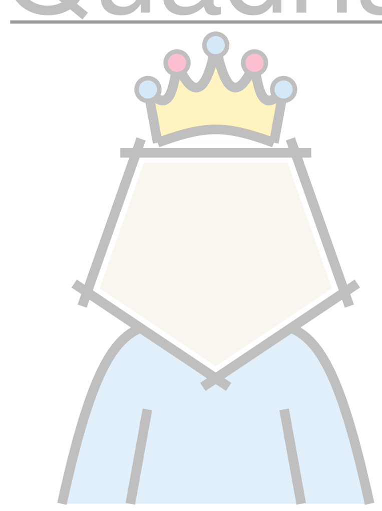
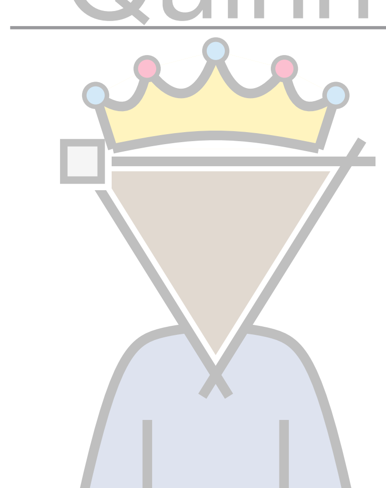
Forbidden cell	Lower Bound	Upper Bound
 Triss	**$7n - 30$** based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07]
 Quadria		
 Quinn		
 Trimon		

Hexagons on a cylinder

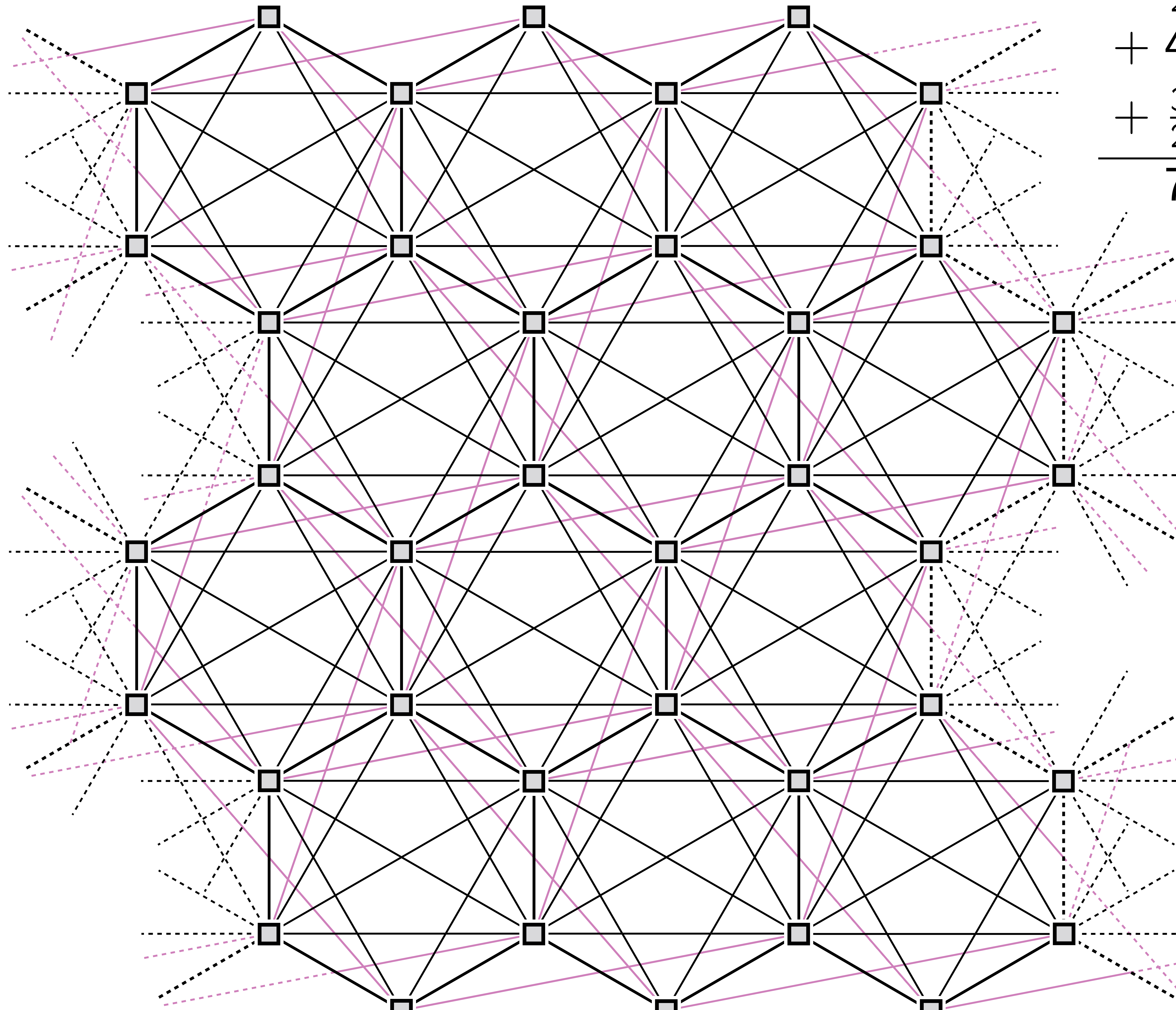


$$\frac{3}{2}n - O(1)$$
$$+\ 4n - O(1)$$
$$+\ \frac{3}{2}n - O(1)$$

Some rulers don't

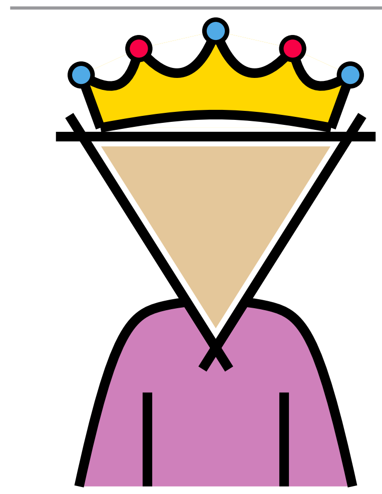
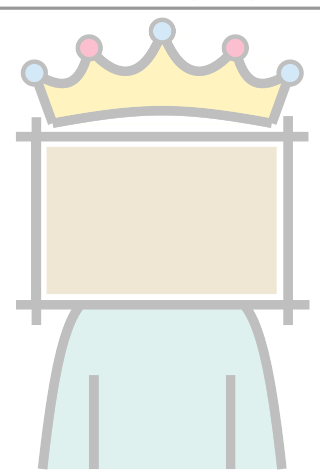
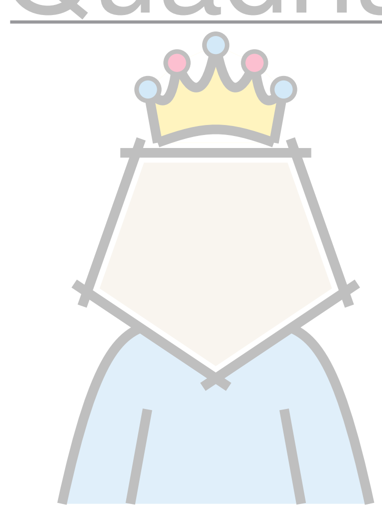
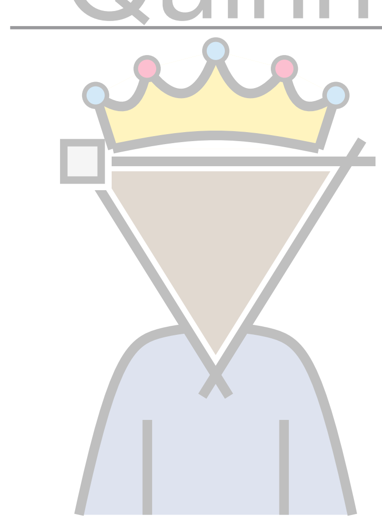
Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n - 30}$ based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07]
 Quadria		
 Quinn		
 Trimon		

Hexagons on a cylinder



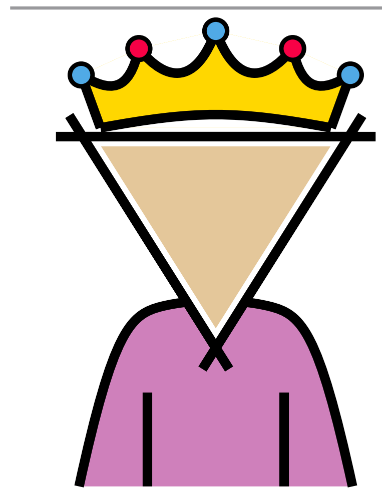
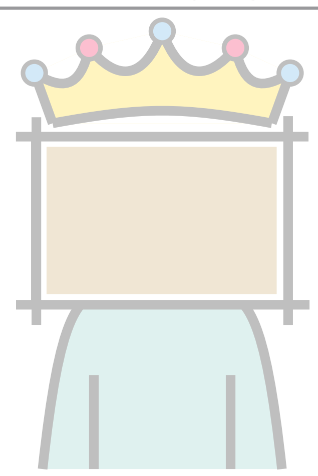
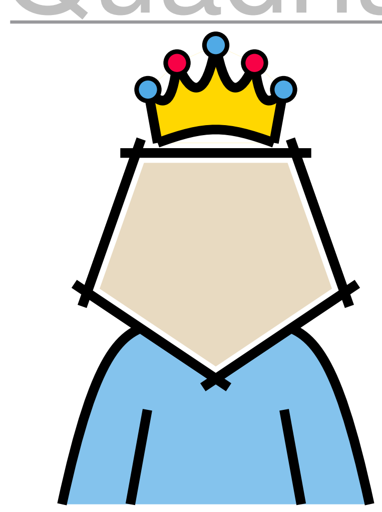
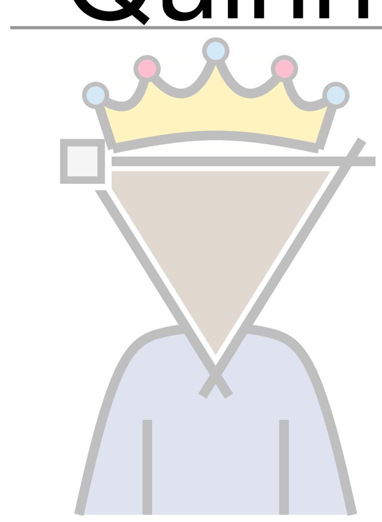
$$\begin{array}{r} \frac{3}{2}n - O(1) \\ +\ 4n - O(1) \\ +\ \frac{3}{2}n - O(1) \\ \hline 7n - O(1) \end{array}$$

Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria		
 Quinn		
 Trimon		

Discharging

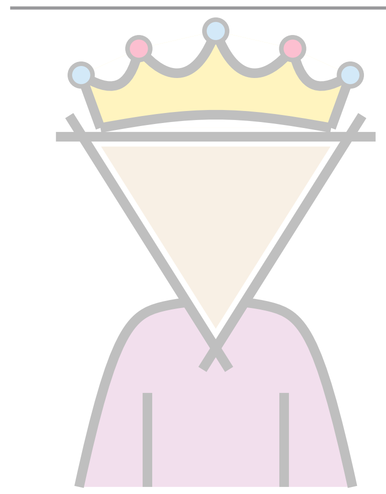
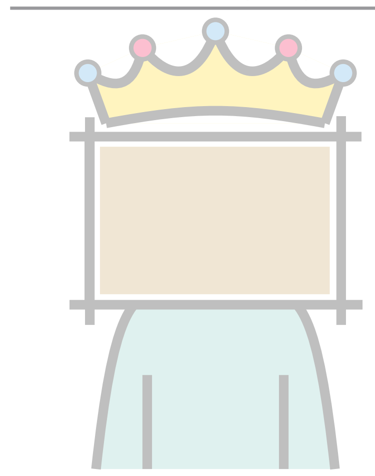
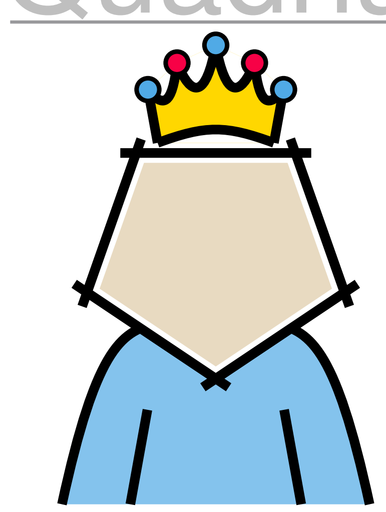
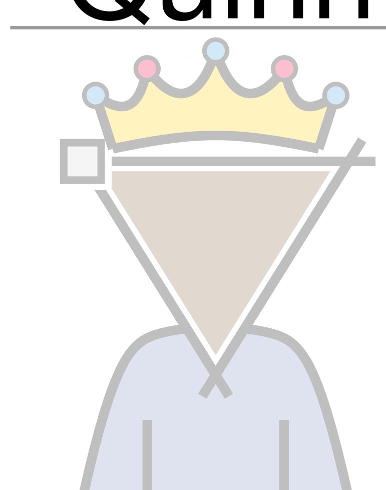
Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria		
 Quinn		$\mathbf{6n-12}$
 Trimon		

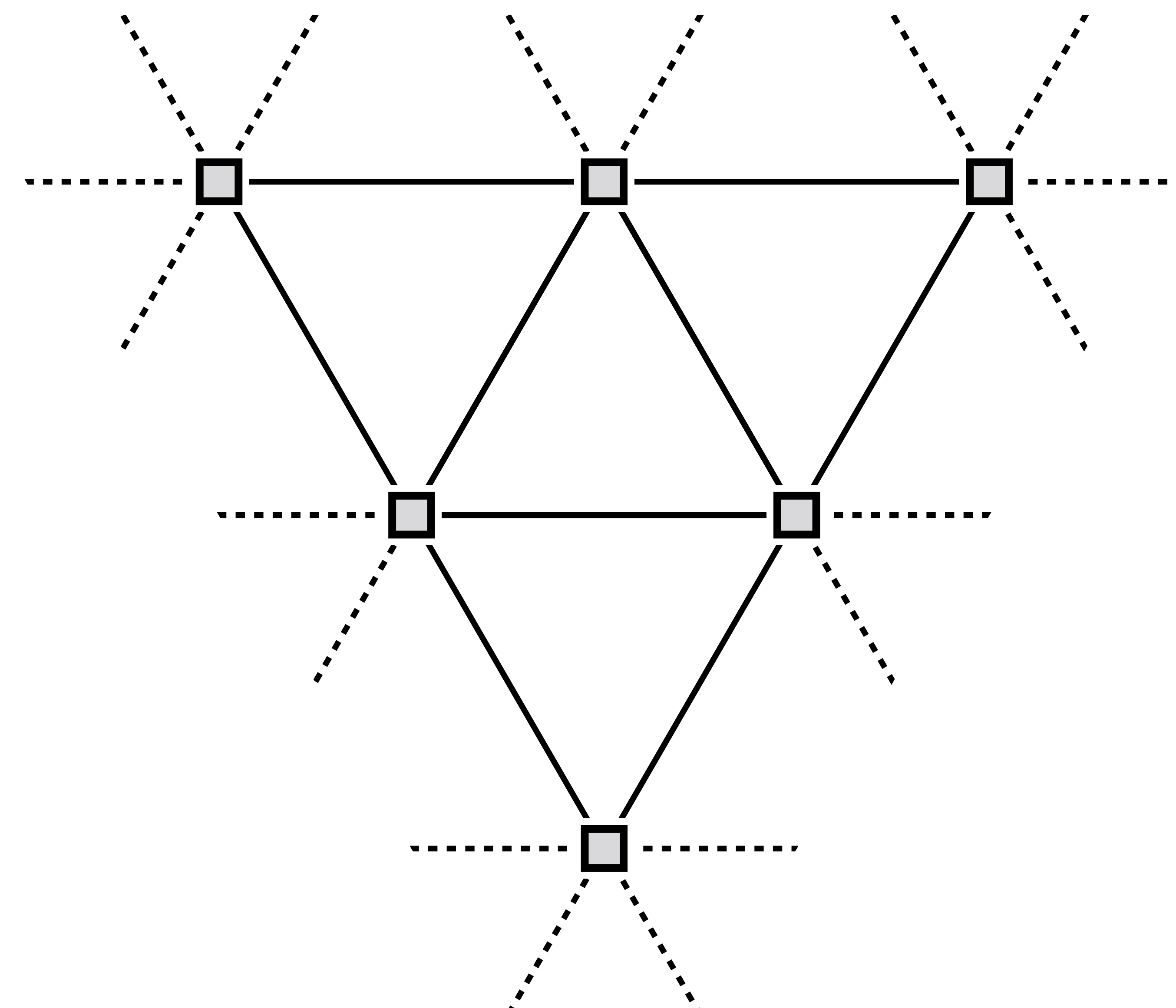
Discharging

Counting cells

Some rulers don't

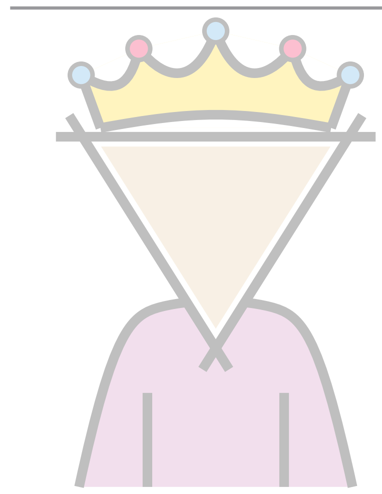
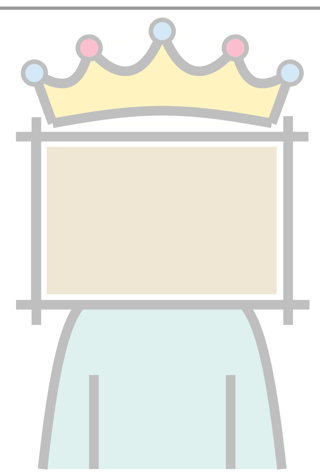
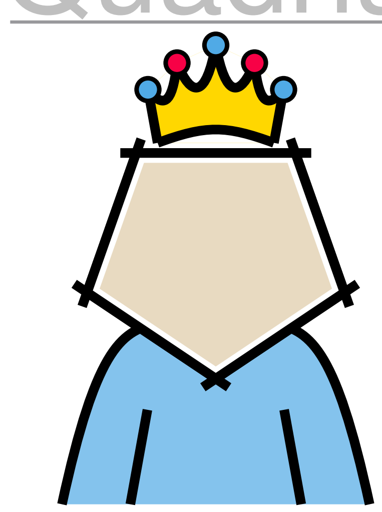
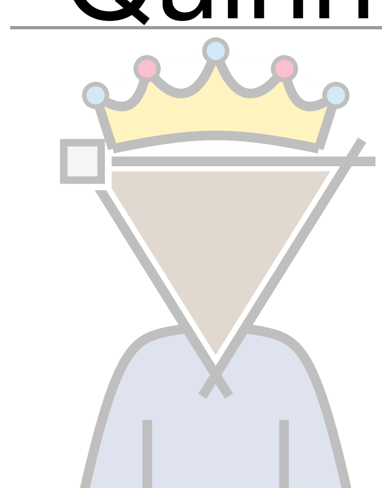
Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria		
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon		

Triangulation augmentation

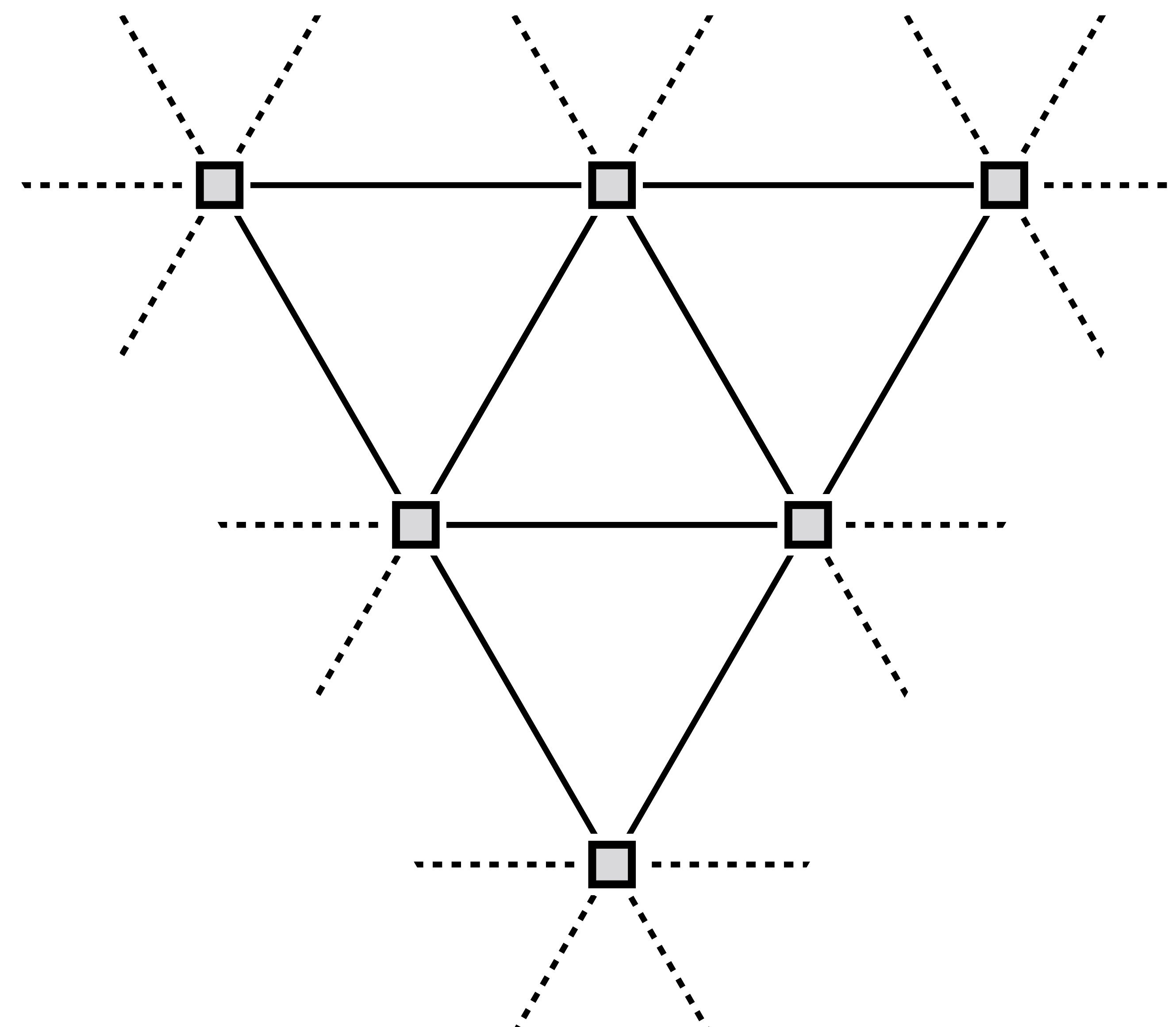


$3n-6$

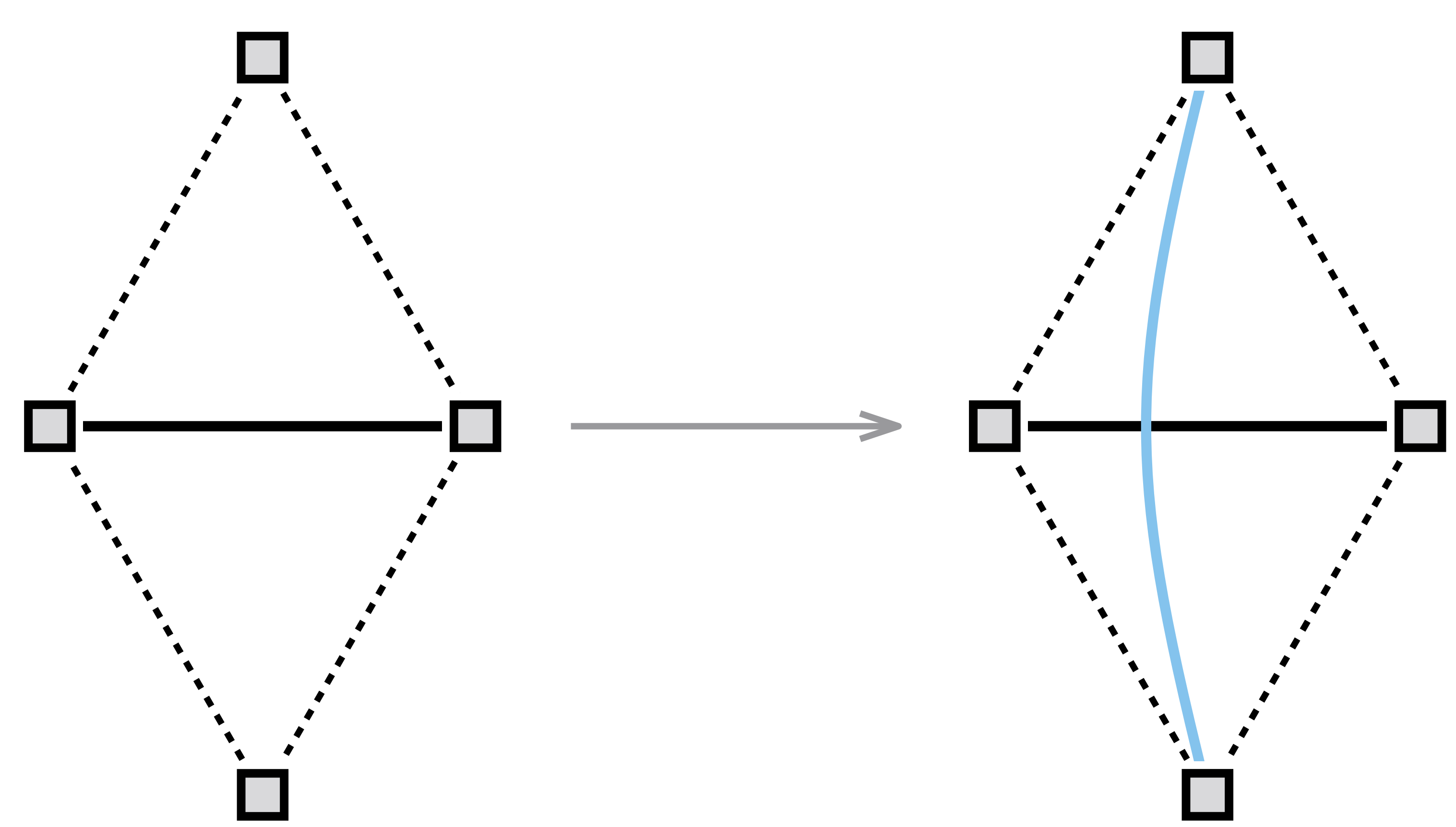
Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria		
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon		

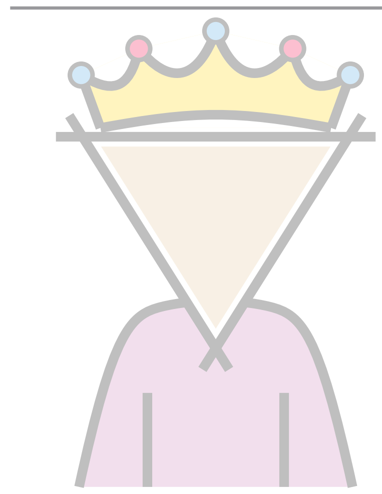
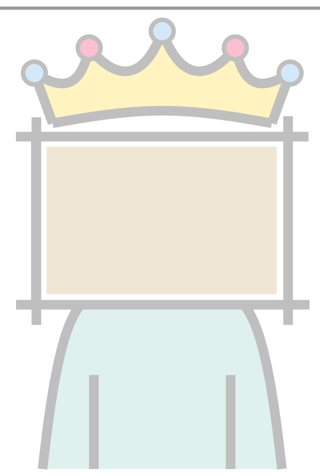
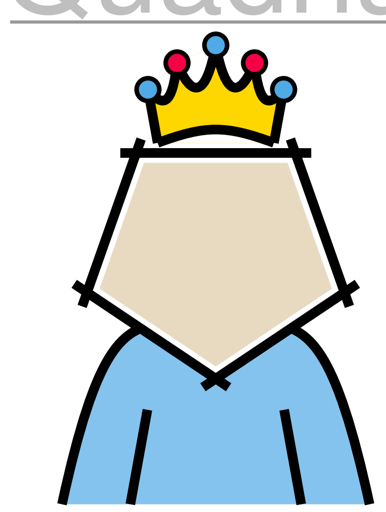
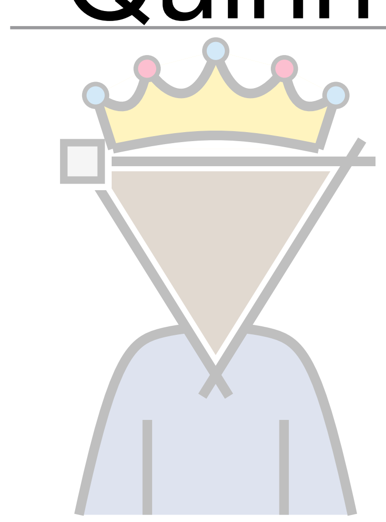
Triangulation augmentation



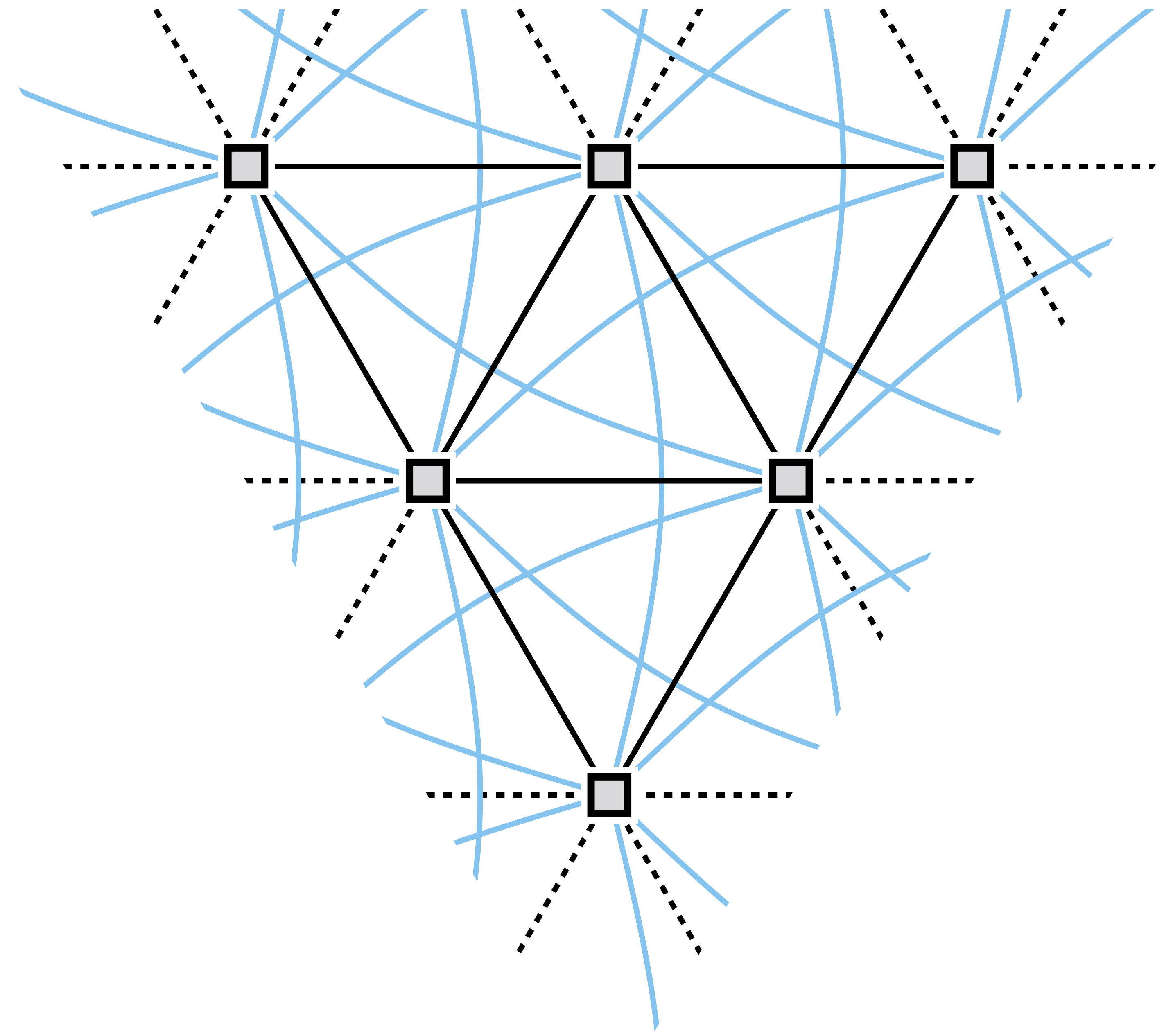
$3n-6$



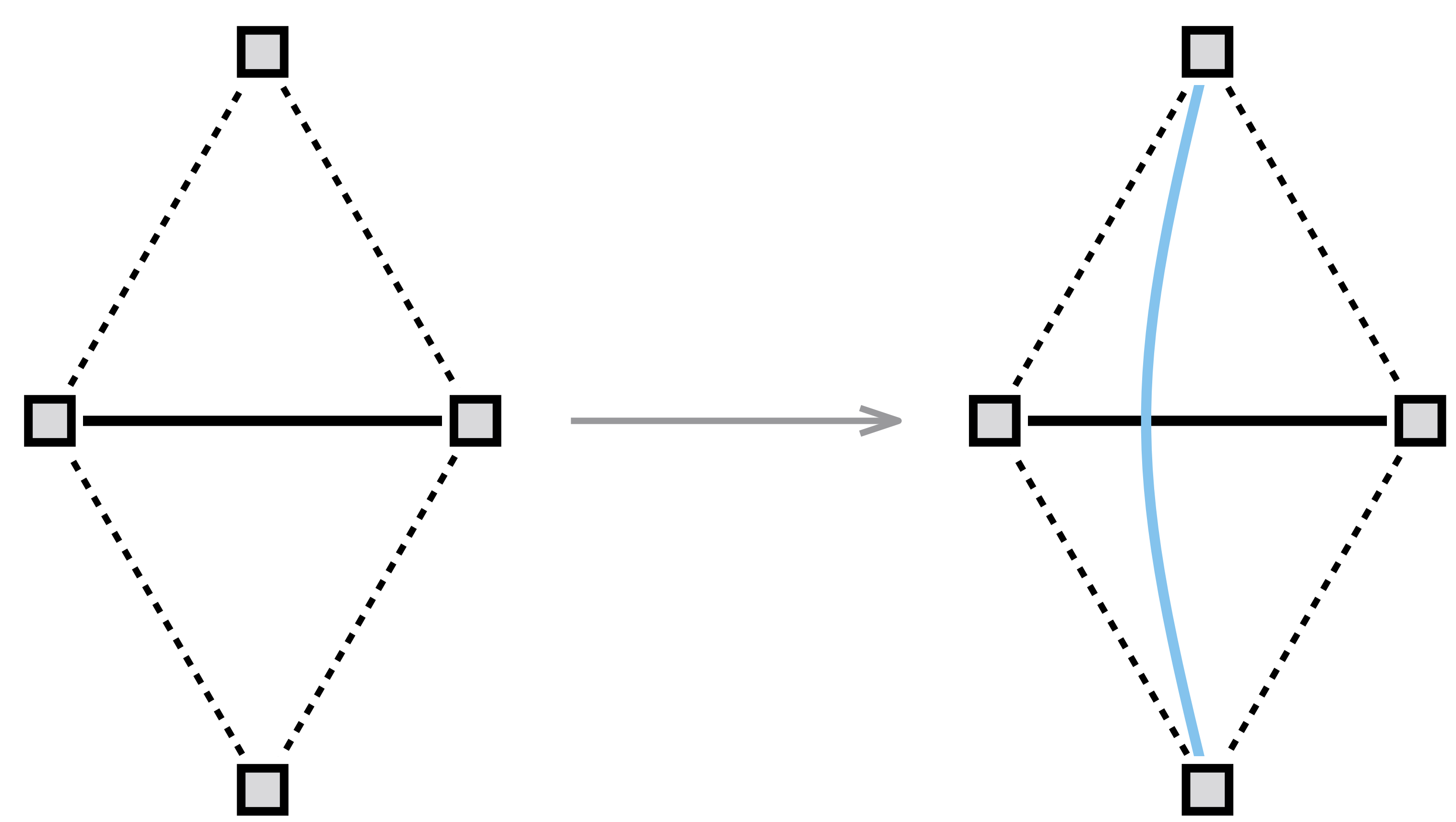
Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n - 30}$ based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07] $\mathbf{7n - 20}$
 Quadria		
 Quinn	$\mathbf{6n - 12}$	$\mathbf{6n - 12}$
 Trimon		

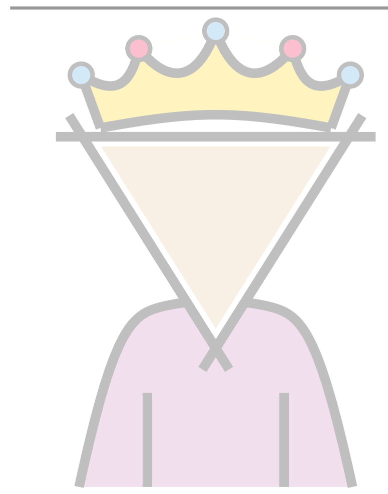
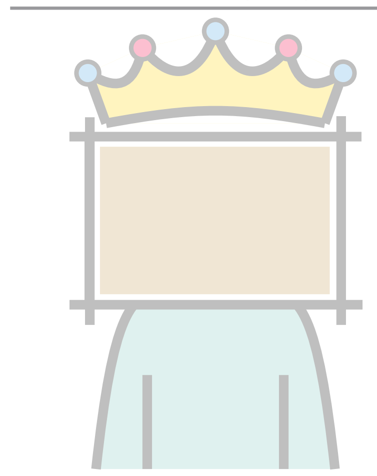
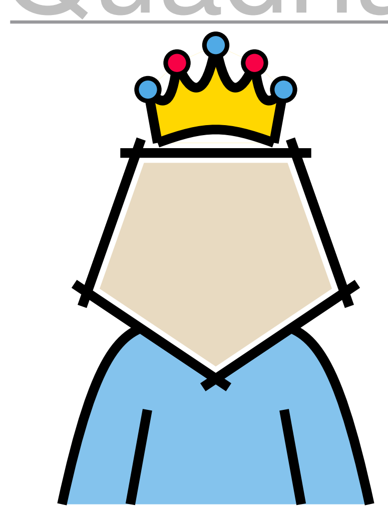
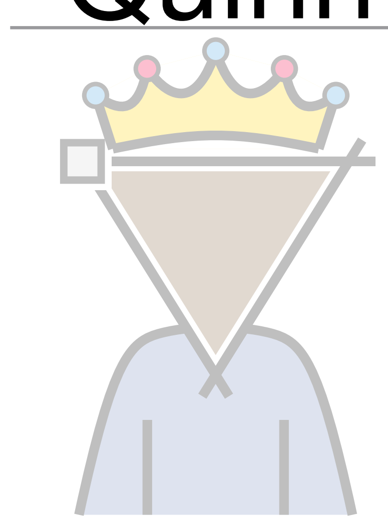
Triangulation augmentation



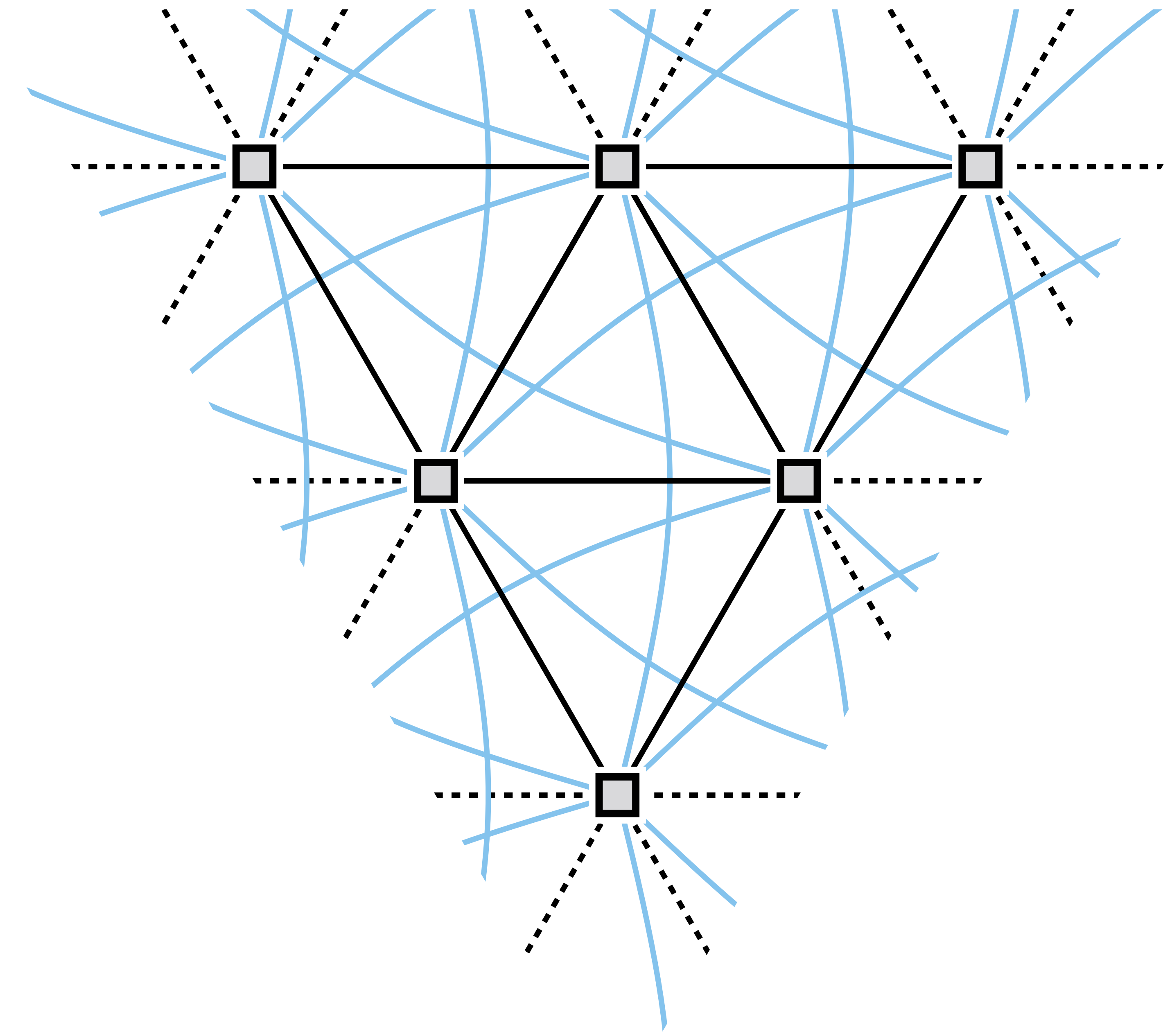
$$\begin{array}{r} 3n - 6 \\ +\ 3n - 6 \\ \hline 6n - 12 \end{array}$$



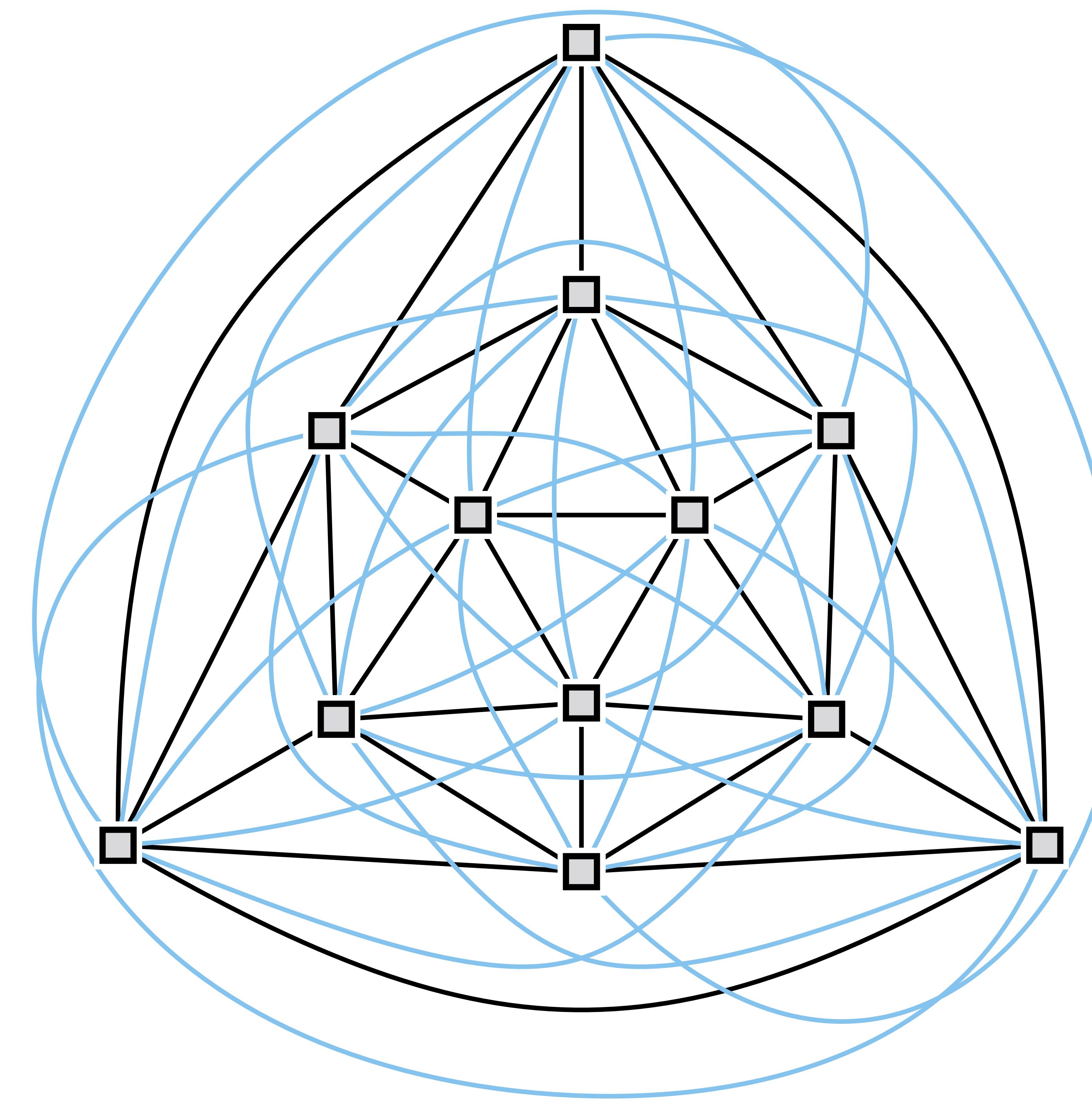
Some rulers don't

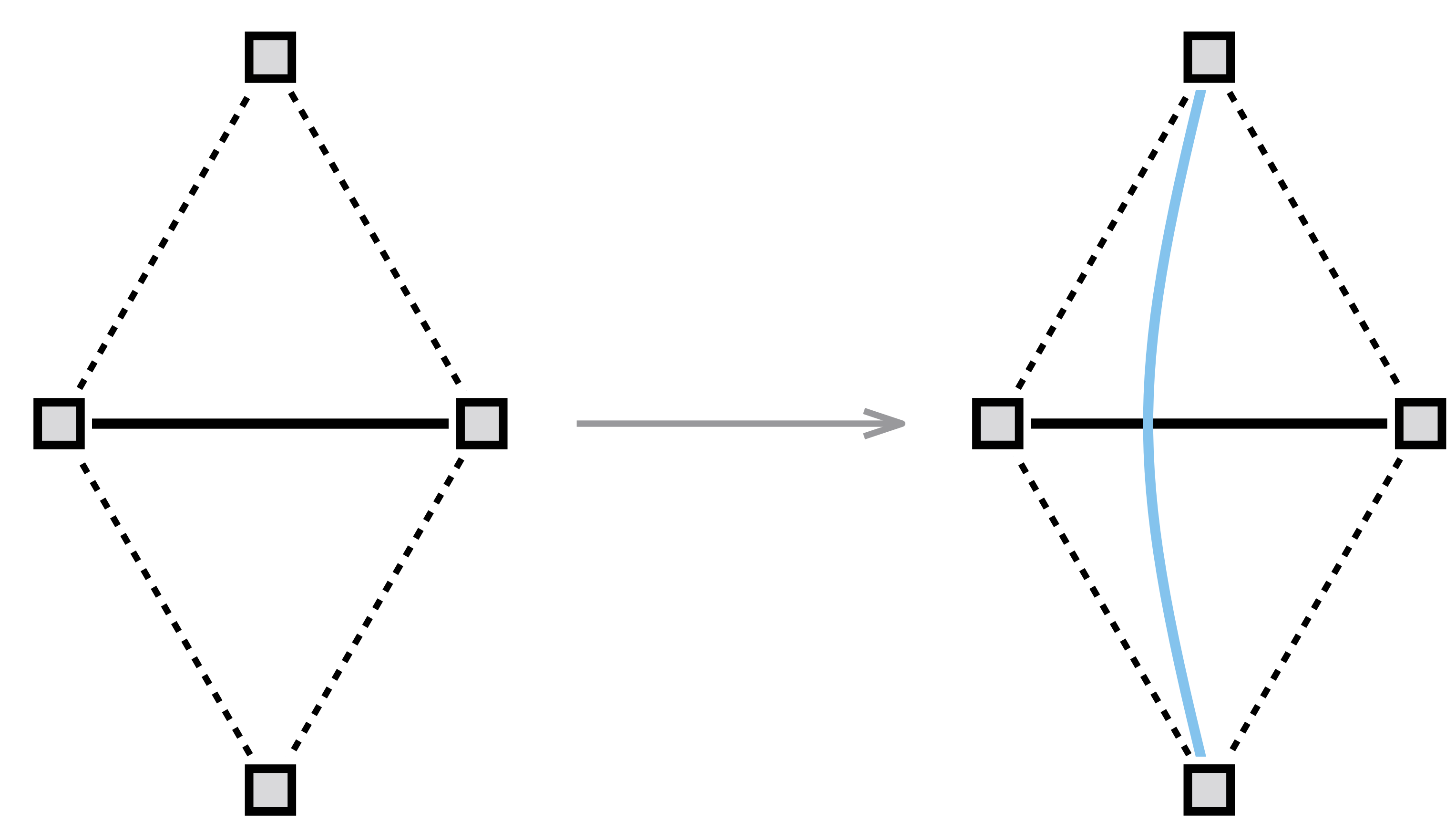
Forbidden cell	Lower Bound	Upper Bound
 Triss	$\boldsymbol{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\boldsymbol{7n-20}$
 Quadria		
 Quinn	$\boldsymbol{6n-12}$	$\boldsymbol{6n-12}$
 Trimon		

Triangulation augmentation



$$\begin{array}{r} 3n-6 \\ +\ 3n-6 \\ \hline 6n-12 \end{array}$$

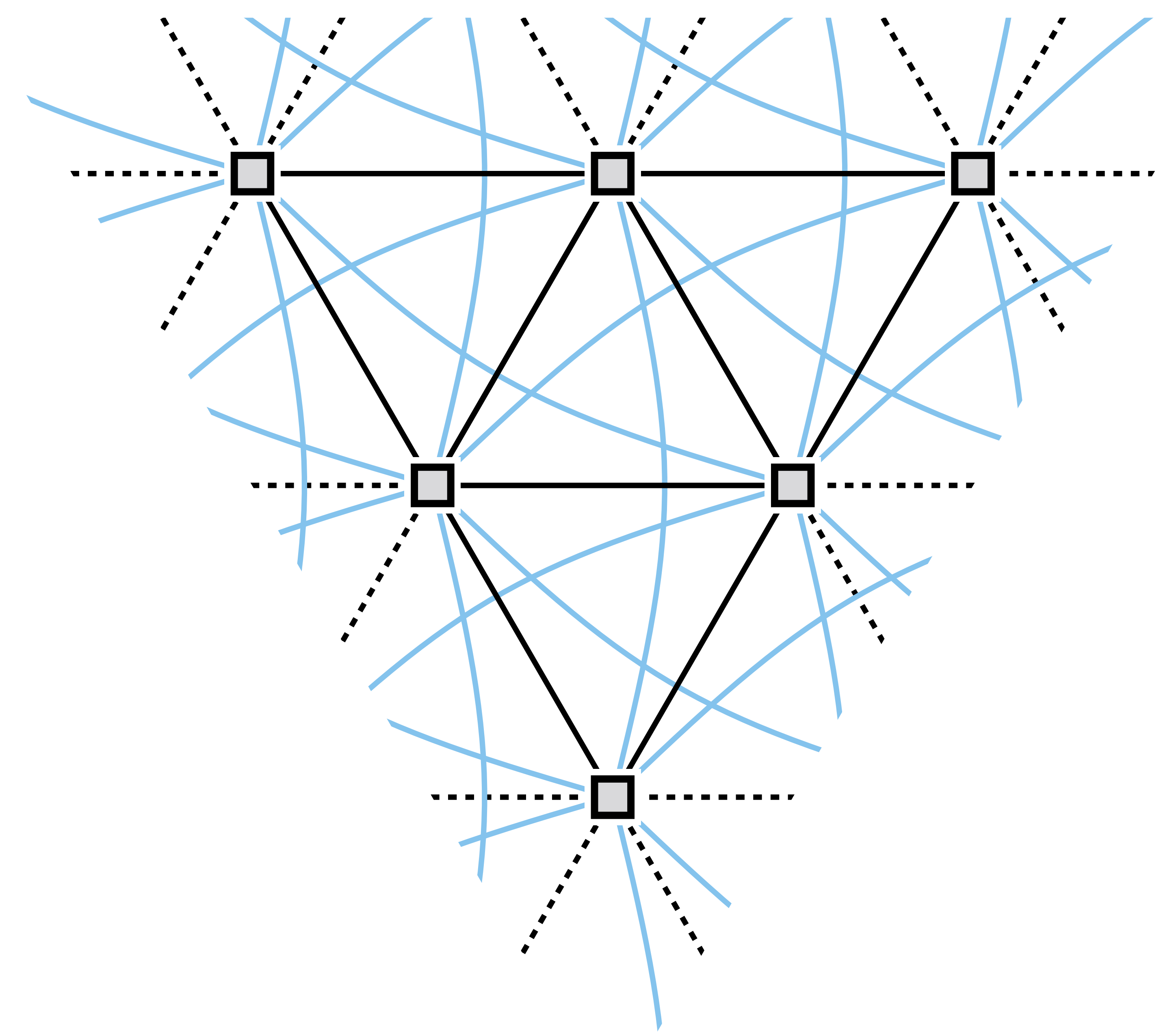




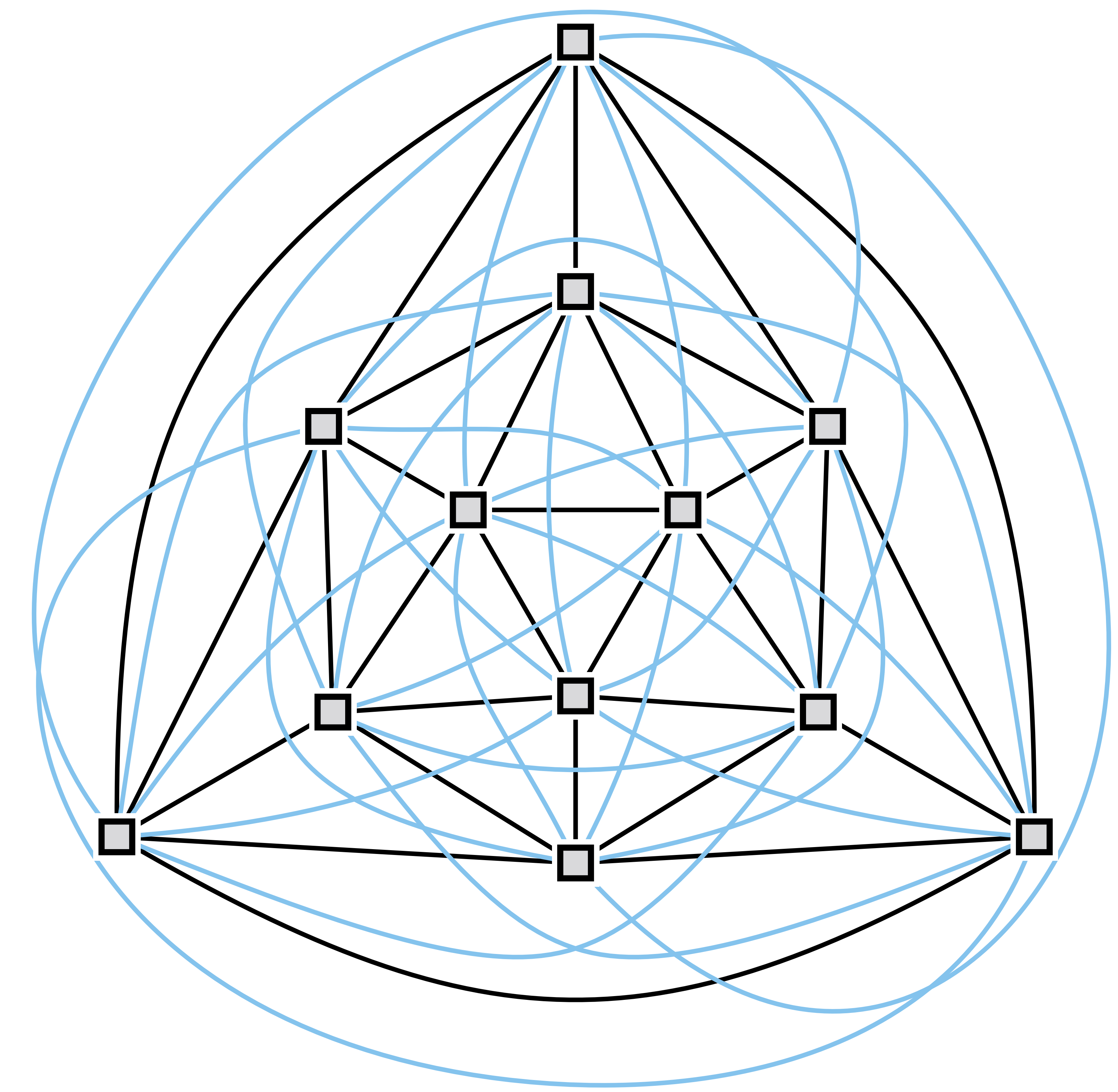
Some rulers don't

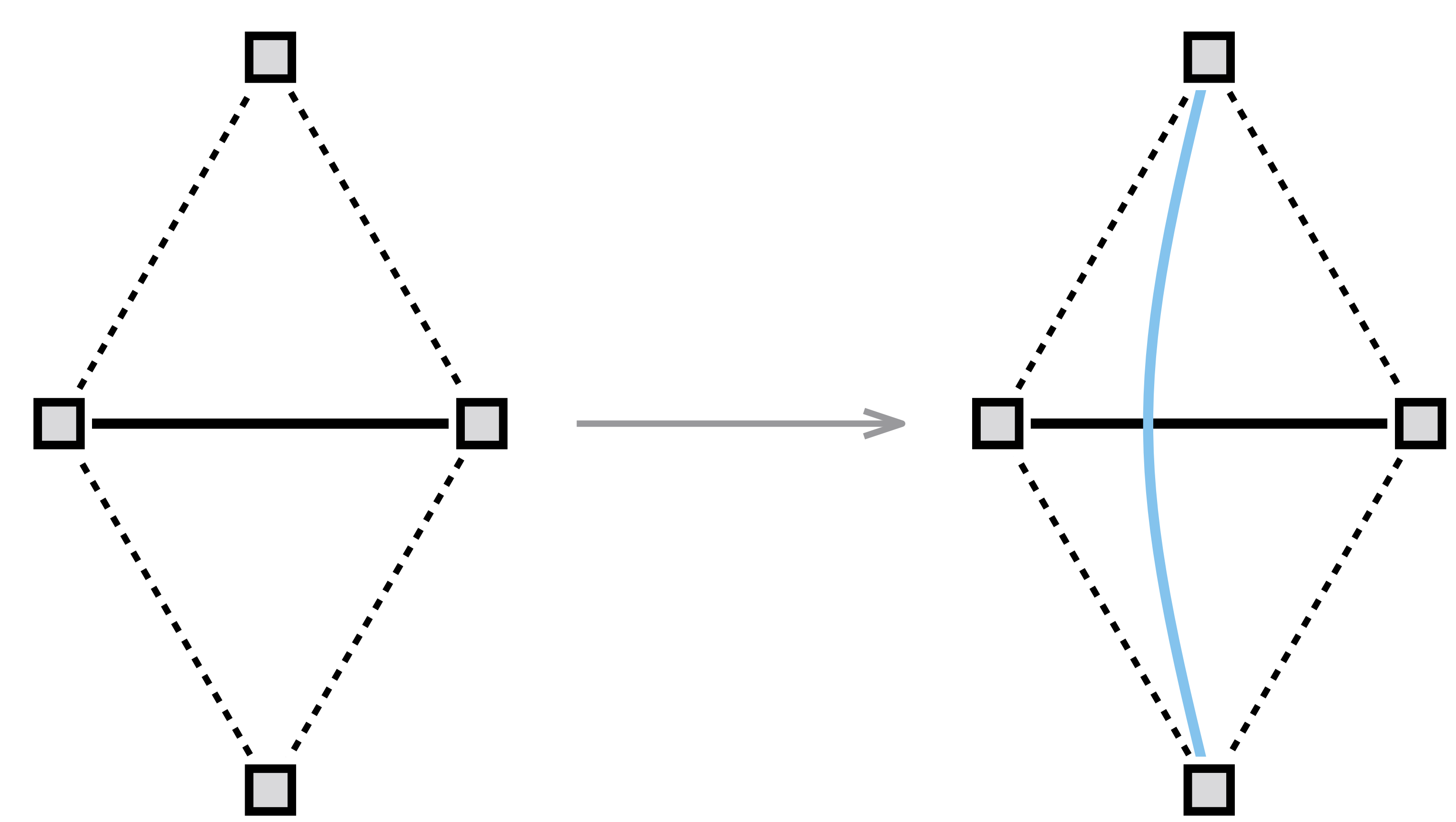
Forbidden cell	Lower Bound	Upper Bound
Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
Quadria	$\mathbf{6n-12}$	
Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
Trimon		

Triangulation augmentation

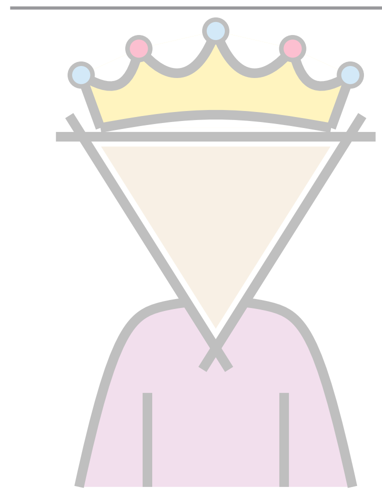
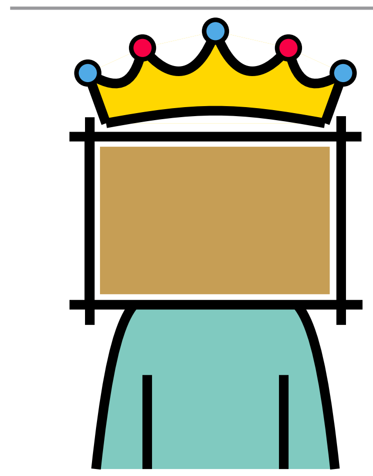
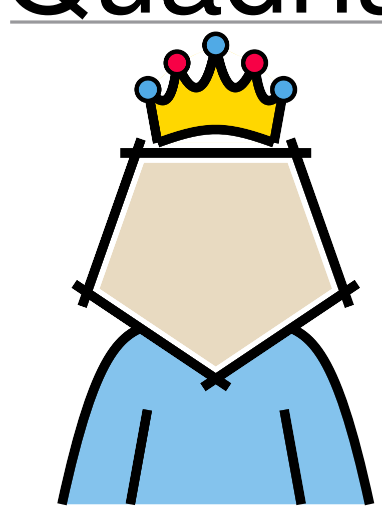
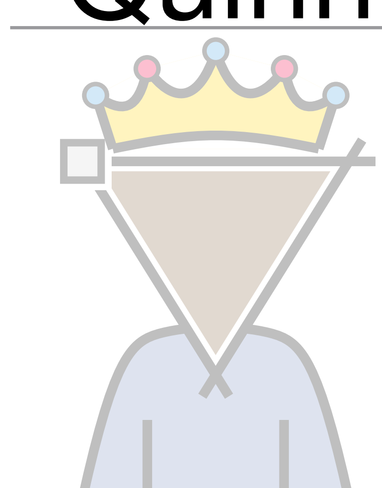


$$\begin{array}{r} 3n-6 \\ +\ 3n-6 \\ \hline 6n-12 \end{array}$$



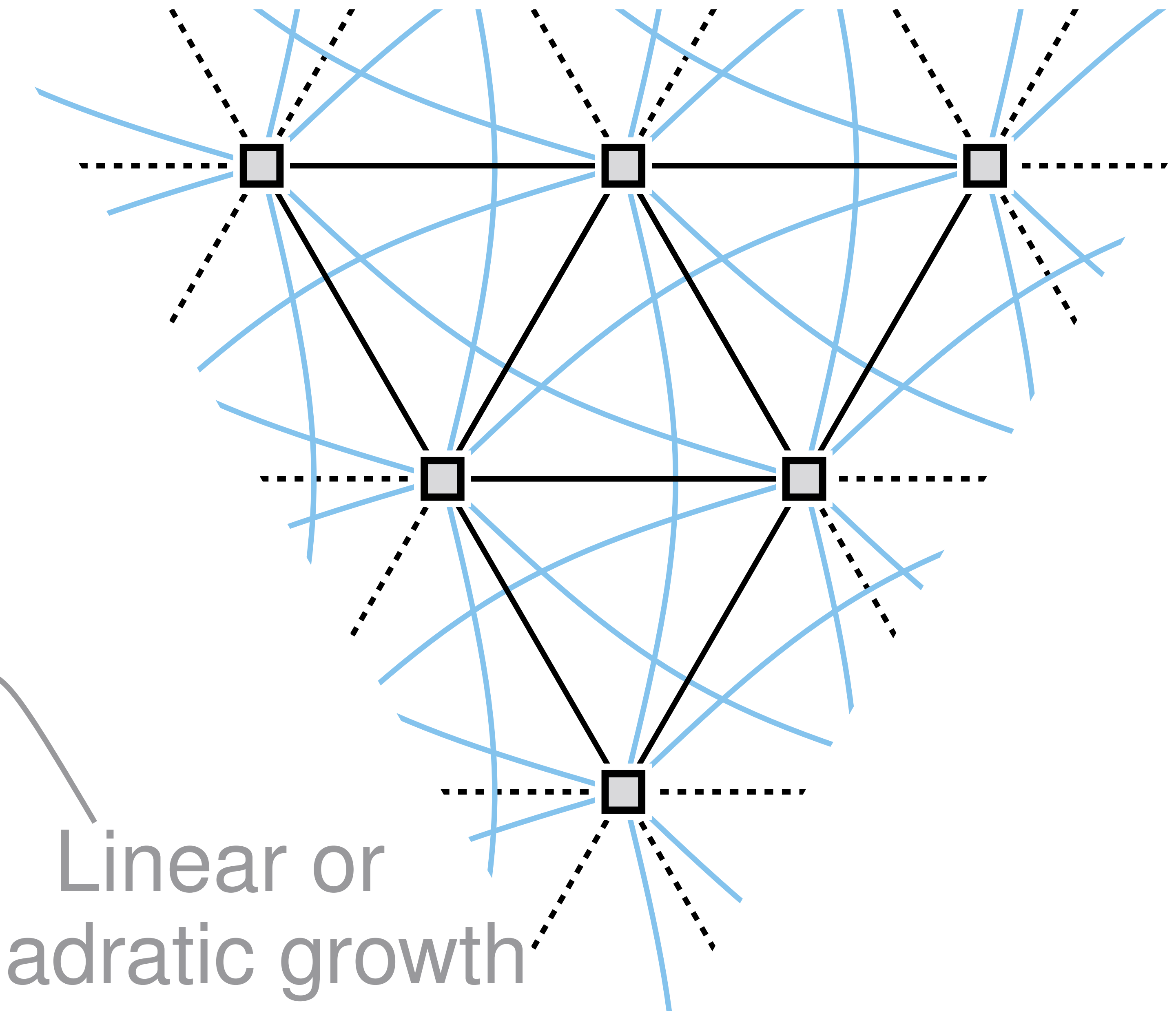


Some rulers don't

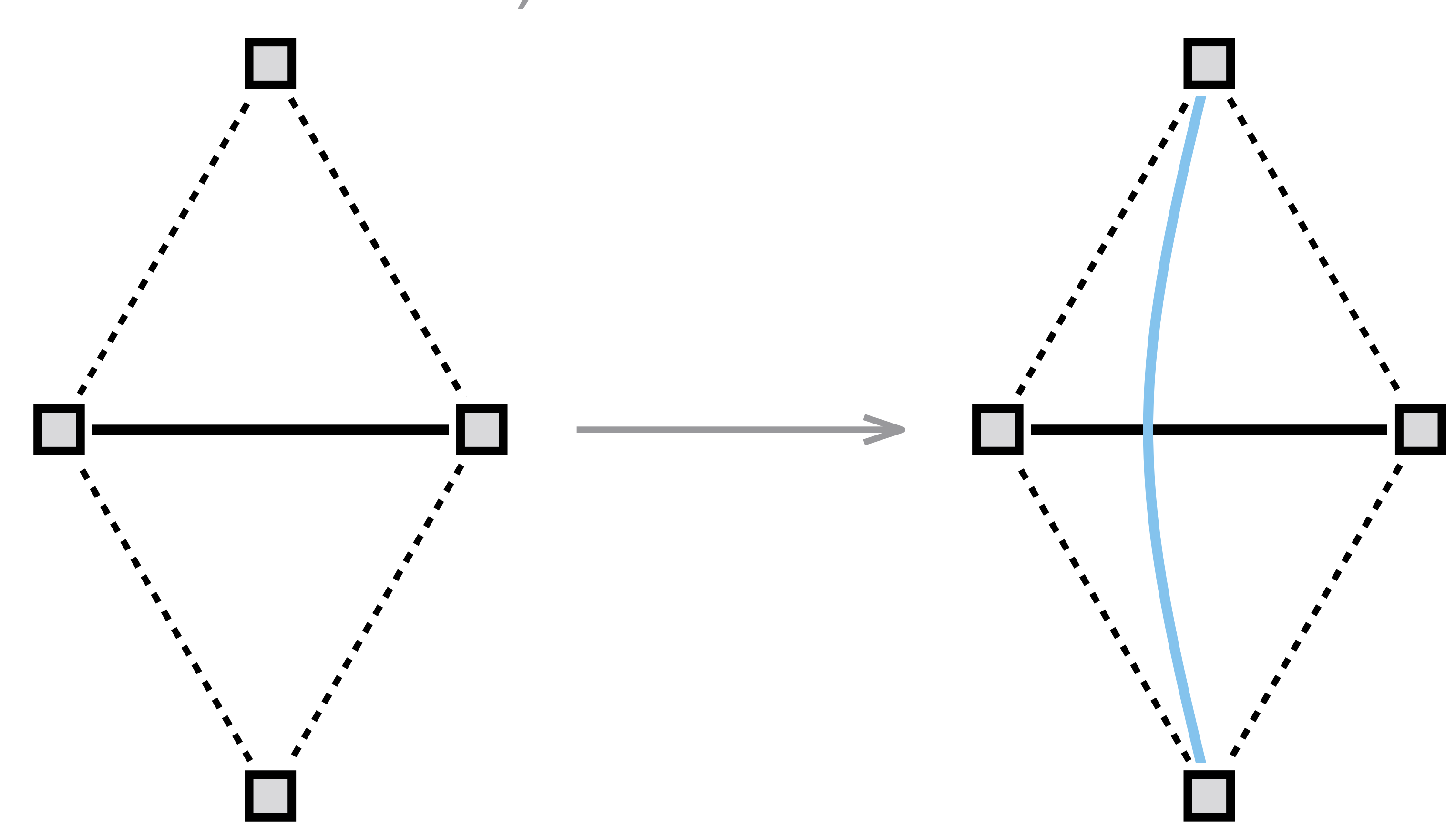
Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon		

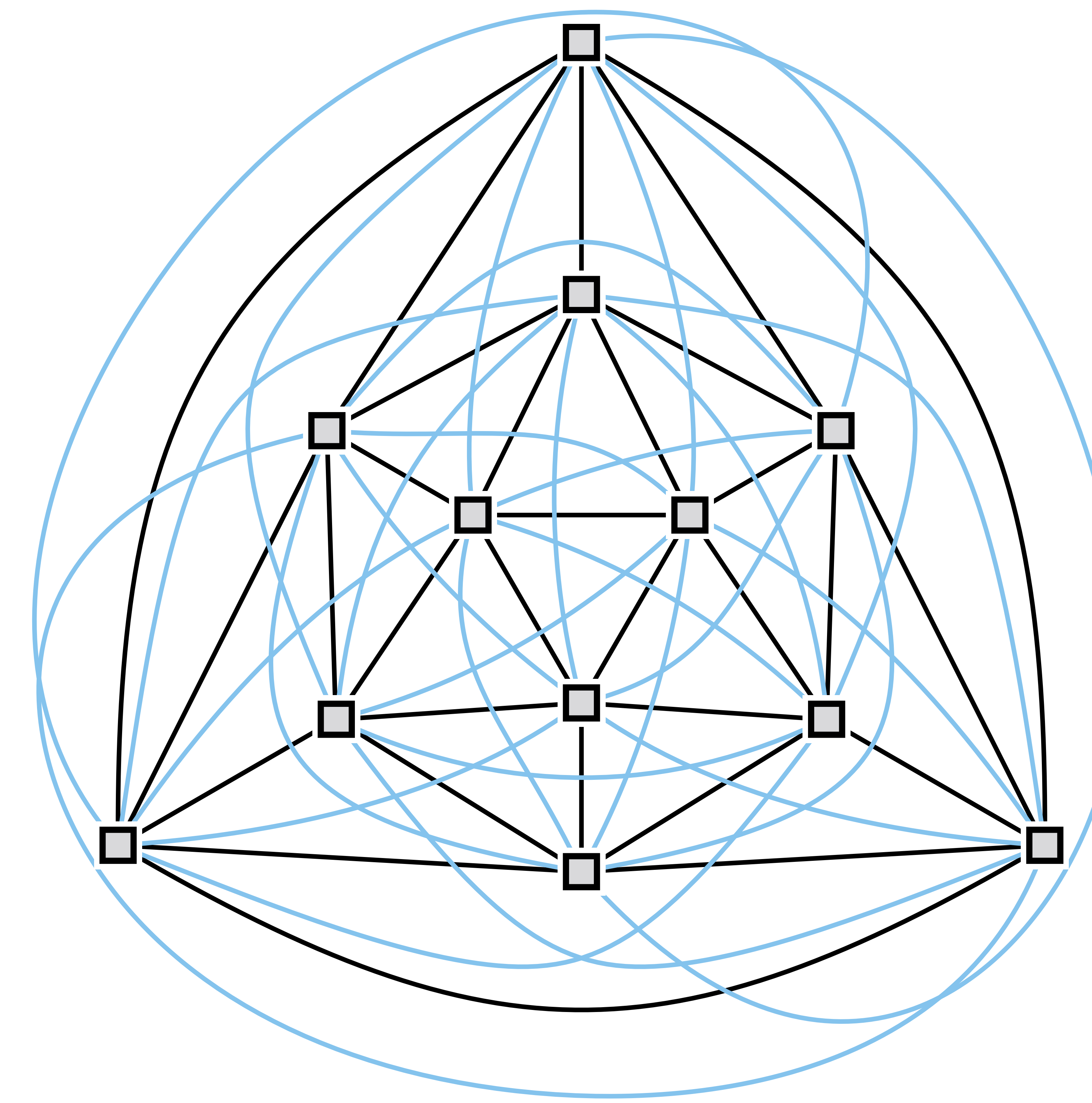
Linear or quadratic growth (or in between)?

Triangulation augmentation

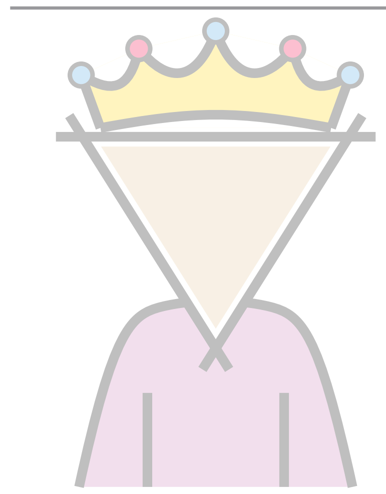
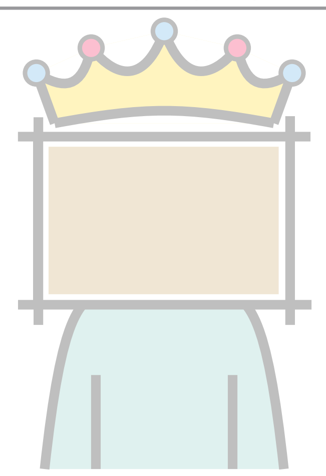
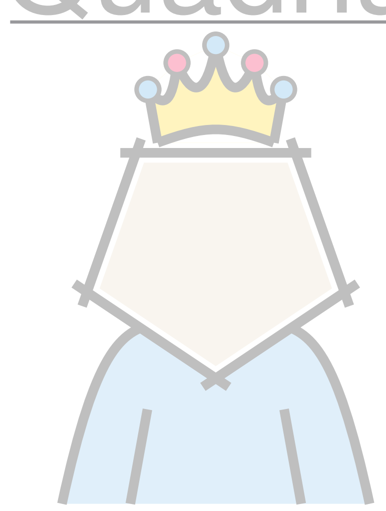
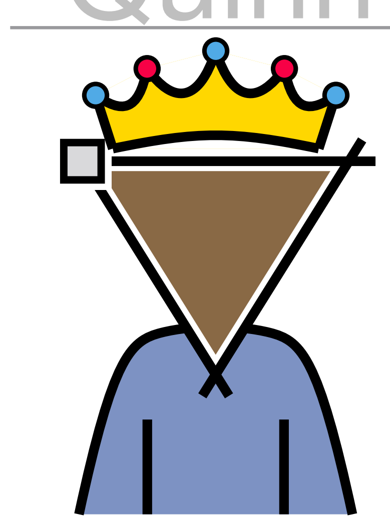


$$\begin{array}{r} 3n-6 \\ +\ 3n-6 \\ \hline 6n-12 \end{array}$$

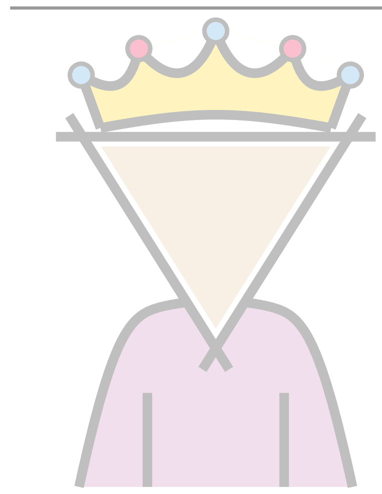
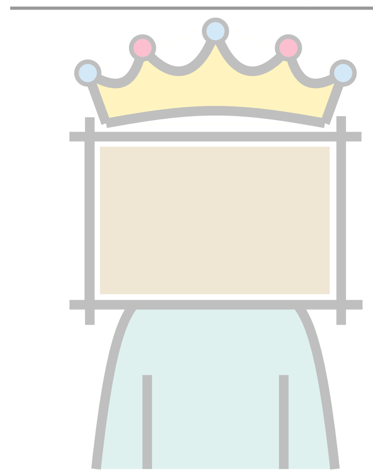
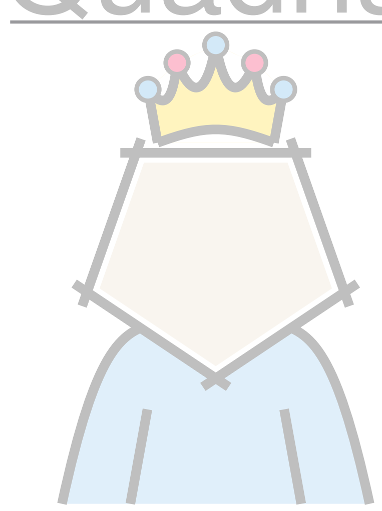
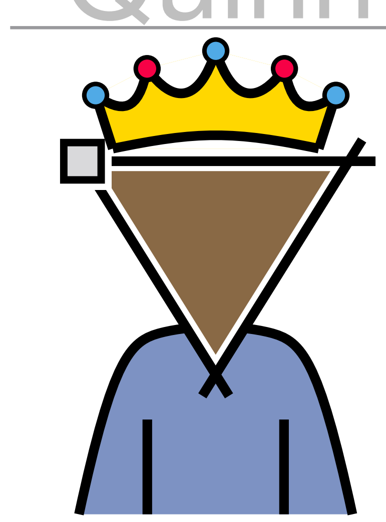




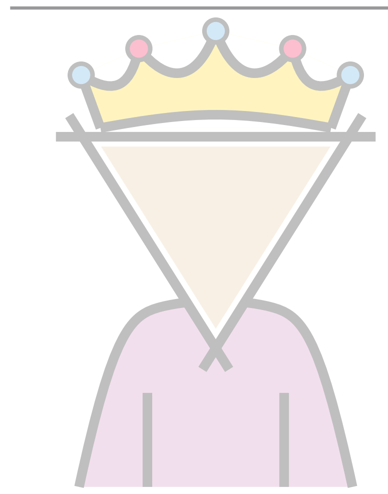
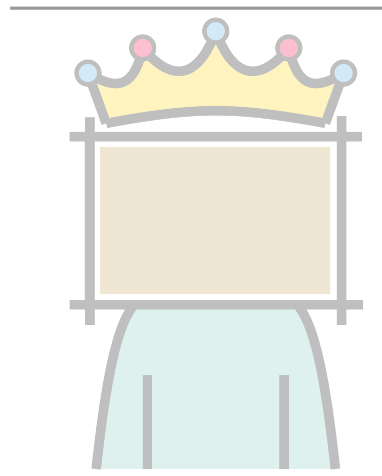
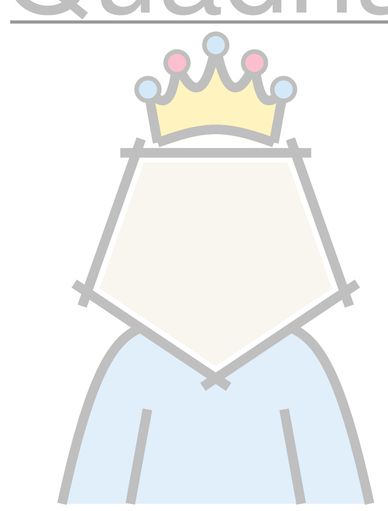
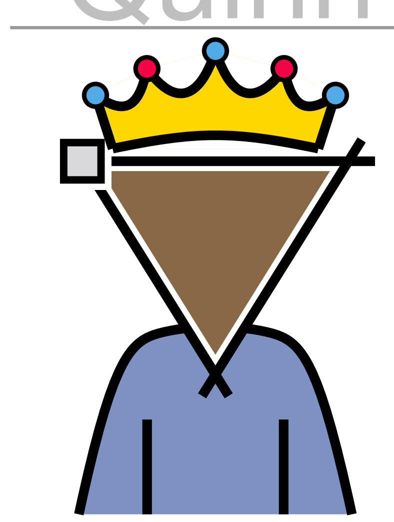
Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon		$\binom{n}{2}$

Some rulers don't

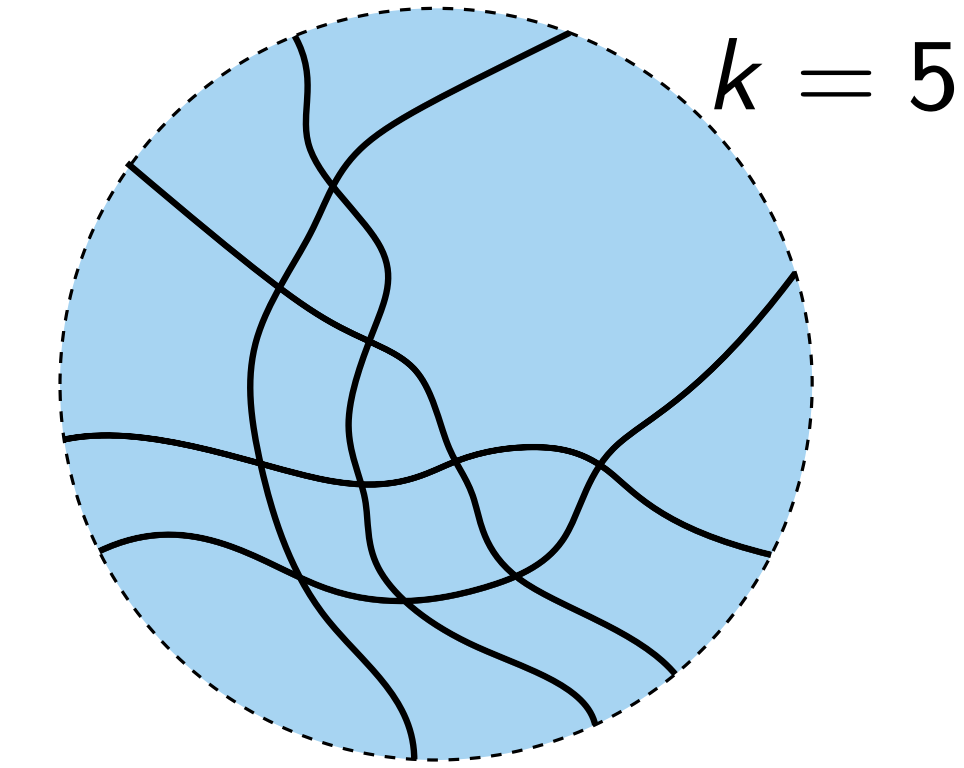
Forbidden cell	Lower Bound	Upper Bound
 Triss	**$7n-30$** based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] **$7n-20$**
 Quadria	**$6n-12$**	$\binom{n}{2}$
 Quinn	**$6n-12$**	**$6n-12$**
 Trimon	**$\Omega(n^2)^*$**	$\binom{n}{2}$

Some rulers don't

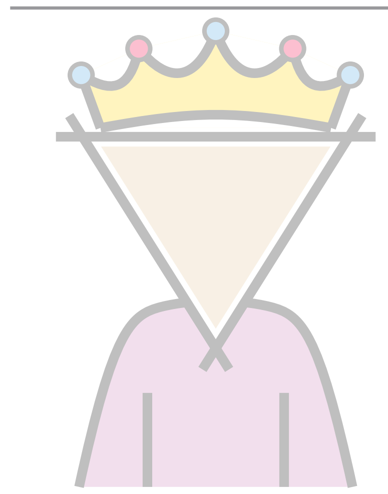
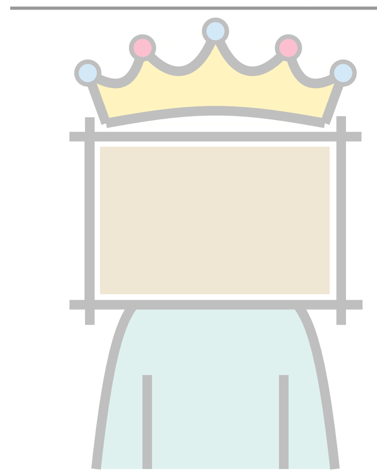
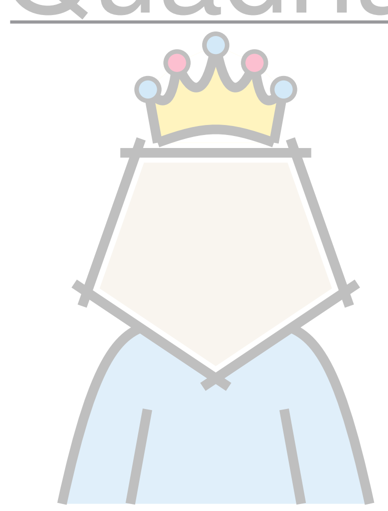
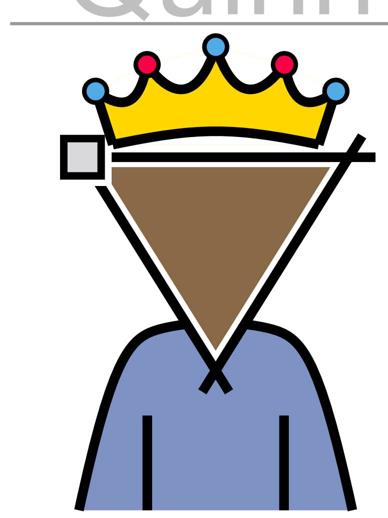
Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}^*$	$\binom{n}{2}$

Supply and demand

- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.



Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}^*$	$\binom{n}{2}$

Supply and demand

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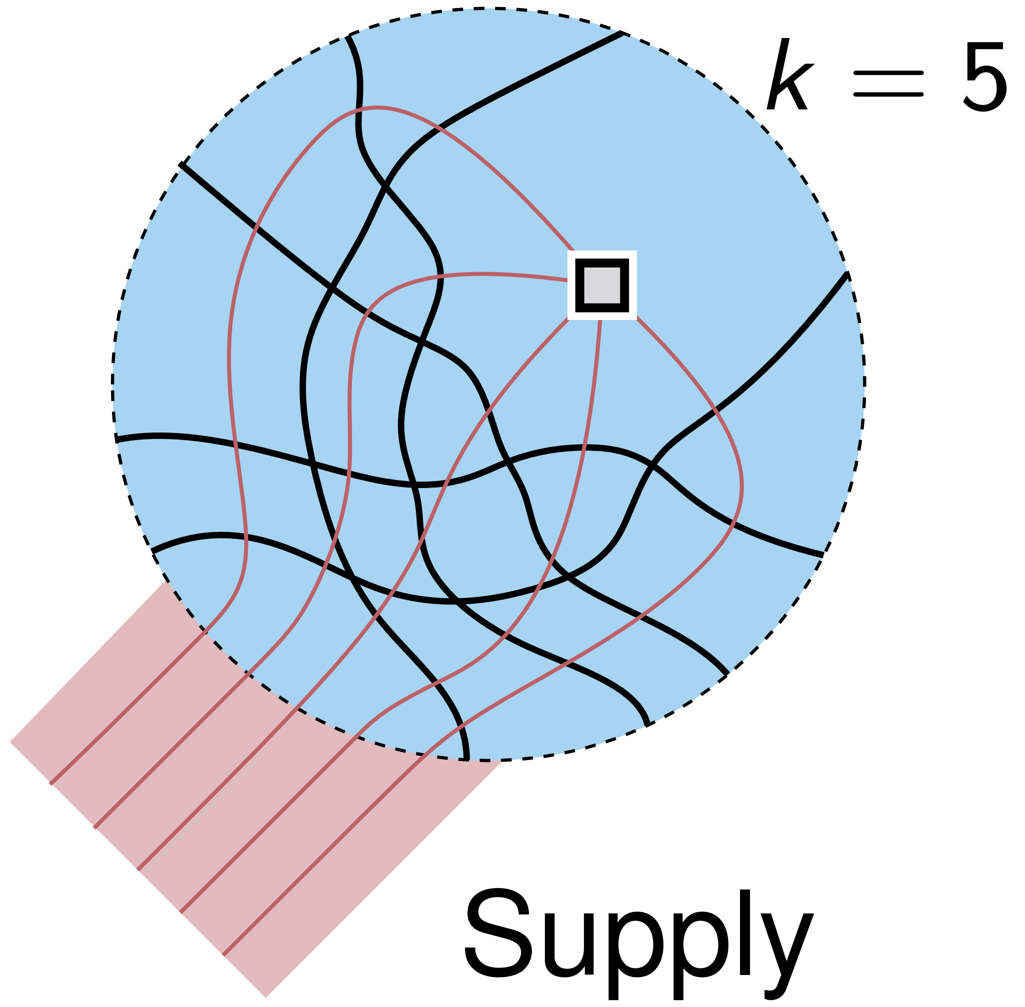
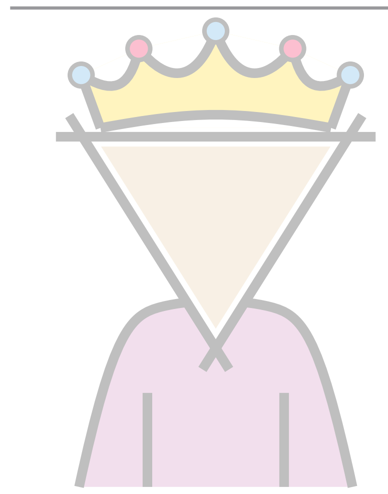
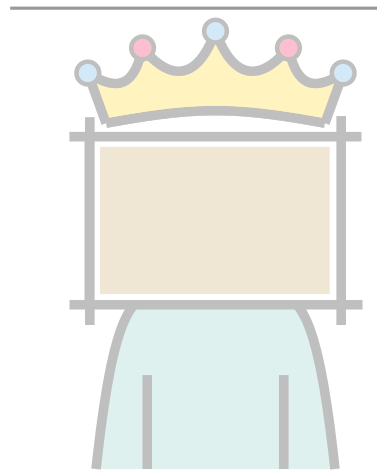
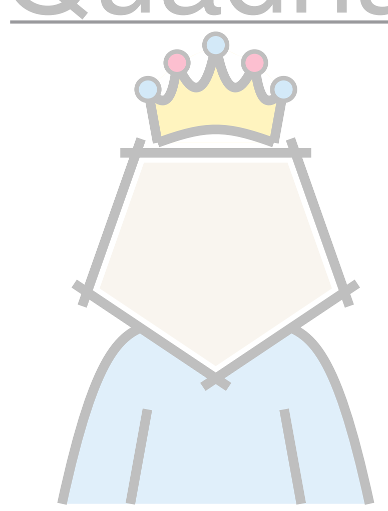
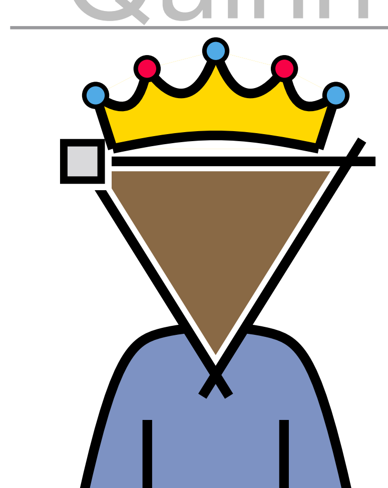
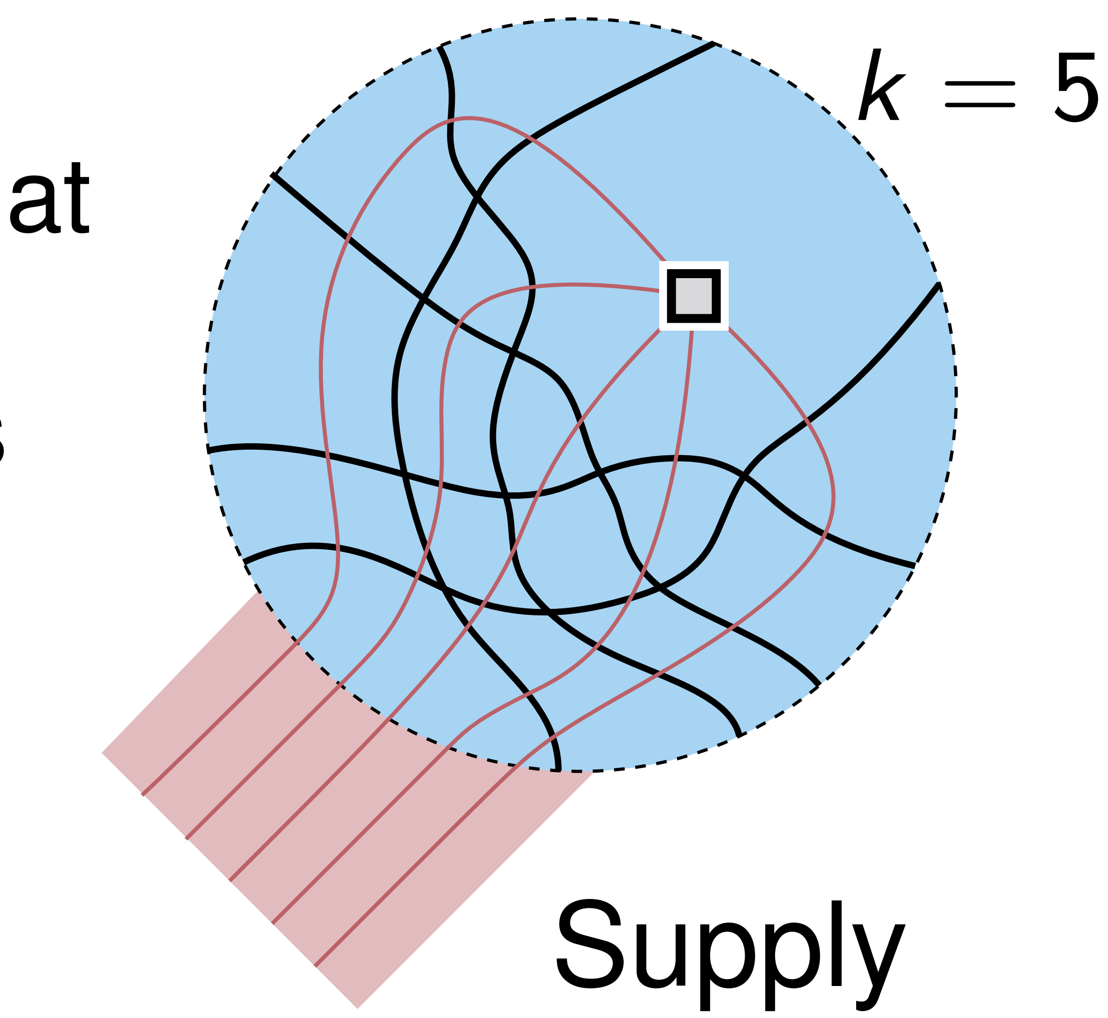


Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\boldsymbol{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\boldsymbol{7n-20}$
 Quadria	$\boldsymbol{6n-12}$	$\binom{n}{2}$
 Quinn	$\boldsymbol{6n-12}$	$\boldsymbol{6n-12}$
 Trimon	$\boldsymbol{\Omega(n^2)}^{*}$	$\binom{n}{2}$

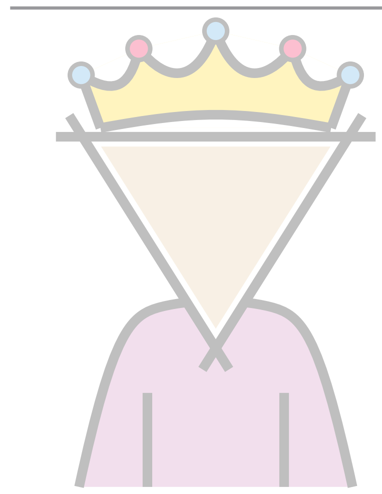
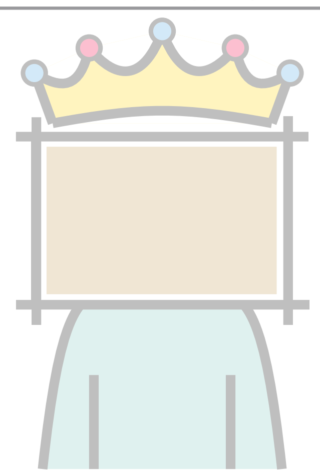
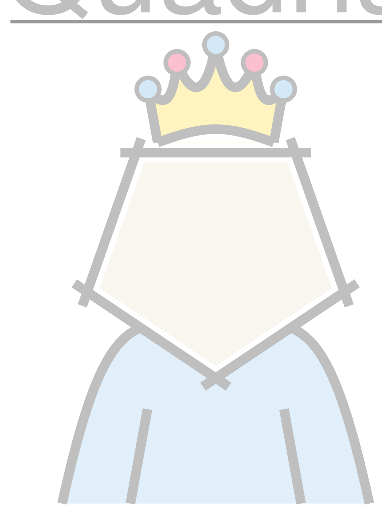
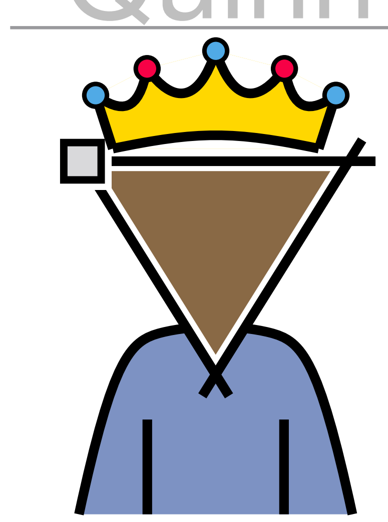
Supply and demand

- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.
- Want many spots of k pairwise crossing edges



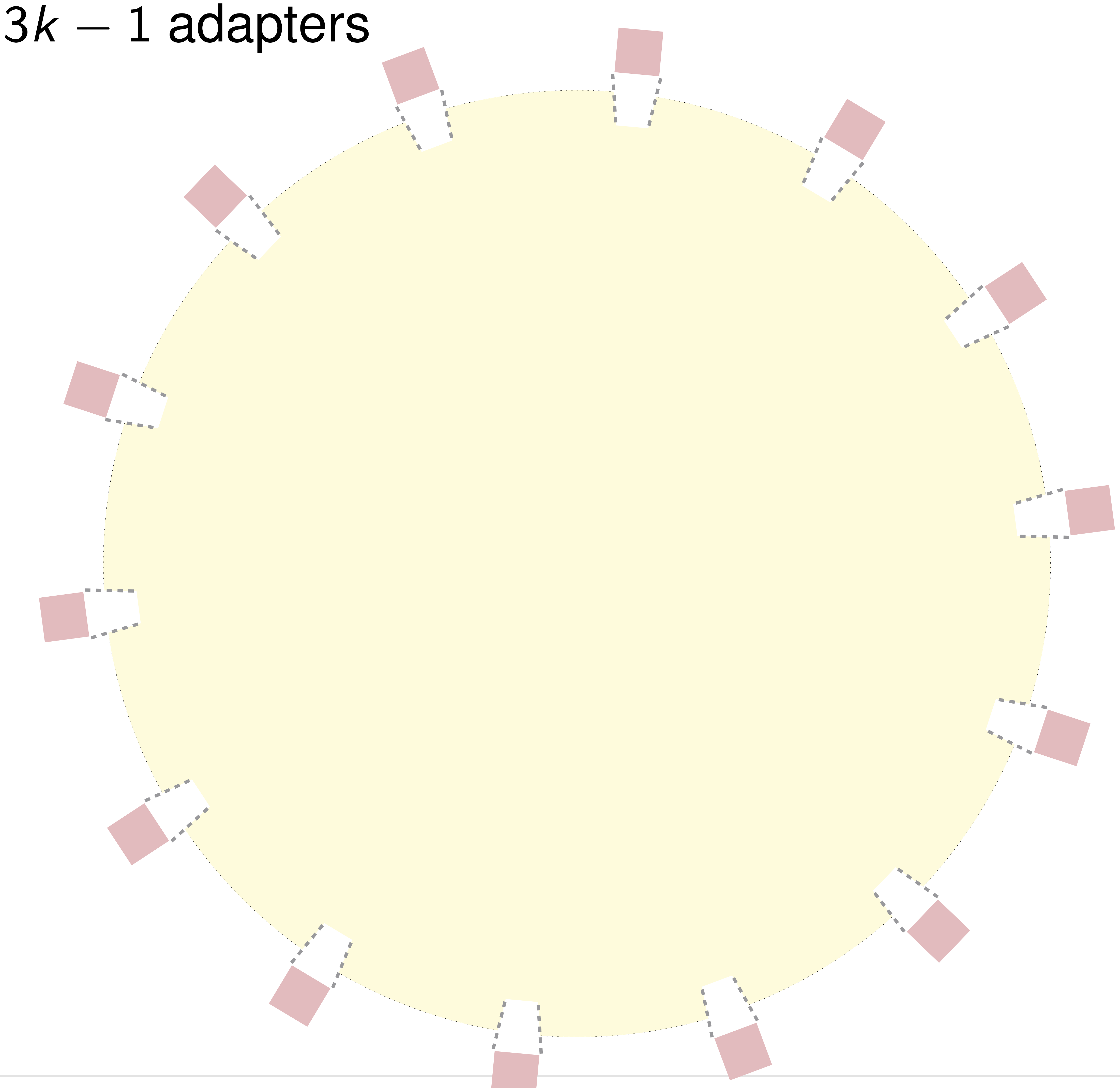
Supply corridor

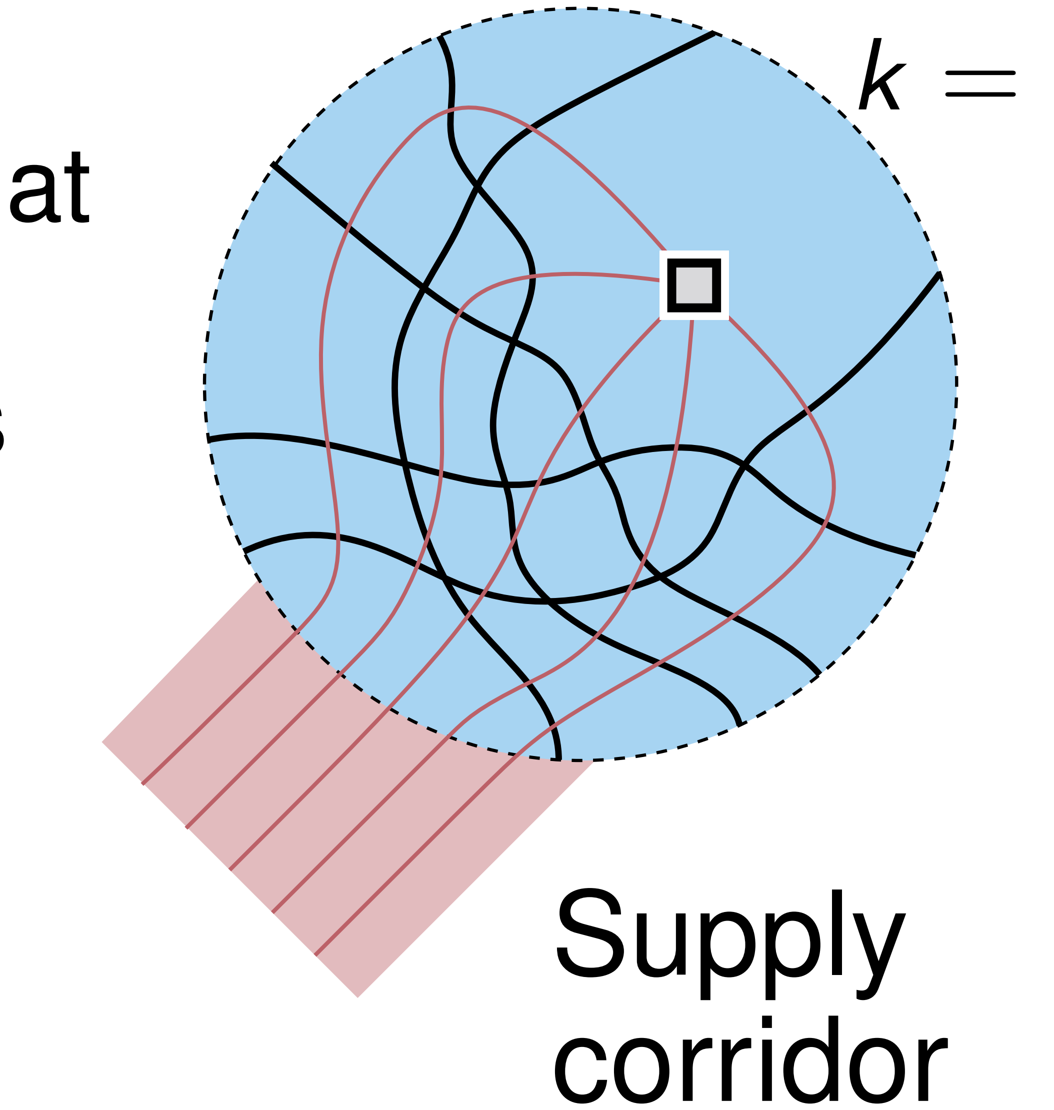
Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	**$7n - 30$** based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07] **$7n - 20$**
 Quadria	**$6n - 12$**	$\binom{n}{2}$
 Quinn	**$6n - 12$**	**$6n - 12$**
 Trimon	**$\Omega(n^2)$***	$\binom{n}{2}$

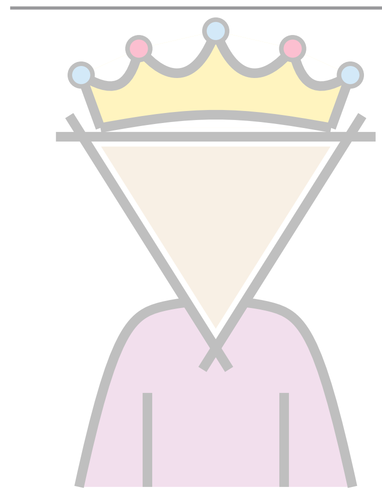
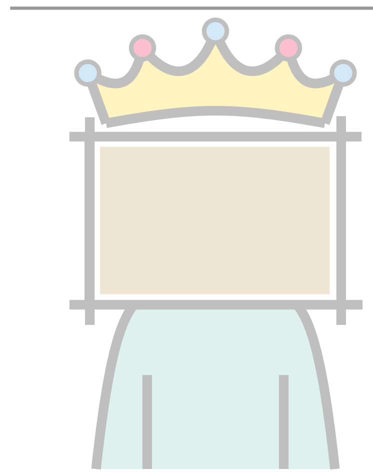
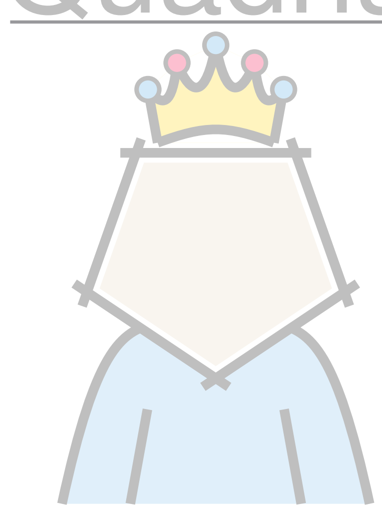
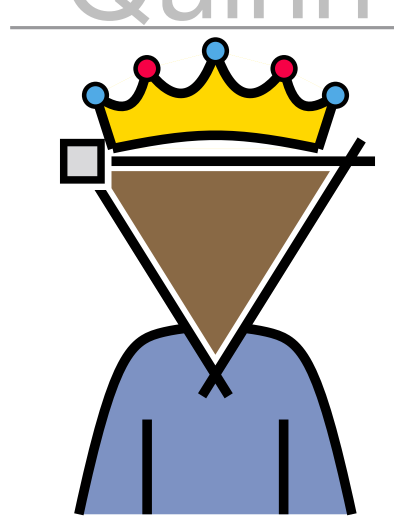
Supply and demand

- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.
- Want many spots of k pairwise crossing edges
- $3k - 1$ adapters



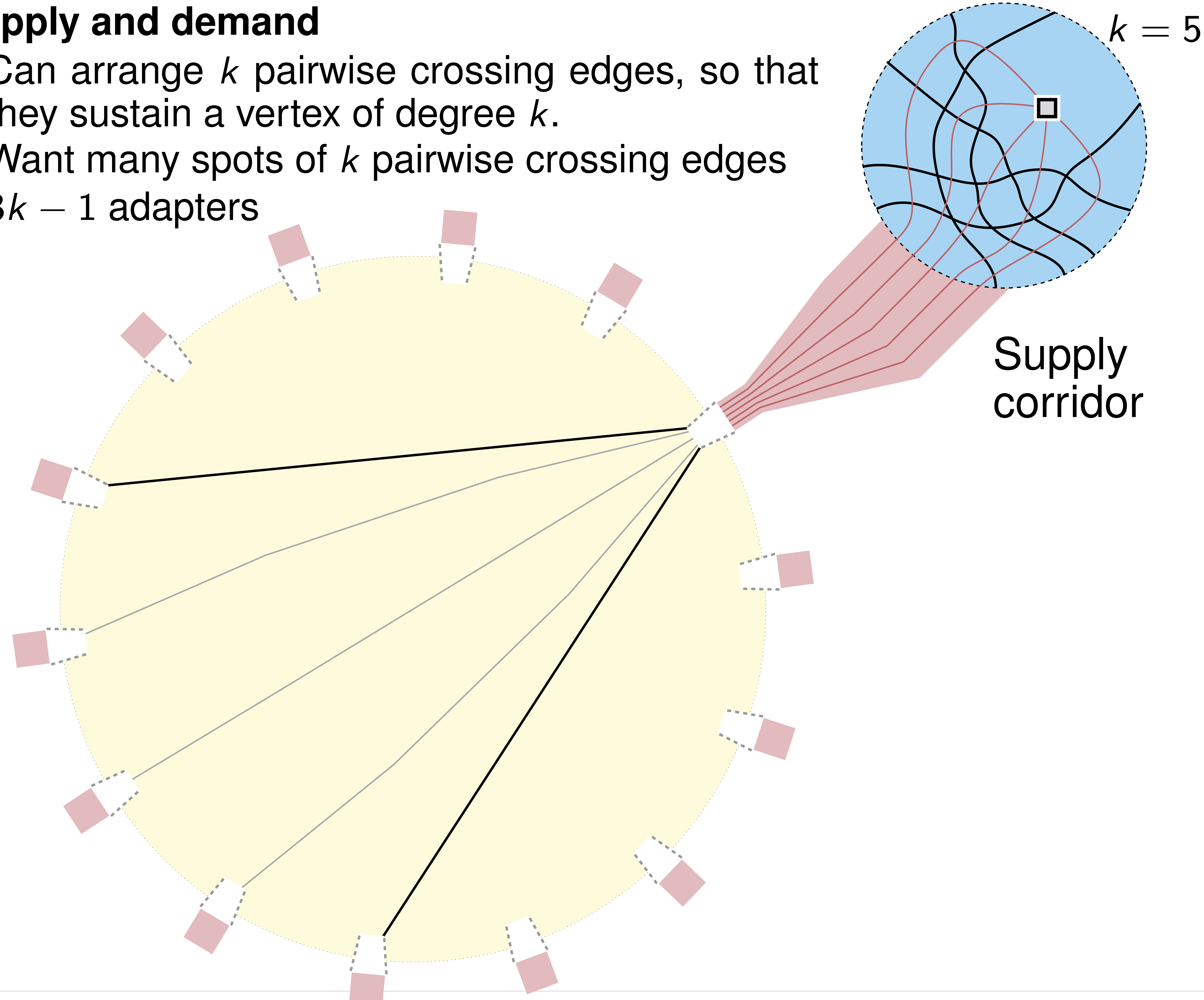


Some rulers don't

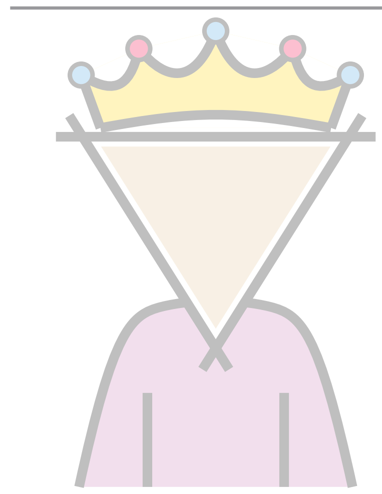
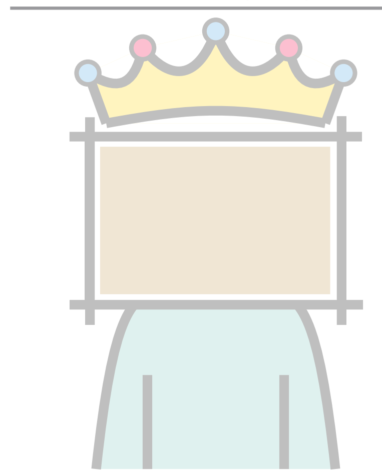
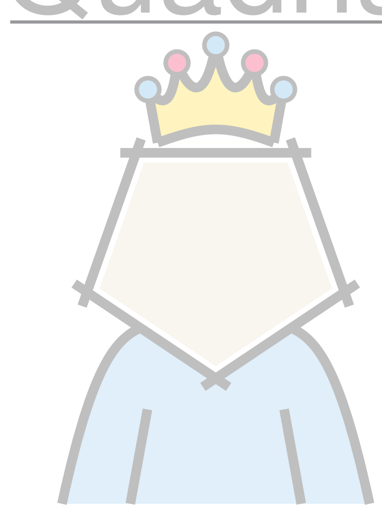
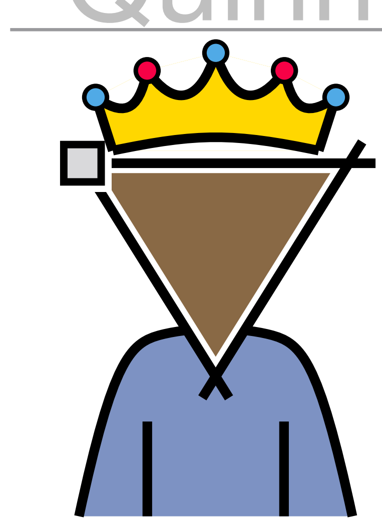
Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)^*}$	$\binom{n}{2}$

Supply and demand

- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.
- Want many spots of k pairwise crossing edges
- $3k-1$ adapters

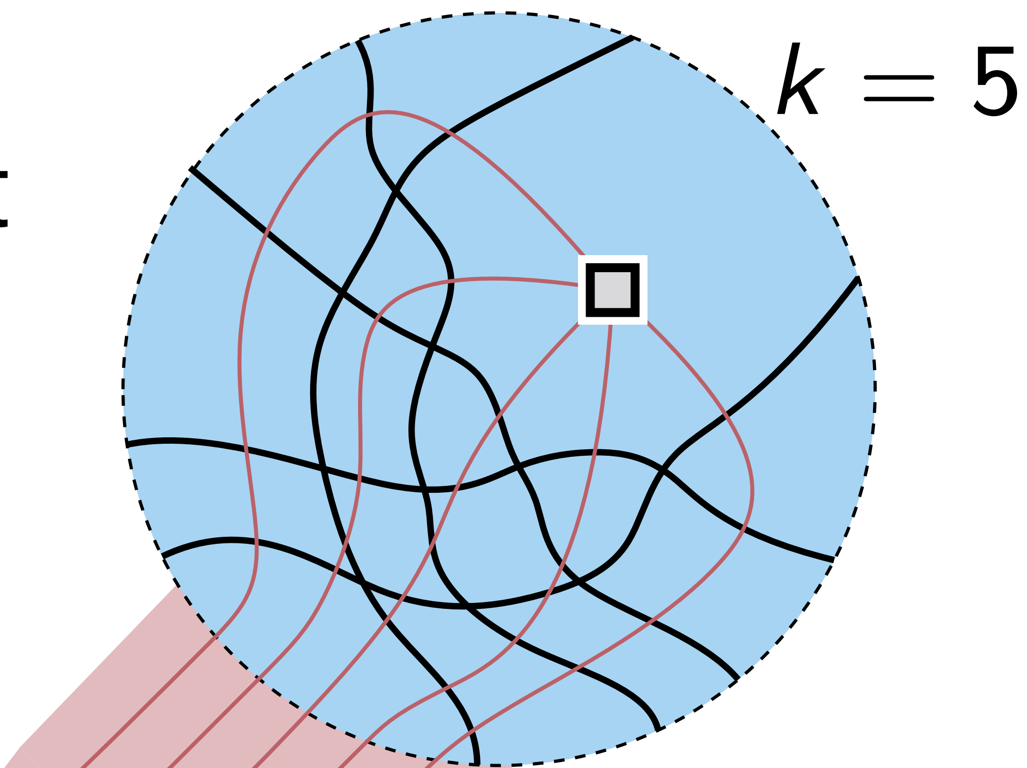


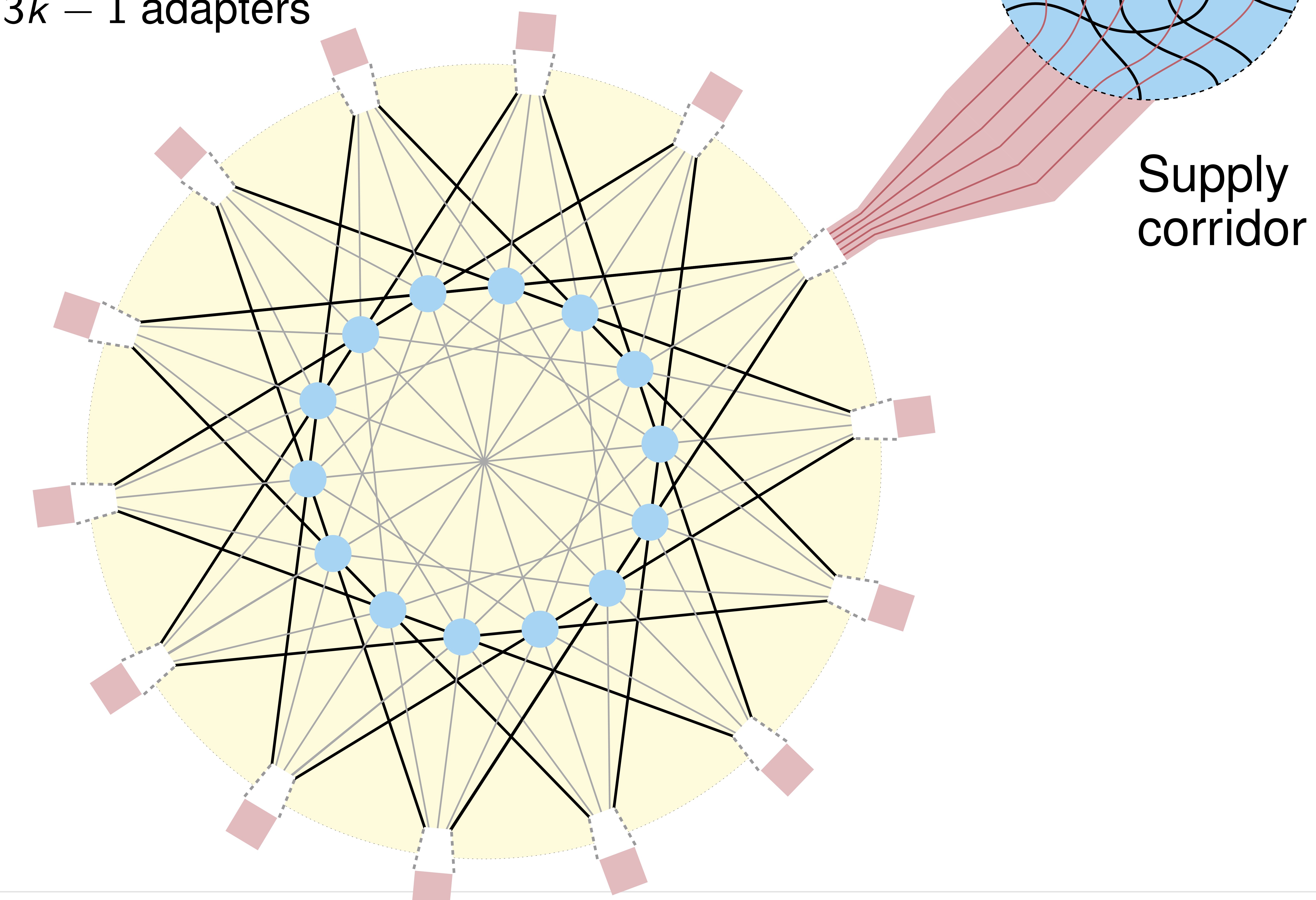
Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}^*$	$\binom{n}{2}$

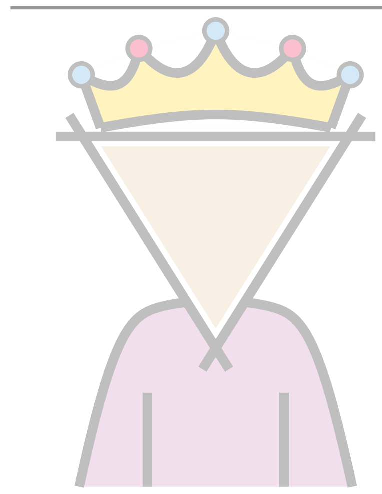
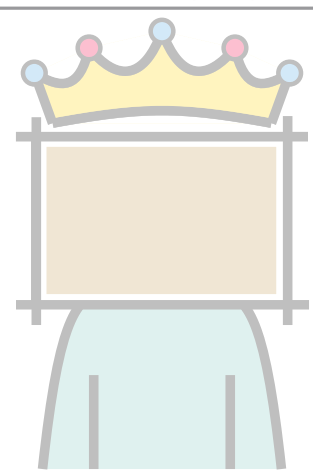
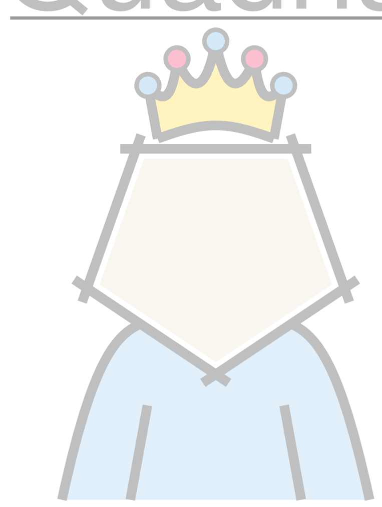
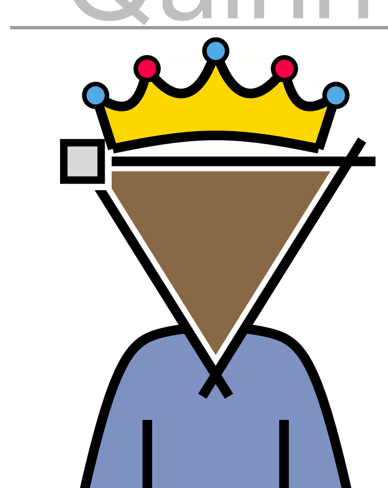
Supply and demand

- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.
- Want many spots of k pairwise crossing edges
- $3k-1$ adapters



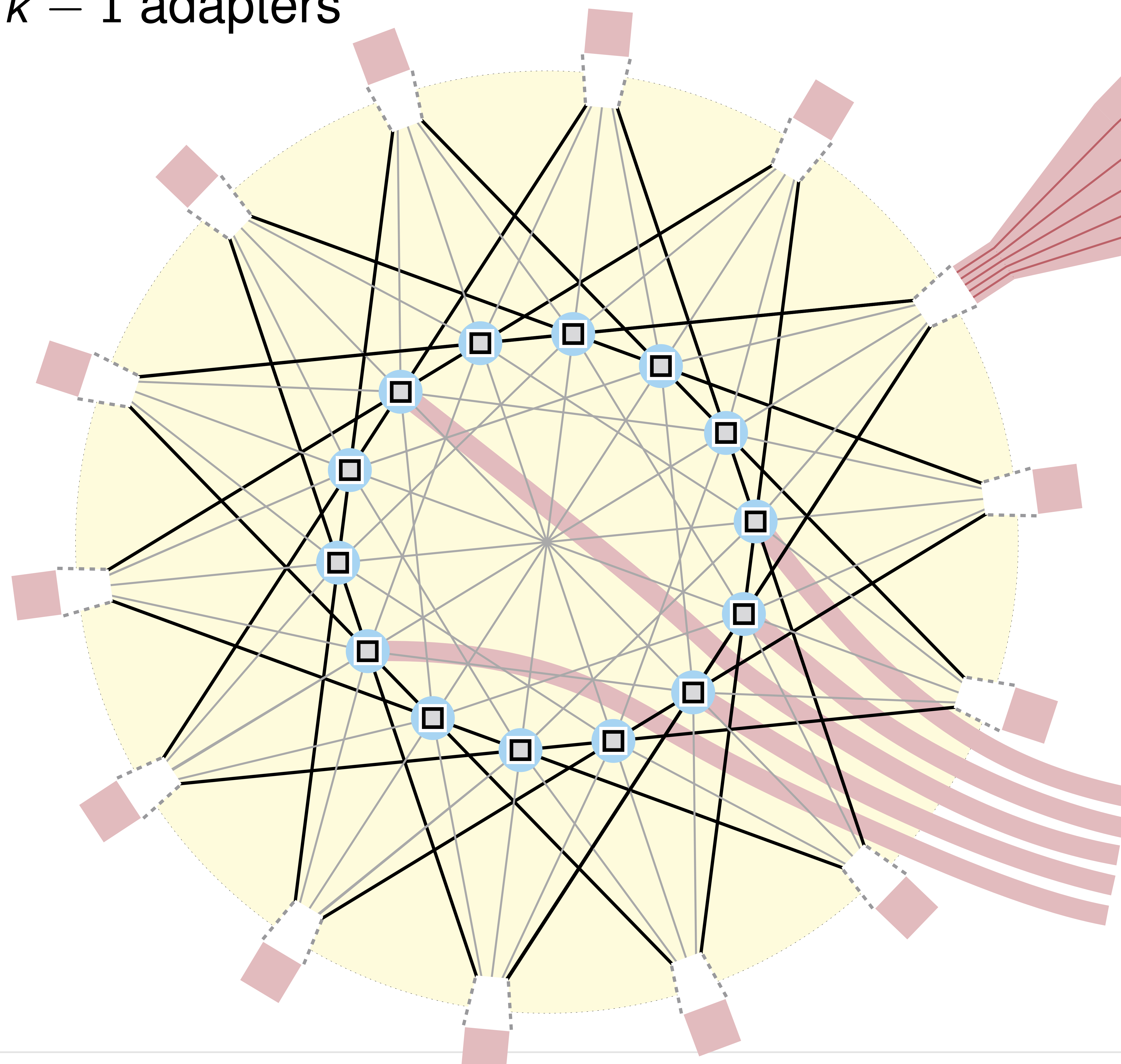


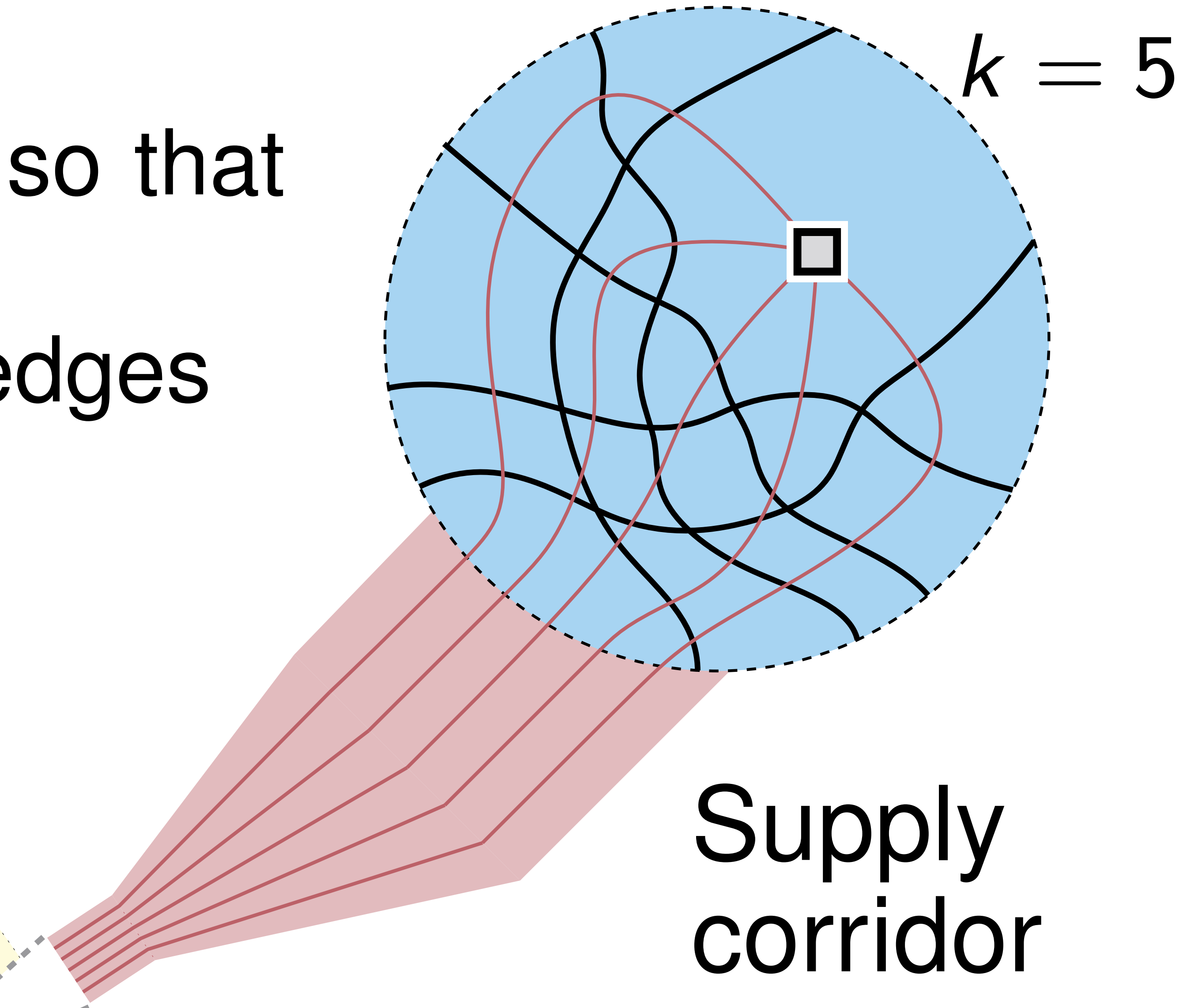
Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$7n-30$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $7n-20$
 Quadria	$6n-12$	$\binom{n}{2}$
 Quinn	$6n-12$	$6n-12$
 Trimon	$\Omega(n^2)^*$	$\binom{n}{2}$

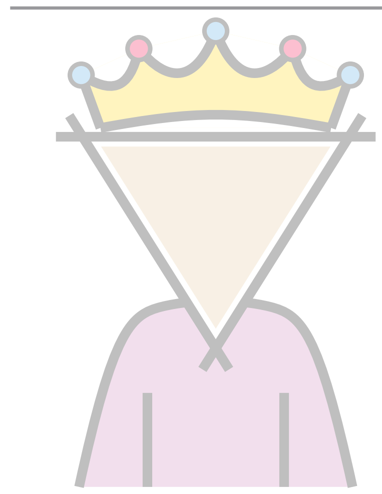
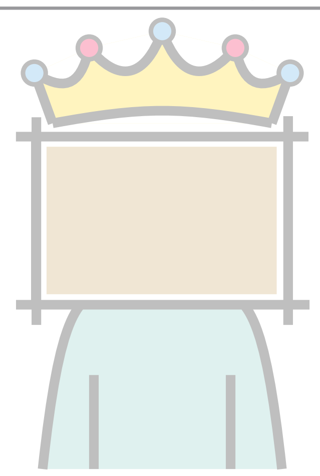
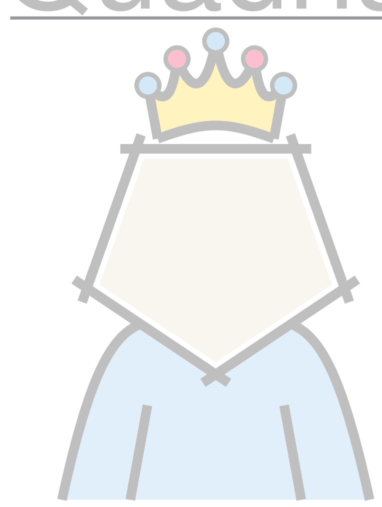
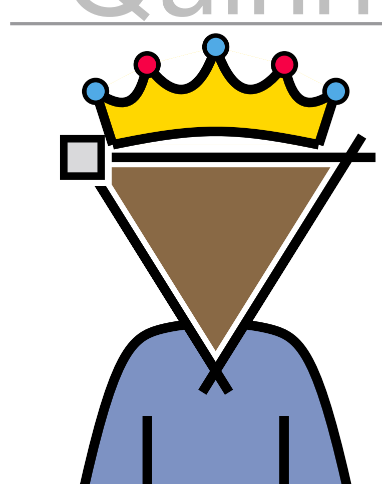
Supply and demand

- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.
- Want many spots of k pairwise crossing edges
- $3k-1$ adapters



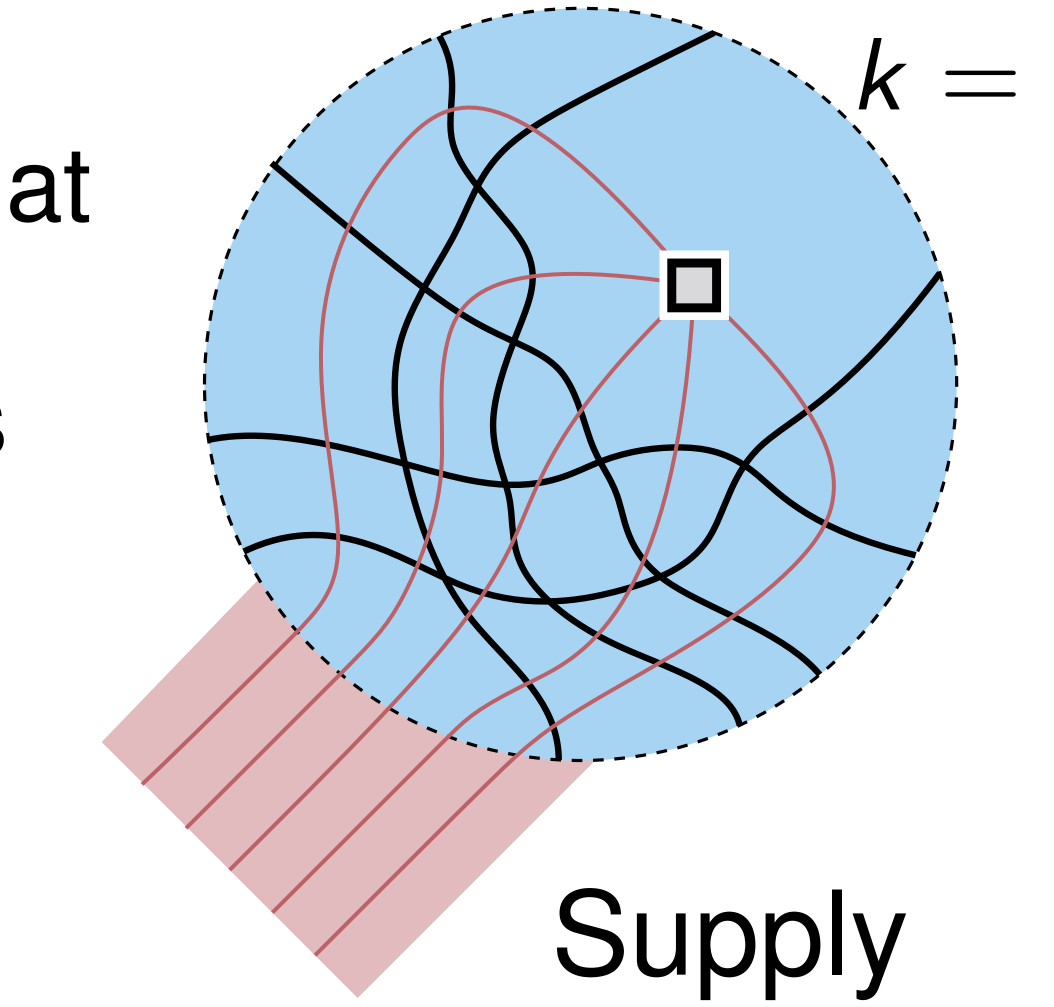


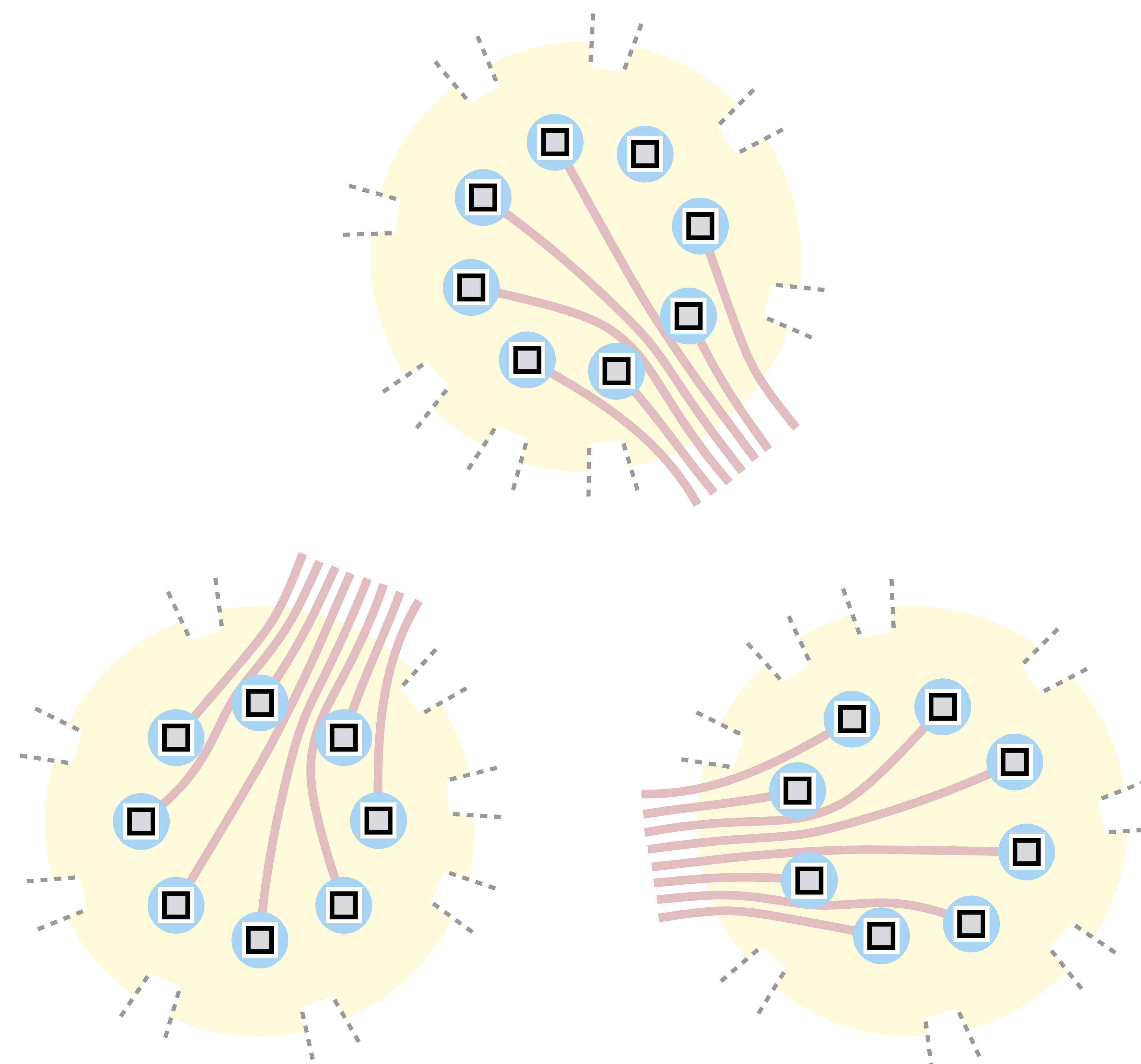
Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}$*	$\binom{n}{2}$

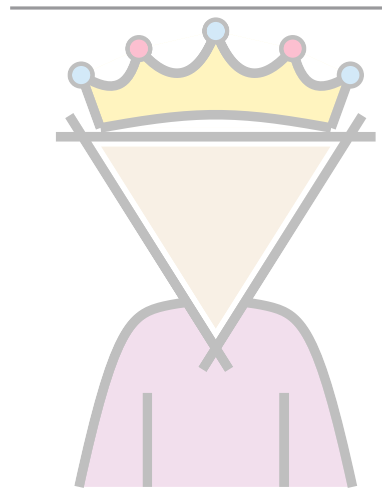
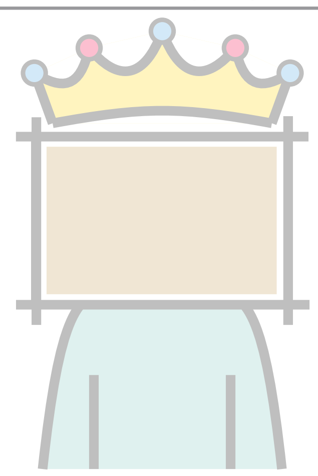
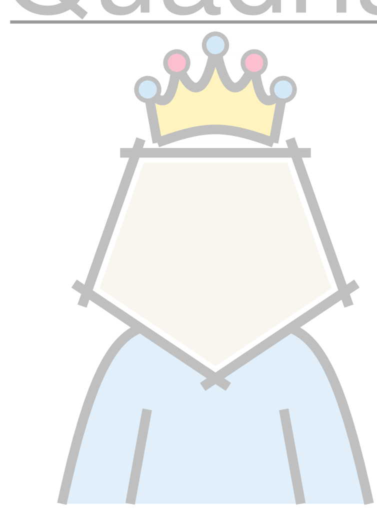
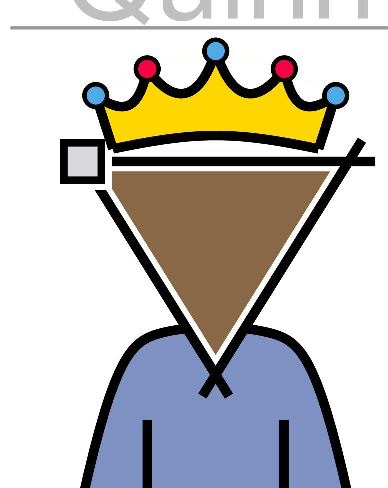
Supply and demand

- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.
- Want many spots of k pairwise crossing edges
- $3k-1$ adapters per copy
- Connect corridors to adapters in another copy





Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}$*	$\binom{n}{2}$

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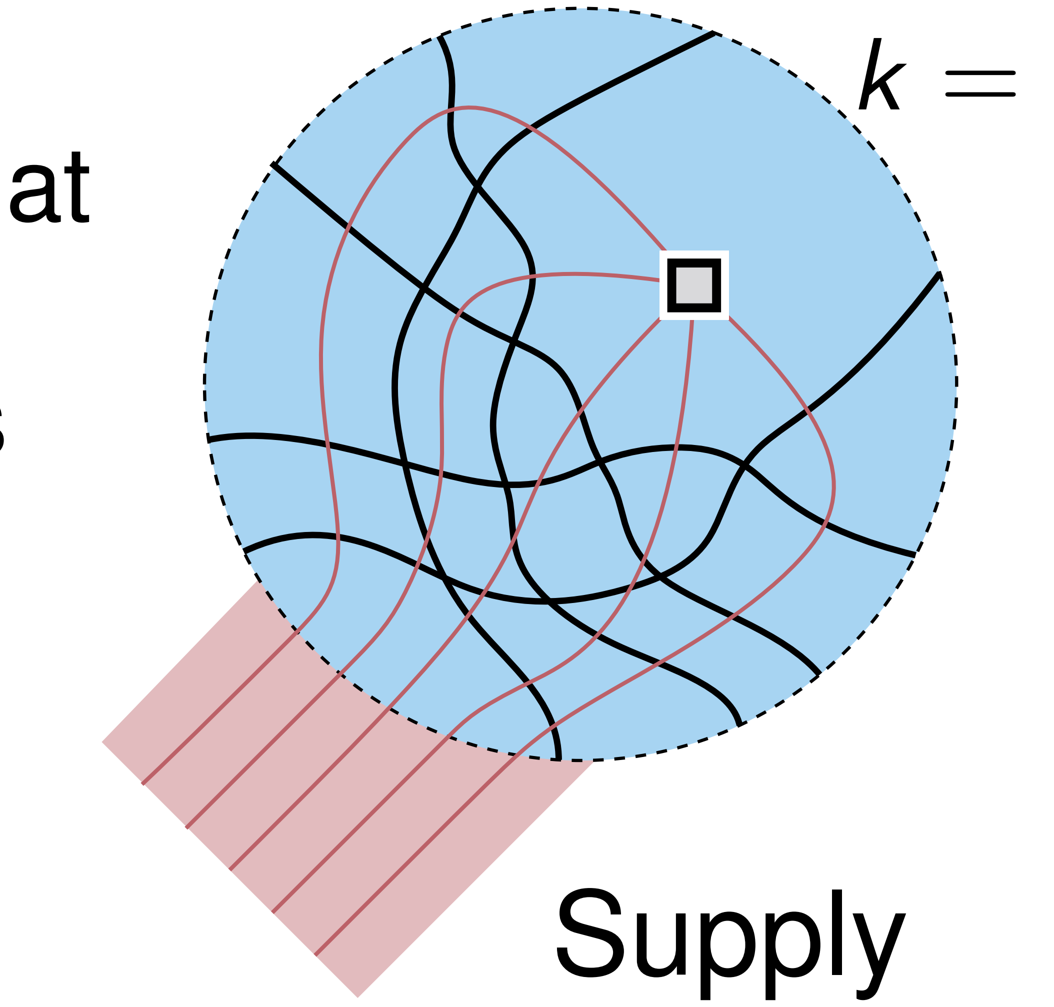
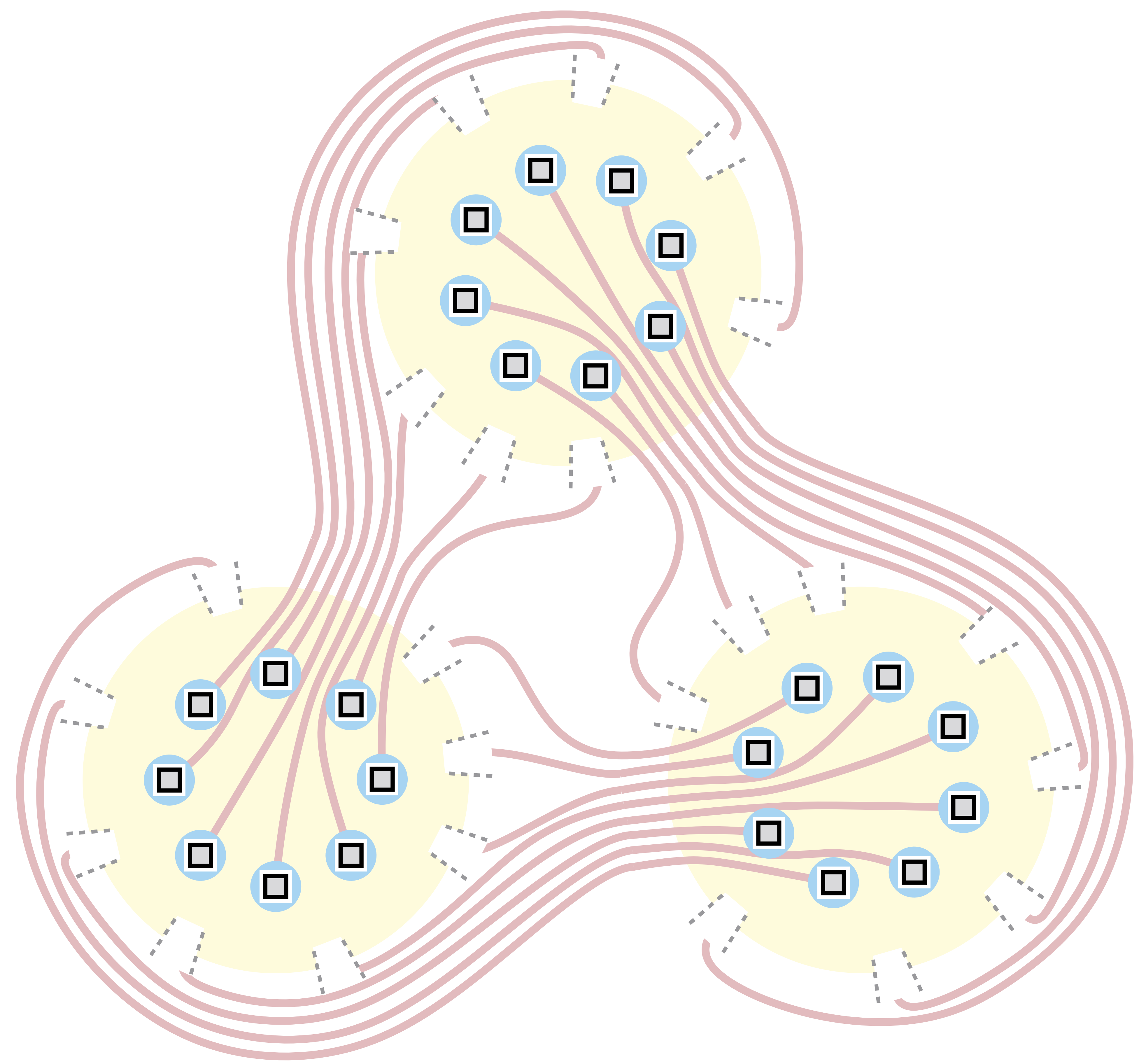
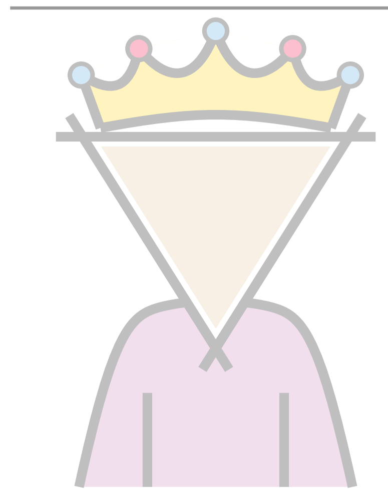
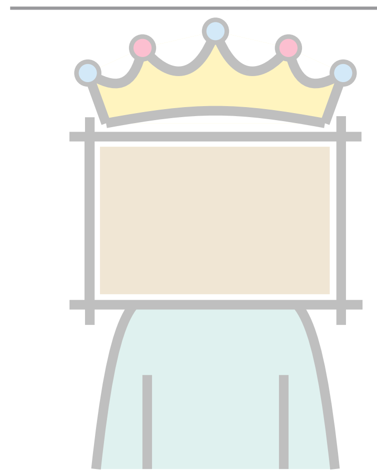
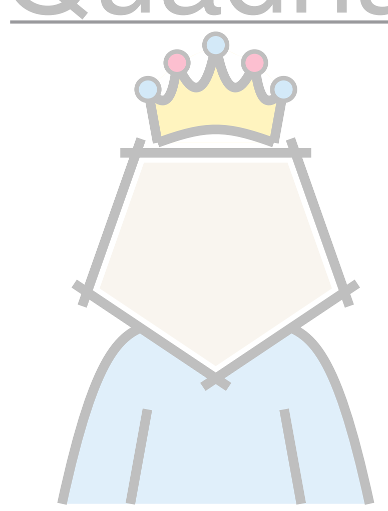
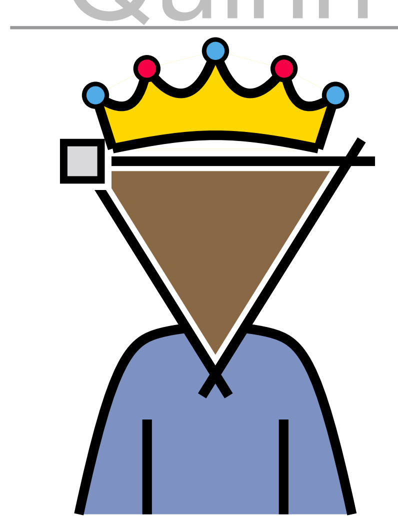
- Want many spots of k pairwise crossing edges
- $3k-1$ adapters per copy
- Connect corridors to adapters in another copy



Supply corridor

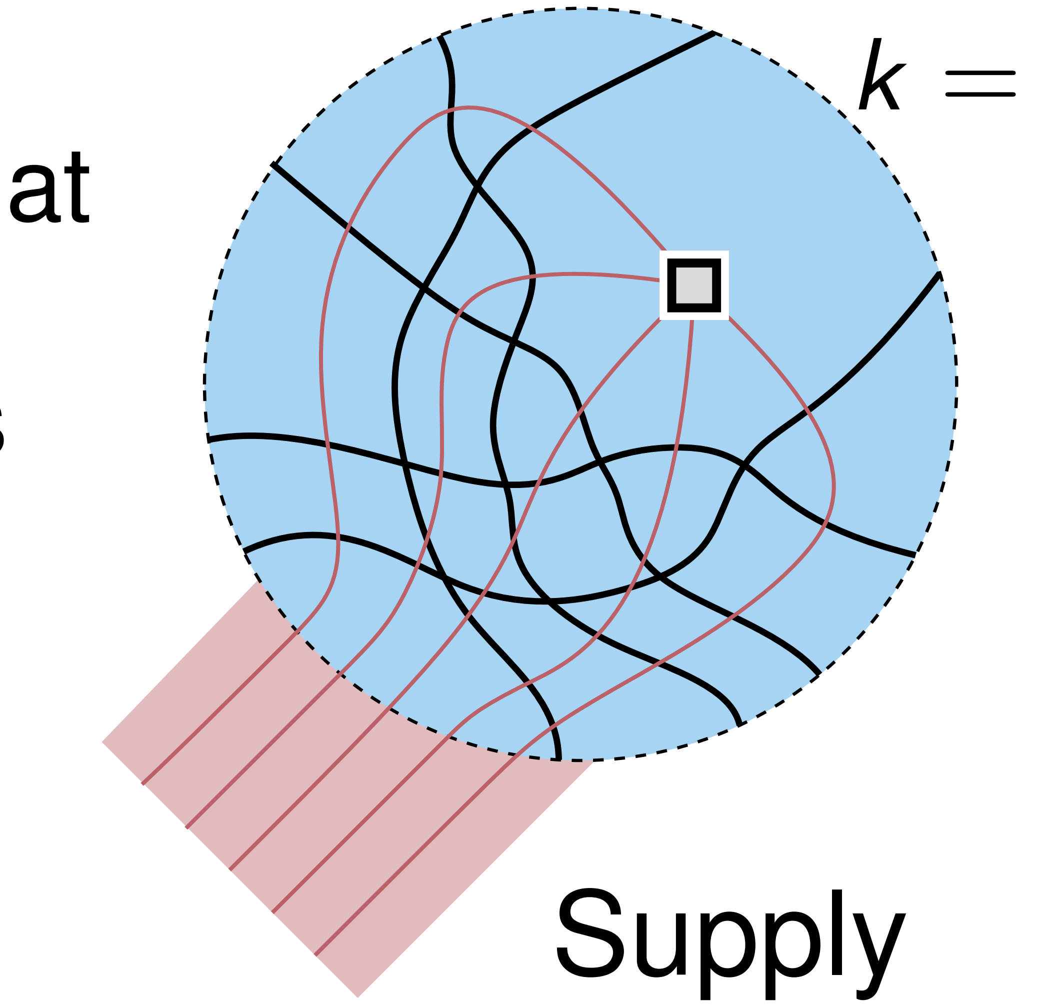


Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}$*	$\binom{n}{2}$

Supply and demand

- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.
- Want many spots of k pairwise crossing edges
- $3k-1$ adapters per copy
- Connect corridors to adapters in another copy

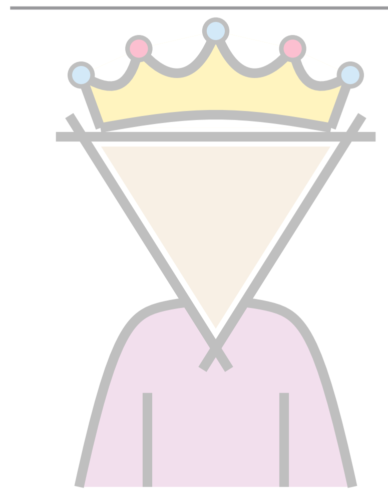
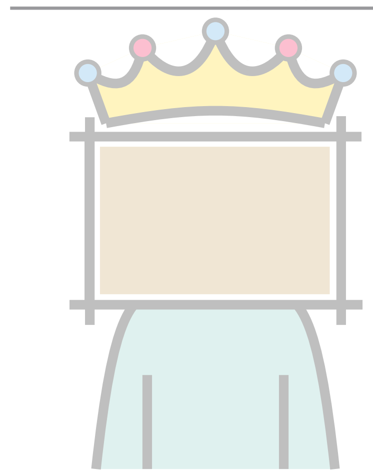
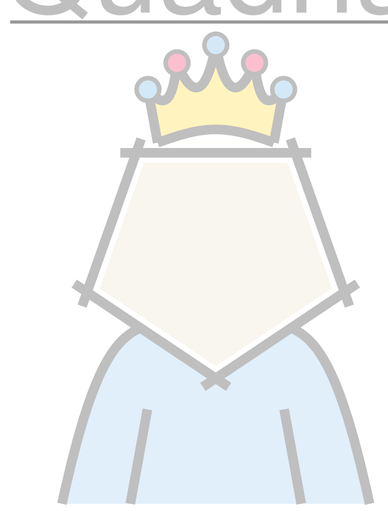
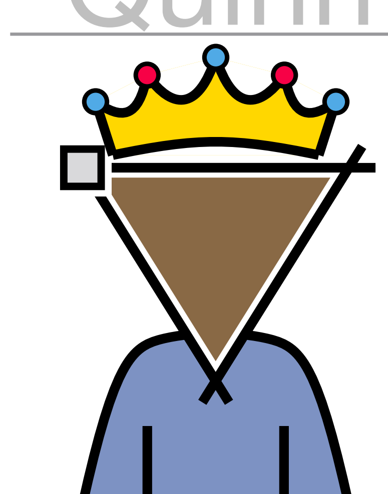


Supply corridor

- $n = 3(3k-1)$ vertices
- k-regular

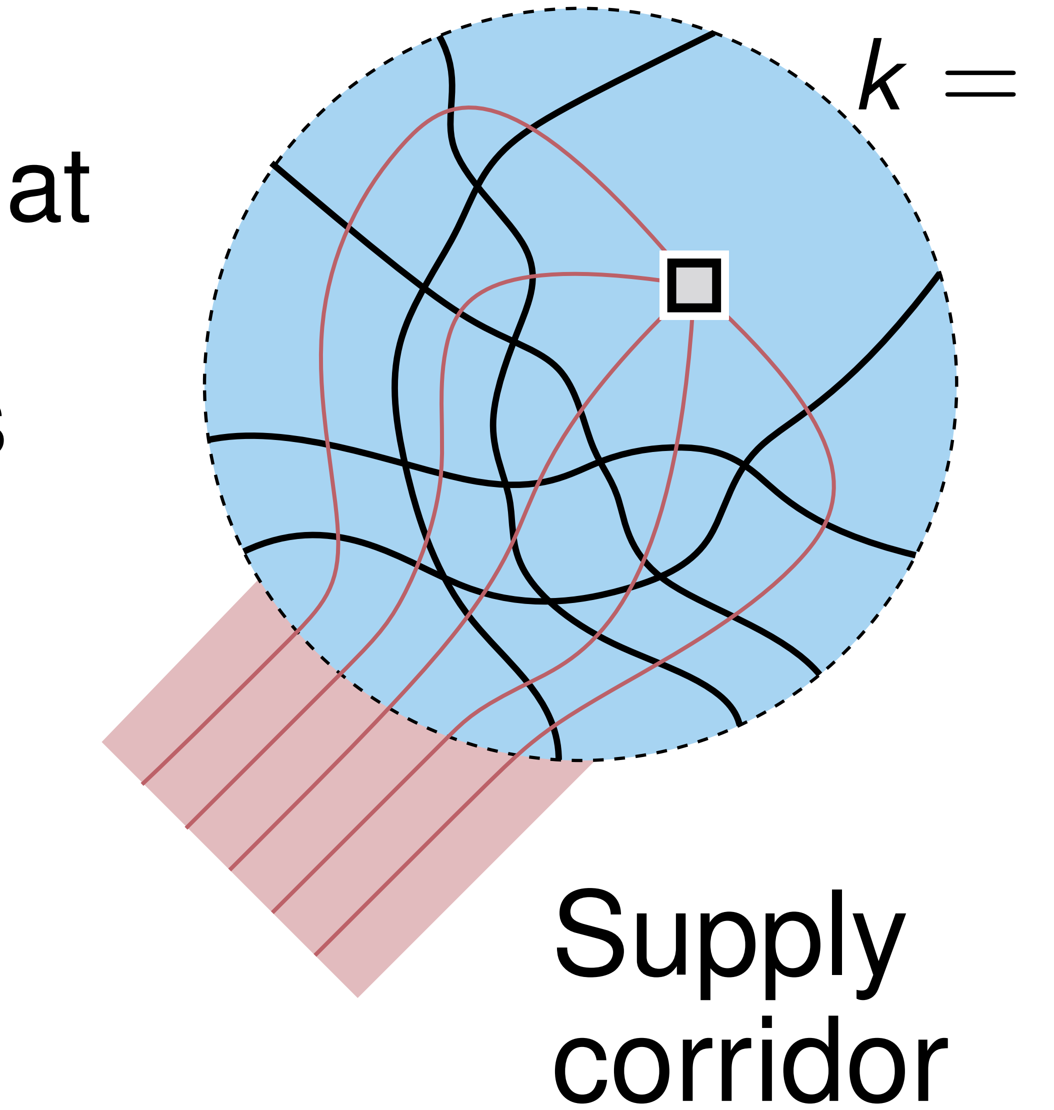
$\Rightarrow \Omega(n^2)$ edges

Some rulers don't

Forbidden cell	Lower Bound	Upper Bound
 Triss	**$7n - 30$** based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07] **$7n - 20$**
 Quadria	**$6n - 12$**	$\binom{n}{2}$
 Quinn	**$6n - 12$**	**$6n - 12$**
 Trimon	**$\Omega(n^2)^*$** *not simple	$\binom{n}{2}$

Supply and demand

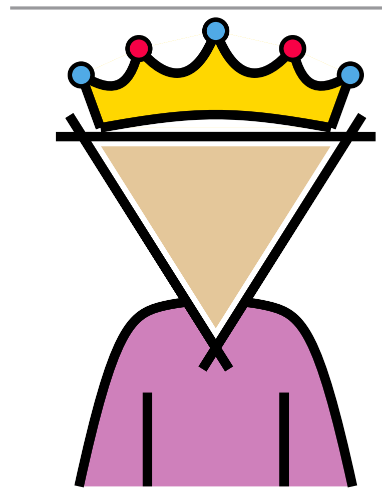
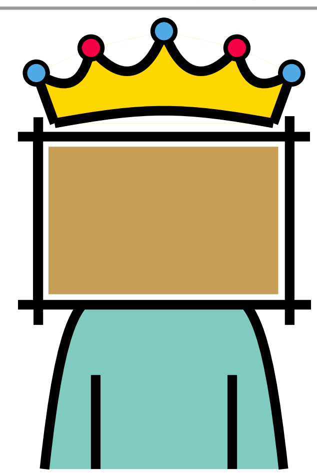
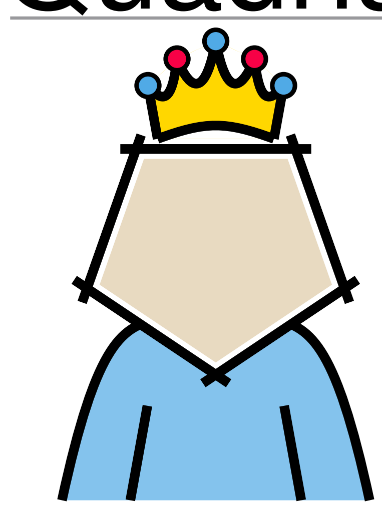
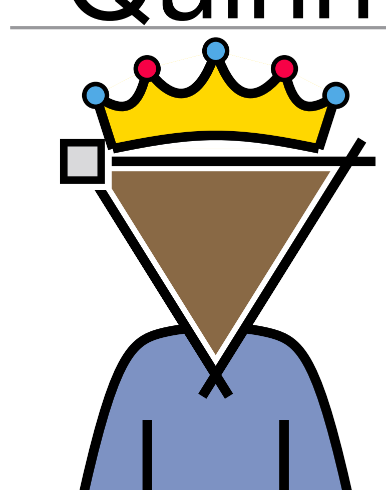
- Can arrange k pairwise crossing edges, so that they sustain a vertex of degree k.
- Want many spots of k pairwise crossing edges
- $3k - 1$ adapters per copy
- Connect corridors to adapters in another copy



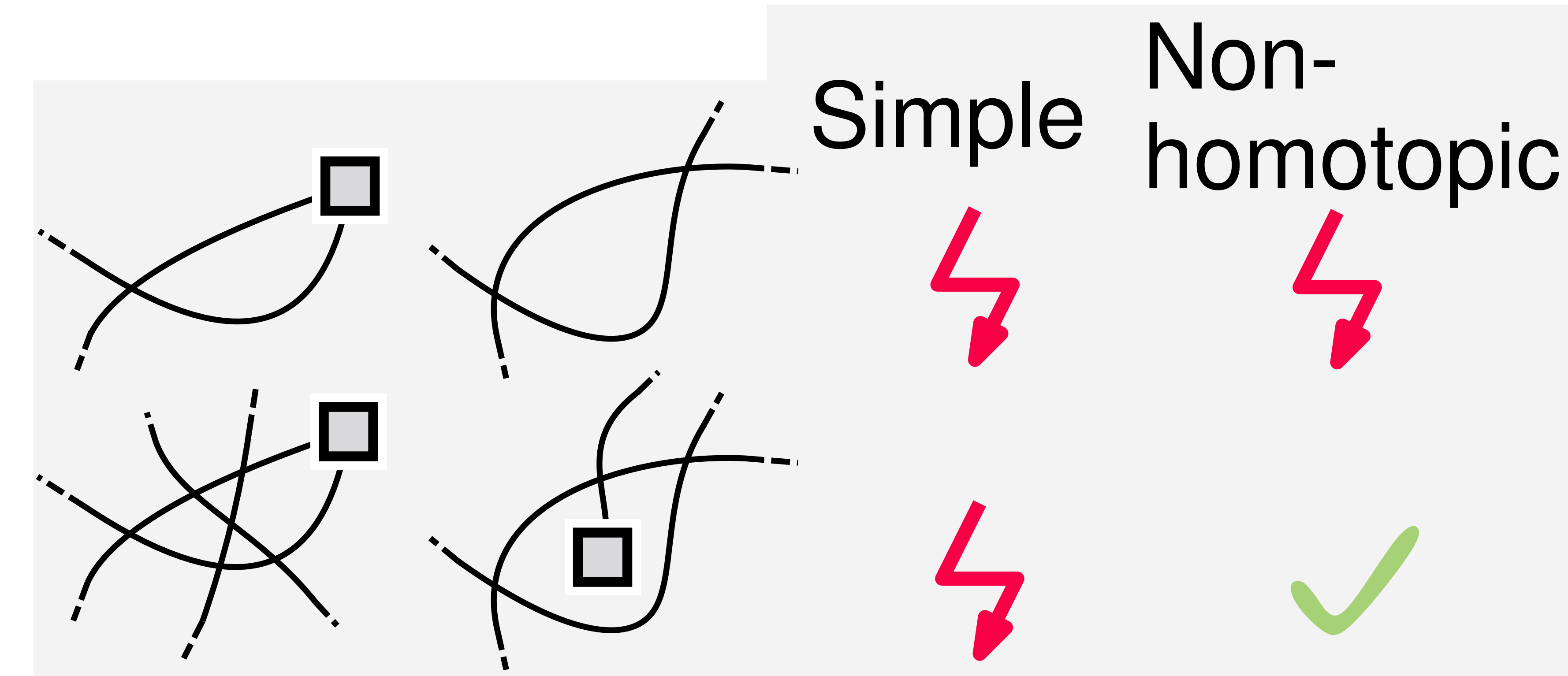
- $n = 3(3k - 1)$ vertices
- k-regular

$\Rightarrow \Omega(n^2)$ edges

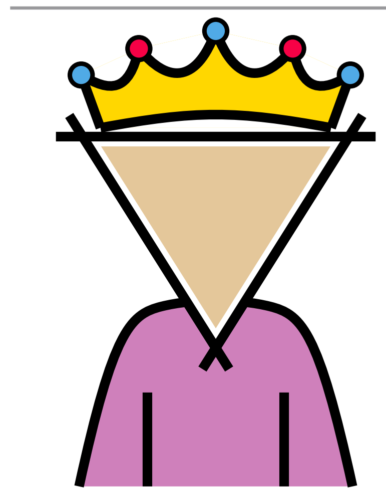
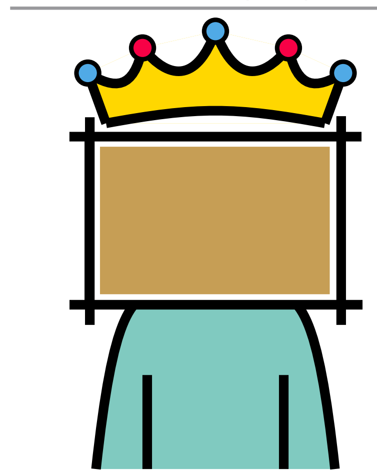
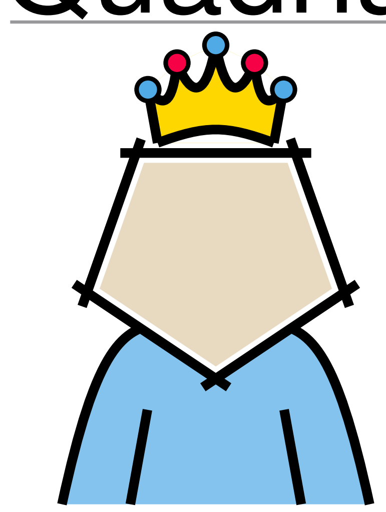
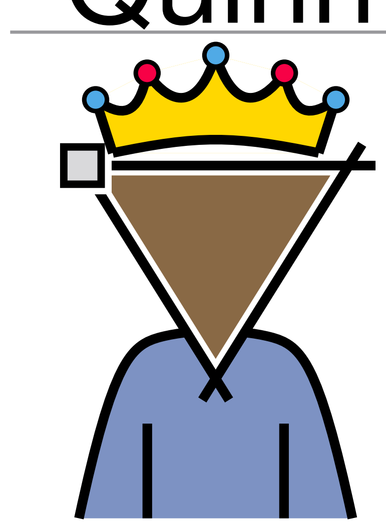
Rulers that play by other rules

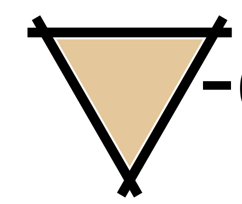
Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}^*$ *non-homotopic	$\binom{n}{2}$

Non-homotopic drawings
(with/without parallel edges)

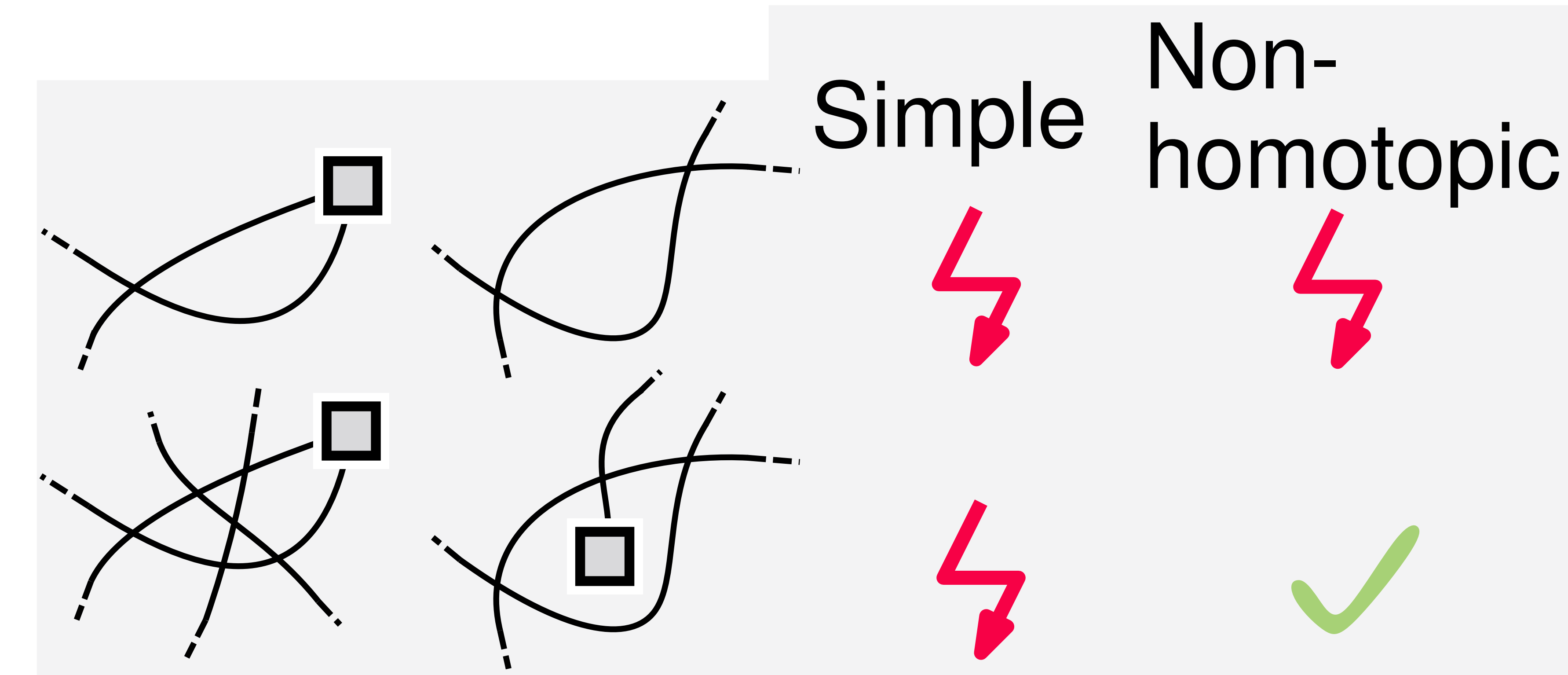


Rulers that play by other rules

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7}\boldsymbol{n}-\mathbf{30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7}\boldsymbol{n}-\mathbf{20}$
 Quadria	$\mathbf{6}\boldsymbol{n}-\mathbf{12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6}\boldsymbol{n}-\mathbf{12}$	$\mathbf{6}\boldsymbol{n}-\mathbf{12}$
 Trimon	$\boldsymbol{\Omega(n^2)}^*$ * non-homotopic	$\binom{n}{2}$

Non-homotopic drawings
(with/without parallel edges)
$\Rightarrow$ Similar, except for  -cells:

- $8n-20$ UB



Rulers that play by other rules

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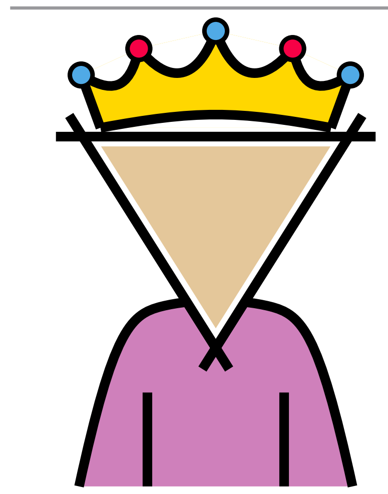
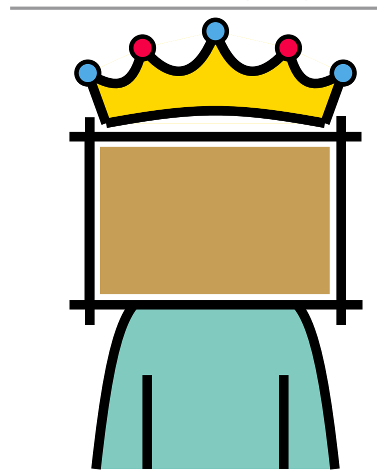
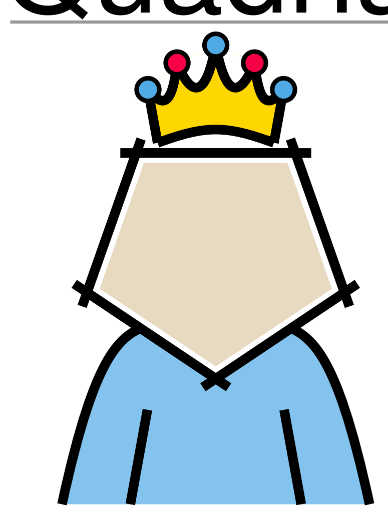
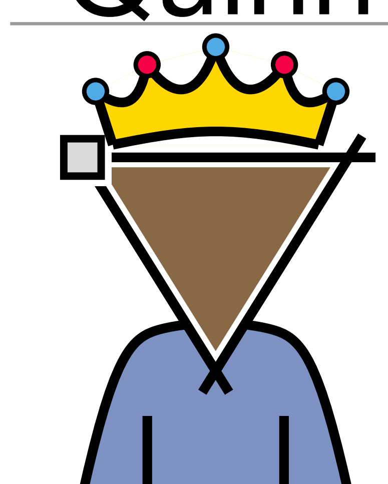
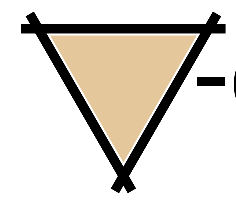
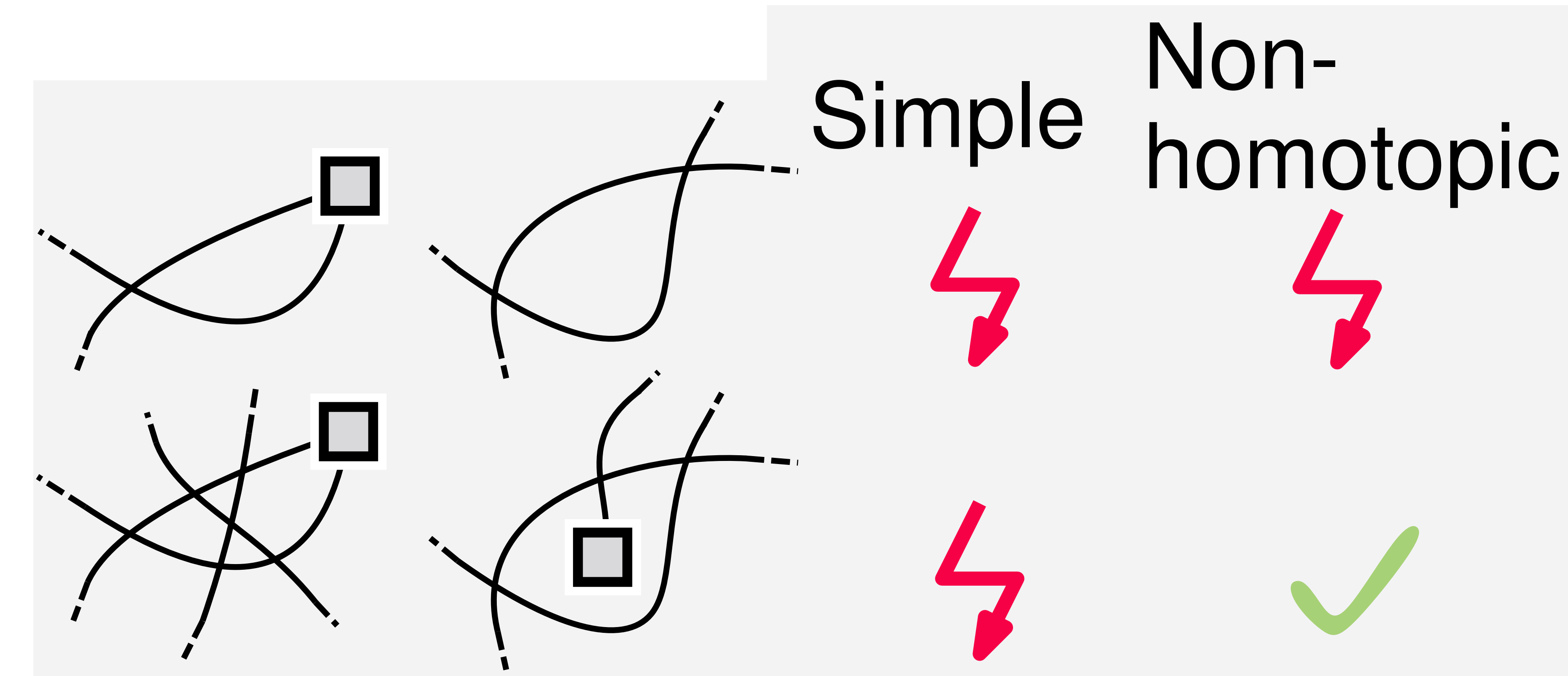
 Triss	$\mathbf{7n - 30}$ based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07] $\mathbf{7n - 20}$
 Quadria	$\mathbf{6n - 12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n - 12}$	$\mathbf{6n - 12}$
 Trimon	$\mathbf{\Omega(n^2)}$* *non-homotopic	$\binom{n}{2}$

Non-homotopic drawings

(with/without parallel edges)

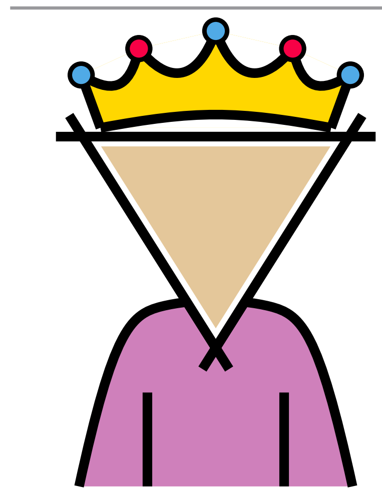
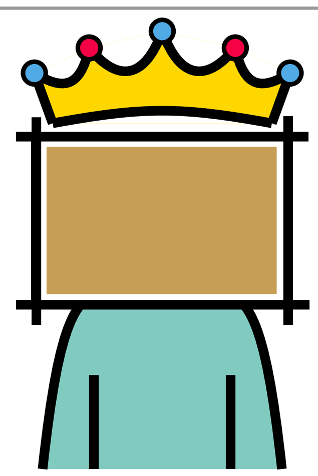
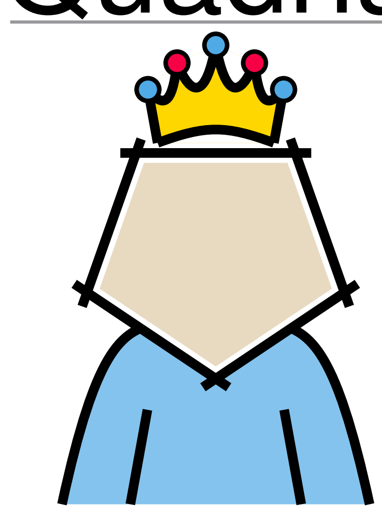
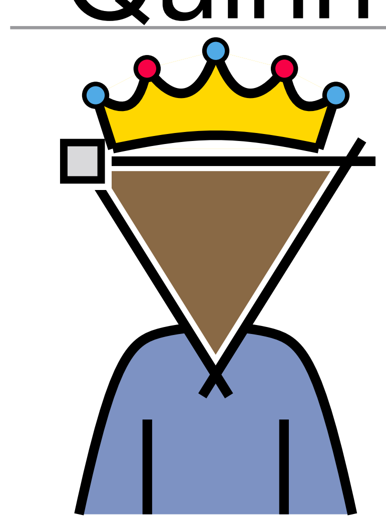
$\Rightarrow$ Similar, except for -cells:

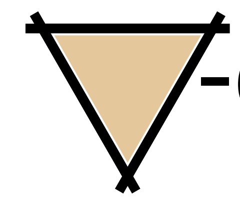
- $8n - 20$ UB
- $8n - 20$ LB (parallel edges)



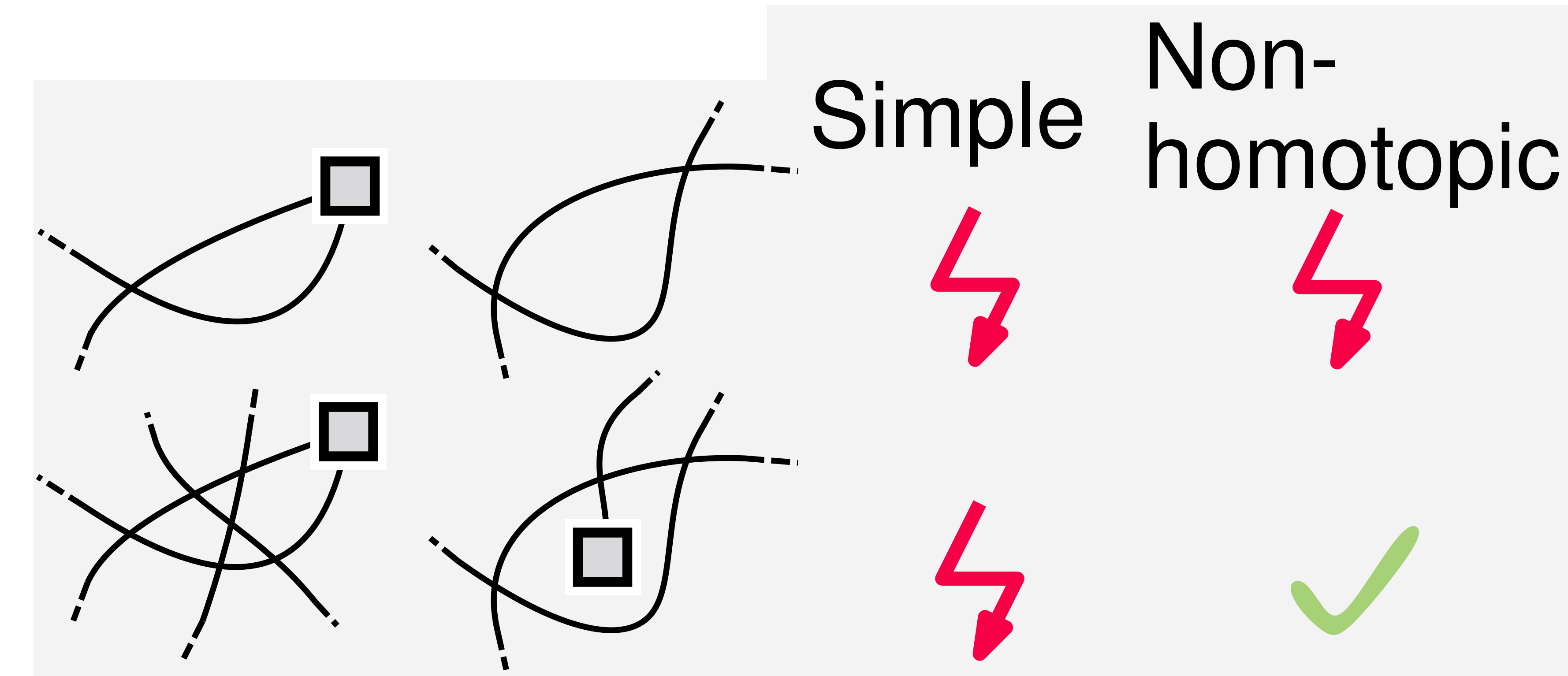
[Kaufmann, Klemz, Knorr, Reddy, Schröder, Ueckerdt; '24]

Rulers that play by other rules

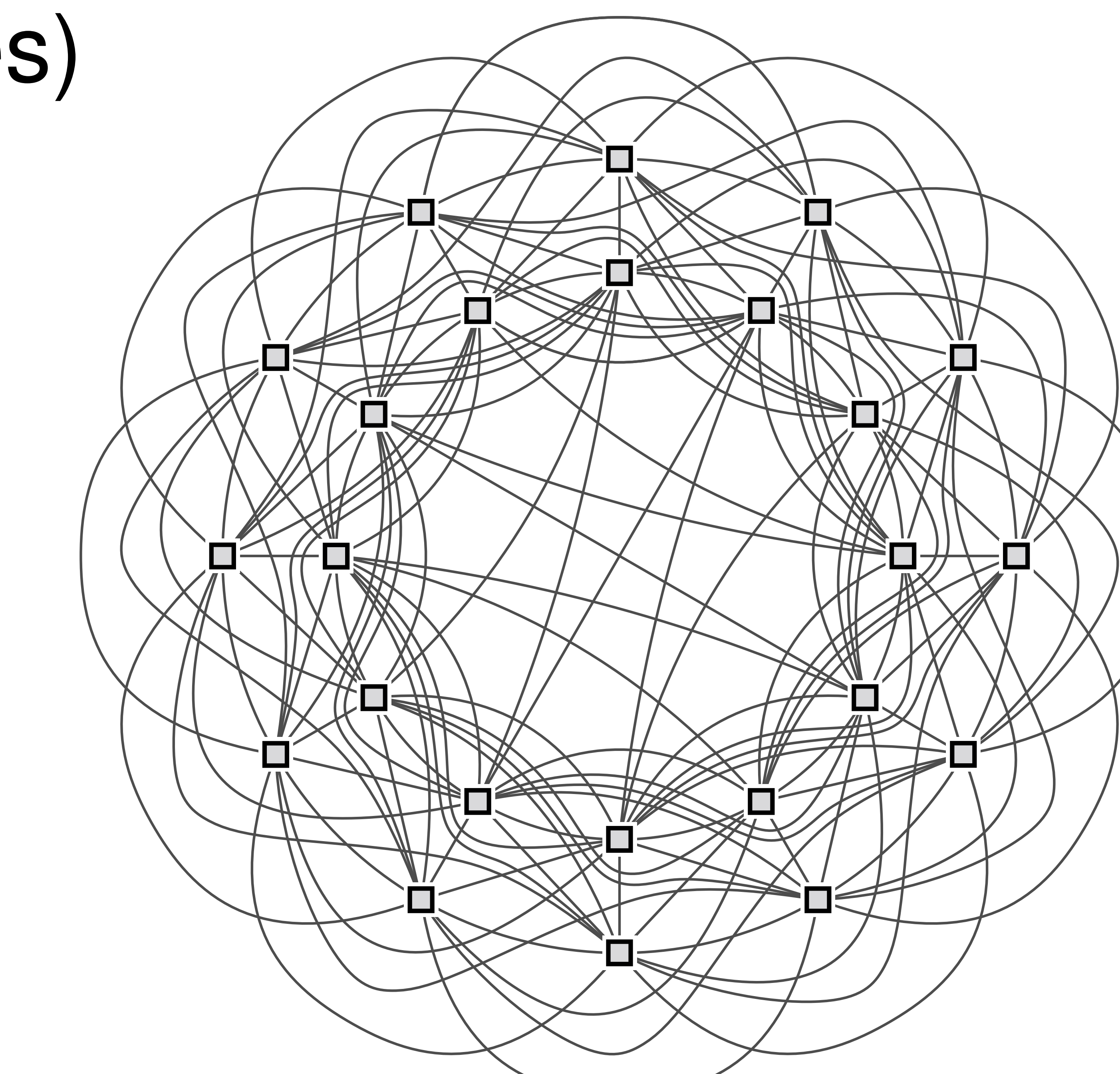
Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7}\boldsymbol{n} - \mathbf{30}$ based on [Ackerman, Tardos; '07]	$8n - 20$ [Ackerman, Tardos; '07] $\mathbf{7}\boldsymbol{n} - \mathbf{20}$
 Quadria	$\mathbf{6}\boldsymbol{n} - \mathbf{12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6}\boldsymbol{n} - \mathbf{12}$	$\mathbf{6}\boldsymbol{n} - \mathbf{12}$
 Trimon	$\boldsymbol{\Omega(n^2)}^*$ * non-homotopic	$\binom{n}{2}$

Non-homotopic drawings
(with/without parallel edges)
$\Rightarrow$ Similar, except for  -cells:

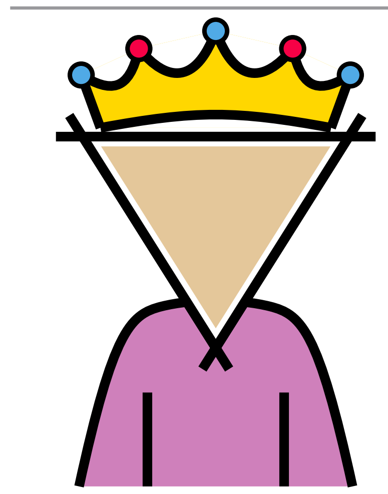
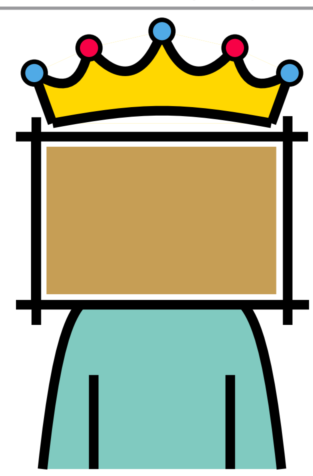
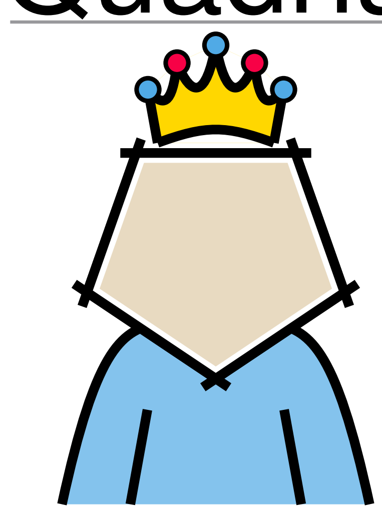
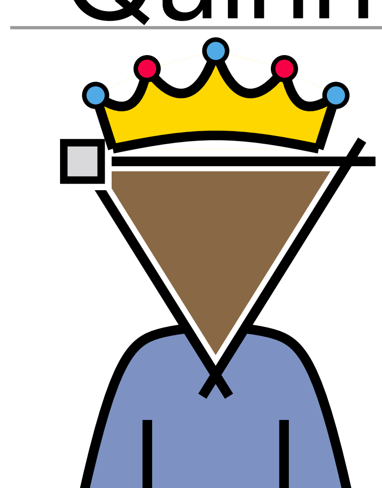
- $8n - 20$ UB
- $8n - 20$ LB (parallel edges)
- $\mathbf{8}\boldsymbol{n} - \mathbf{28}$ **LB** (without parallel edges)



[Kaufmann, Klemz, Knorr, Reddy, Schröder, Ueckerdt; '24]

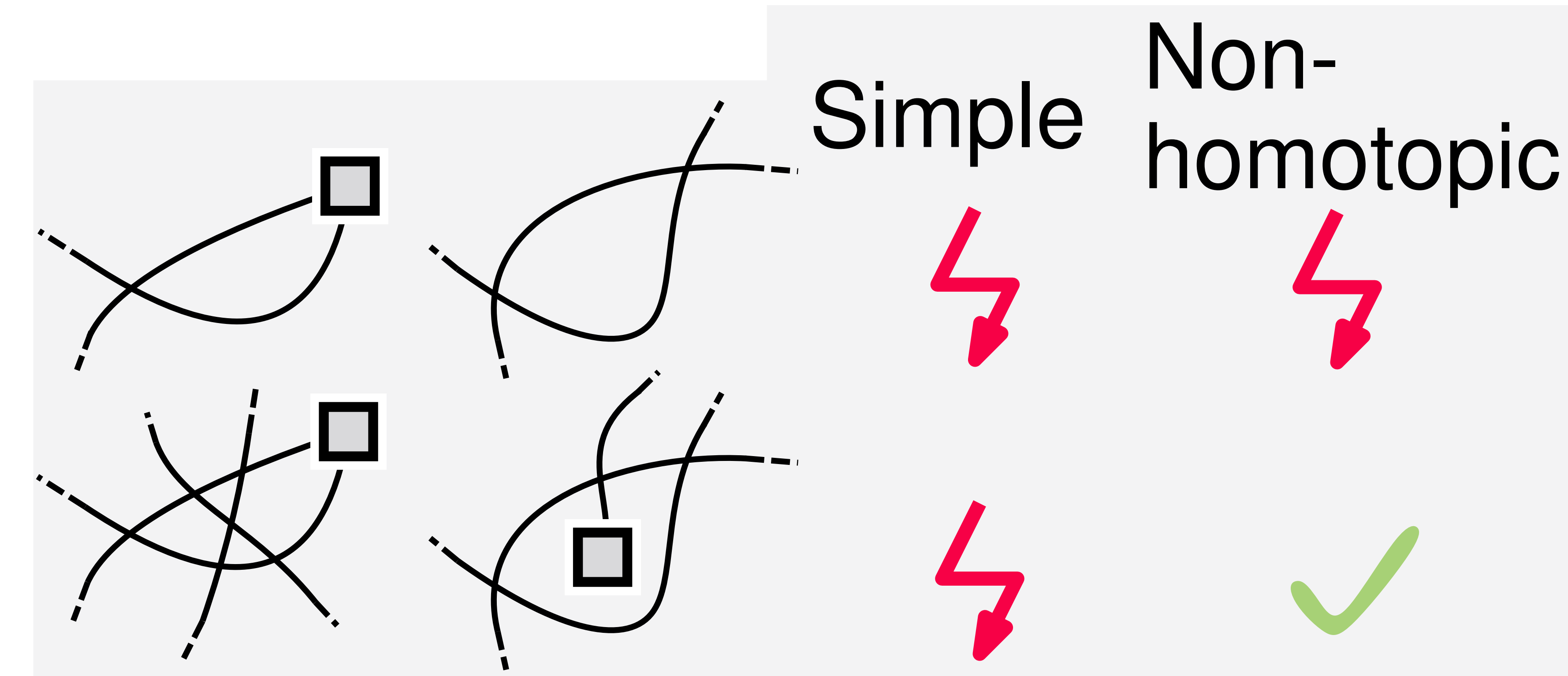


Rulers that play by other rules

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}$* *non-homotopic	$\binom{n}{2}$

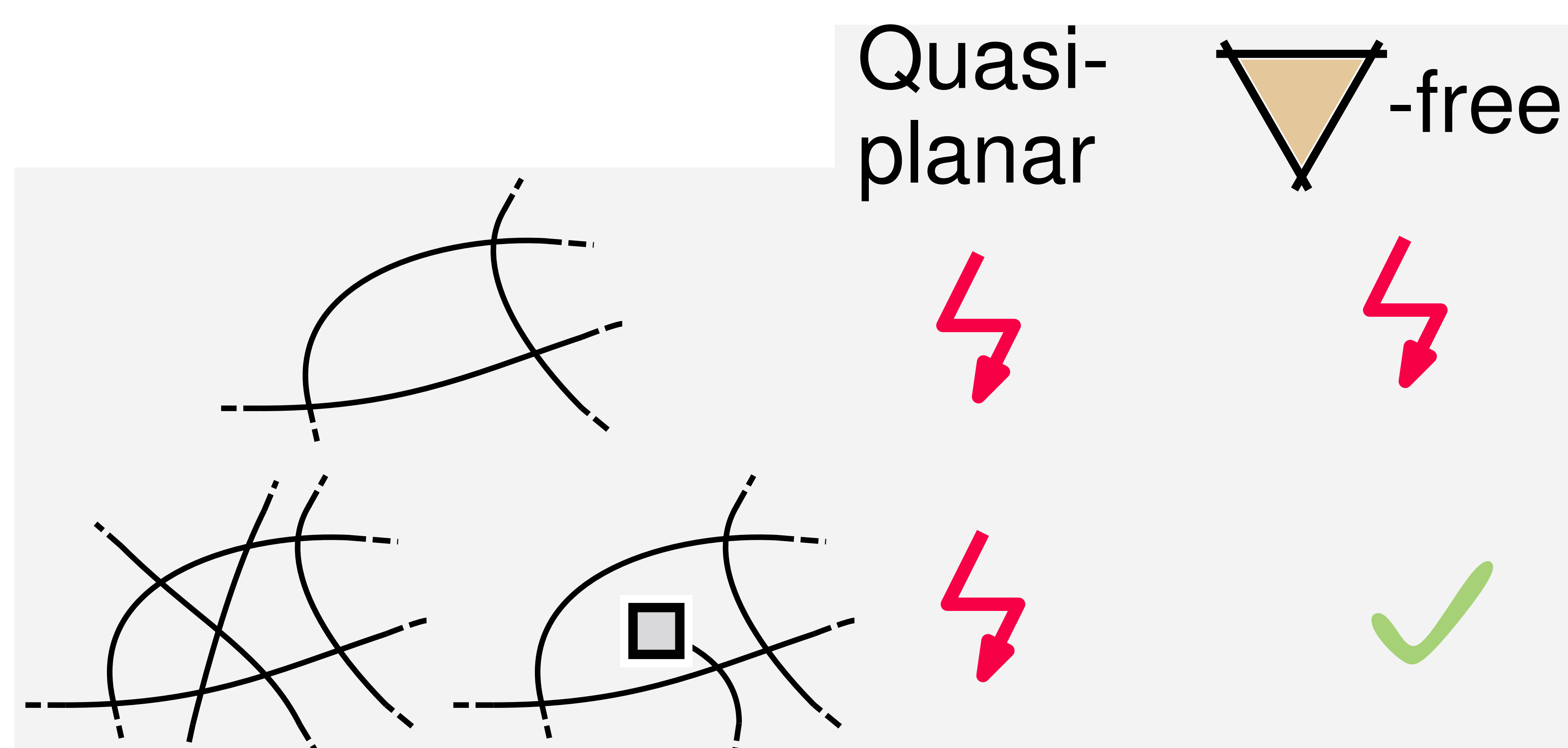
Non-homotopic drawings
(with/without parallel edges)
⇒ Similar, except for ▽-cells:

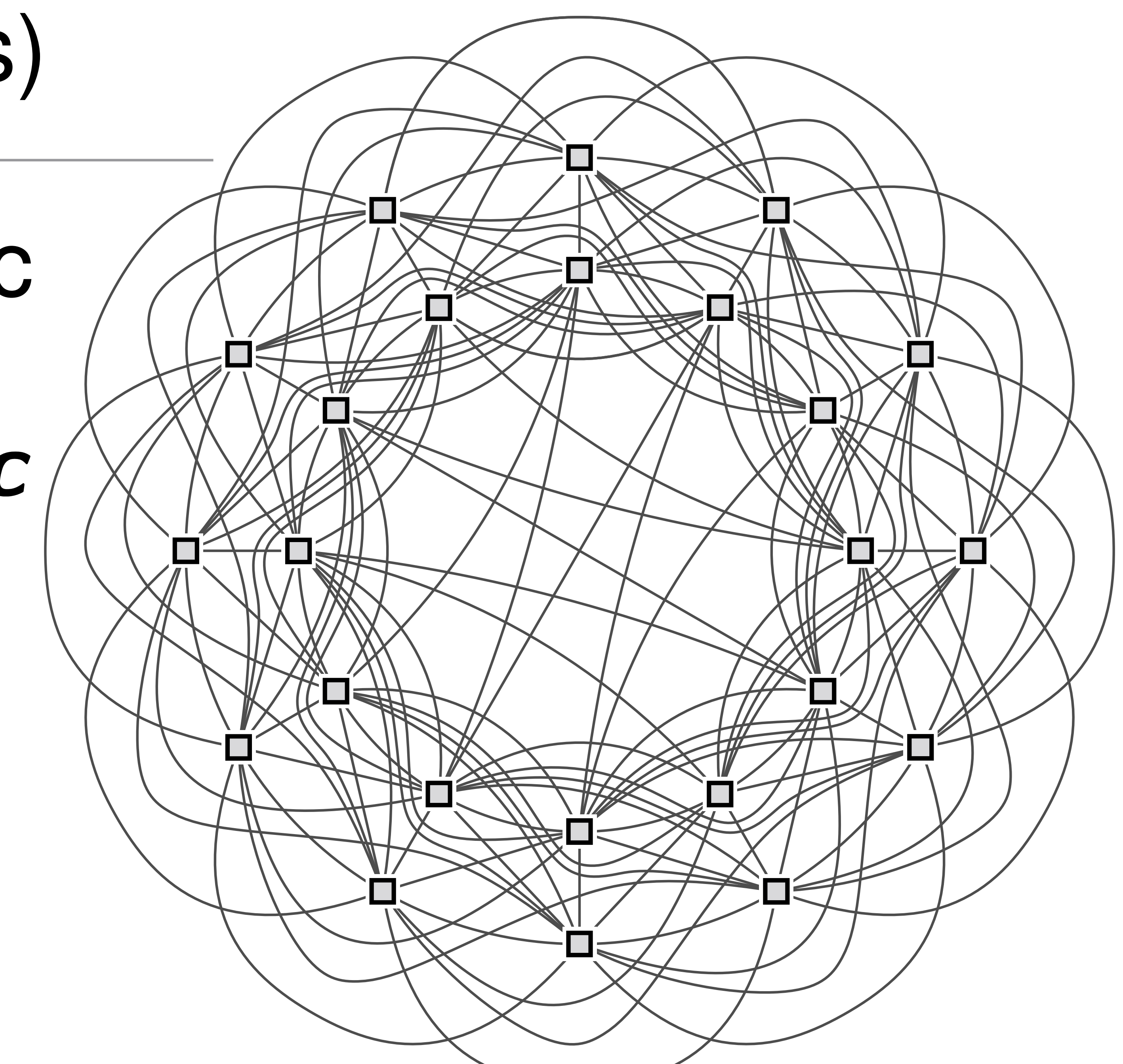
- $8n-20$ UB
- $8n-20$ LB (parallel edges)
- **$\mathbf{8n-28}$ LB** (without parallel edges)



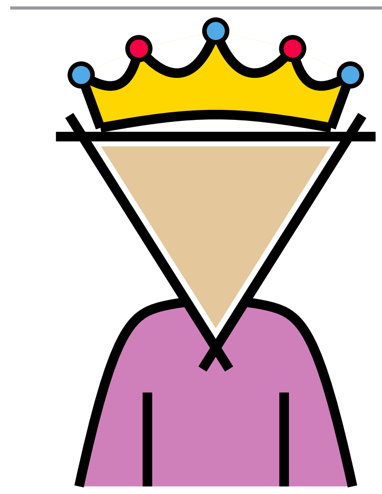
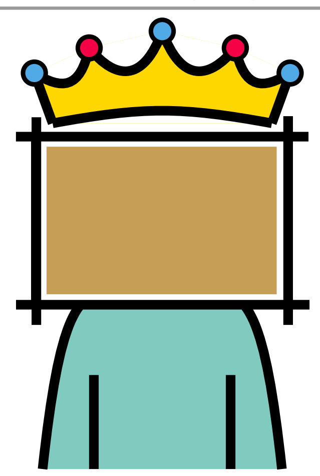
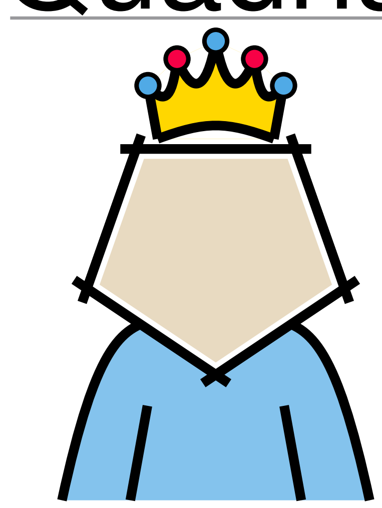
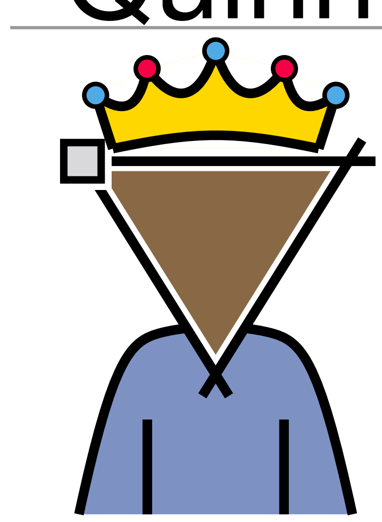
[Kaufmann, Klemz, Knorr, Reddy, Schröder, Ueckerdt; '24]

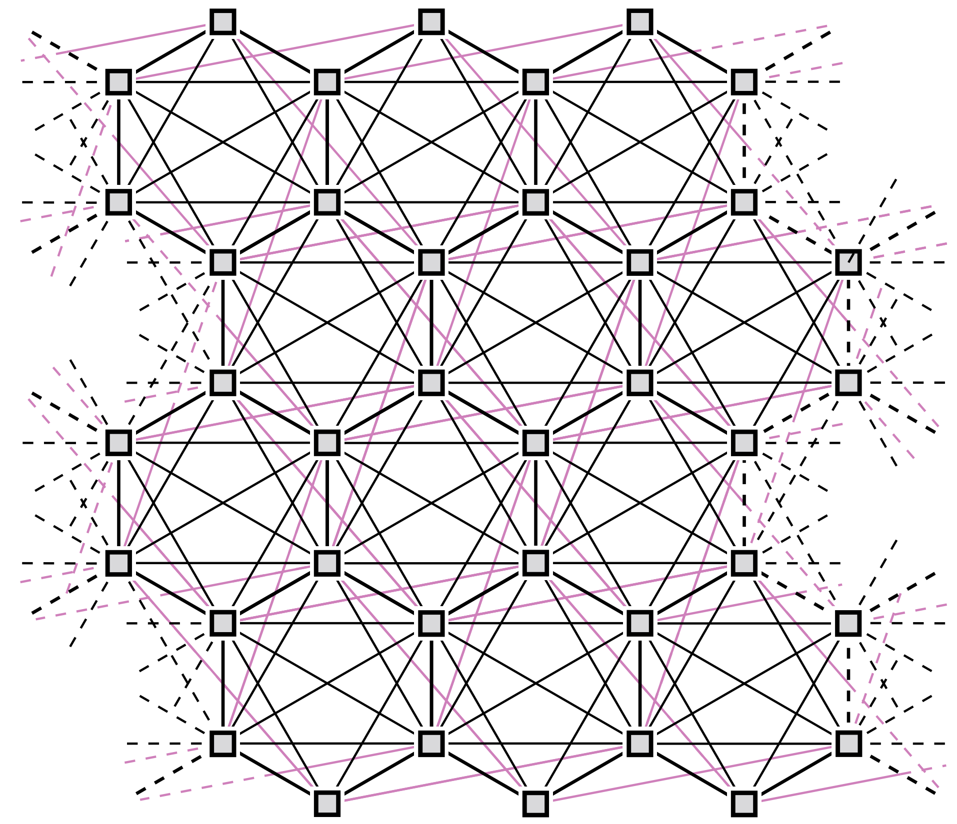
- Improved LB for non-homotopic **quasiplanar** drawings without parallel edges: $7n-c \to \mathbf{7.5n-c}$

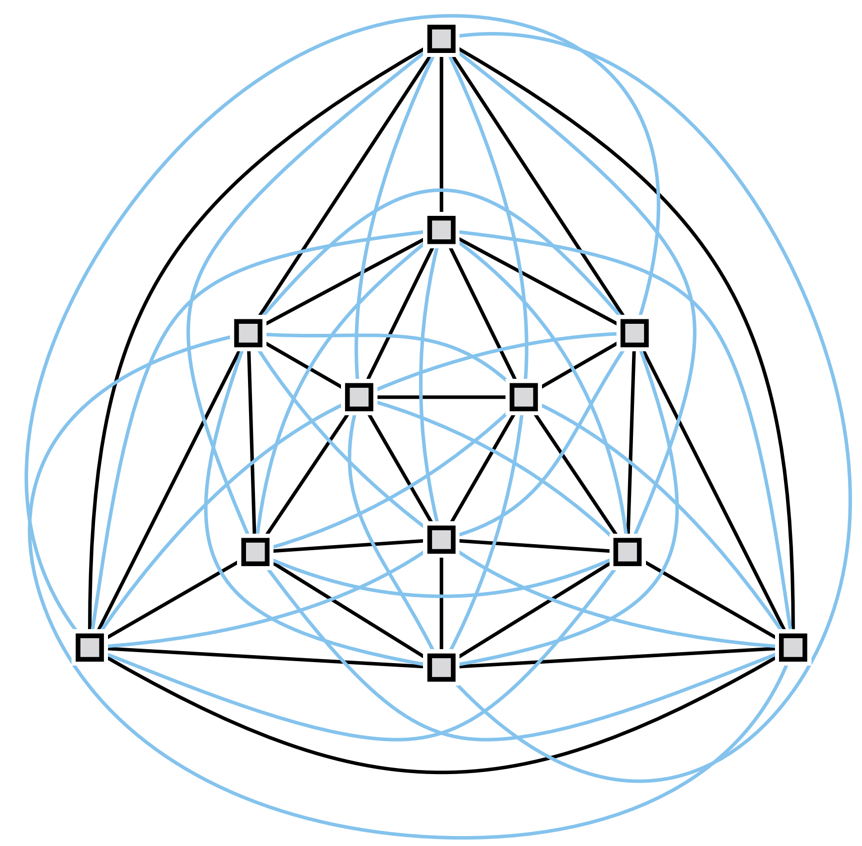


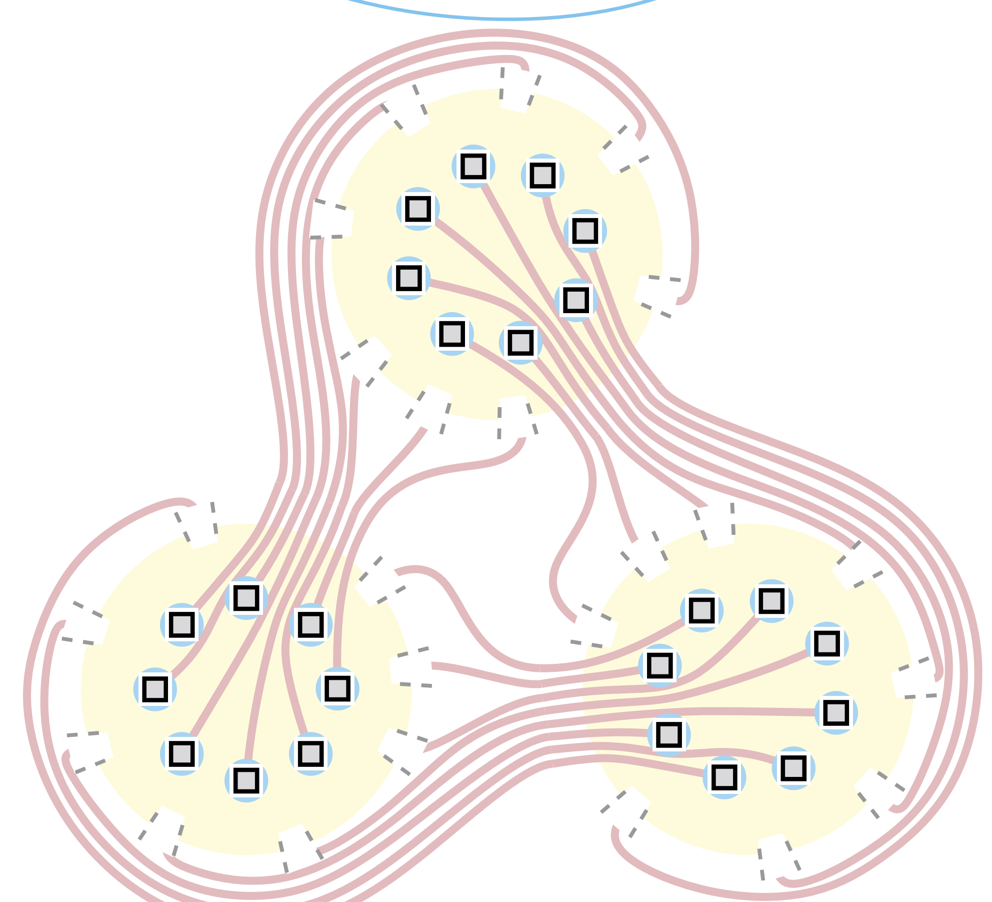


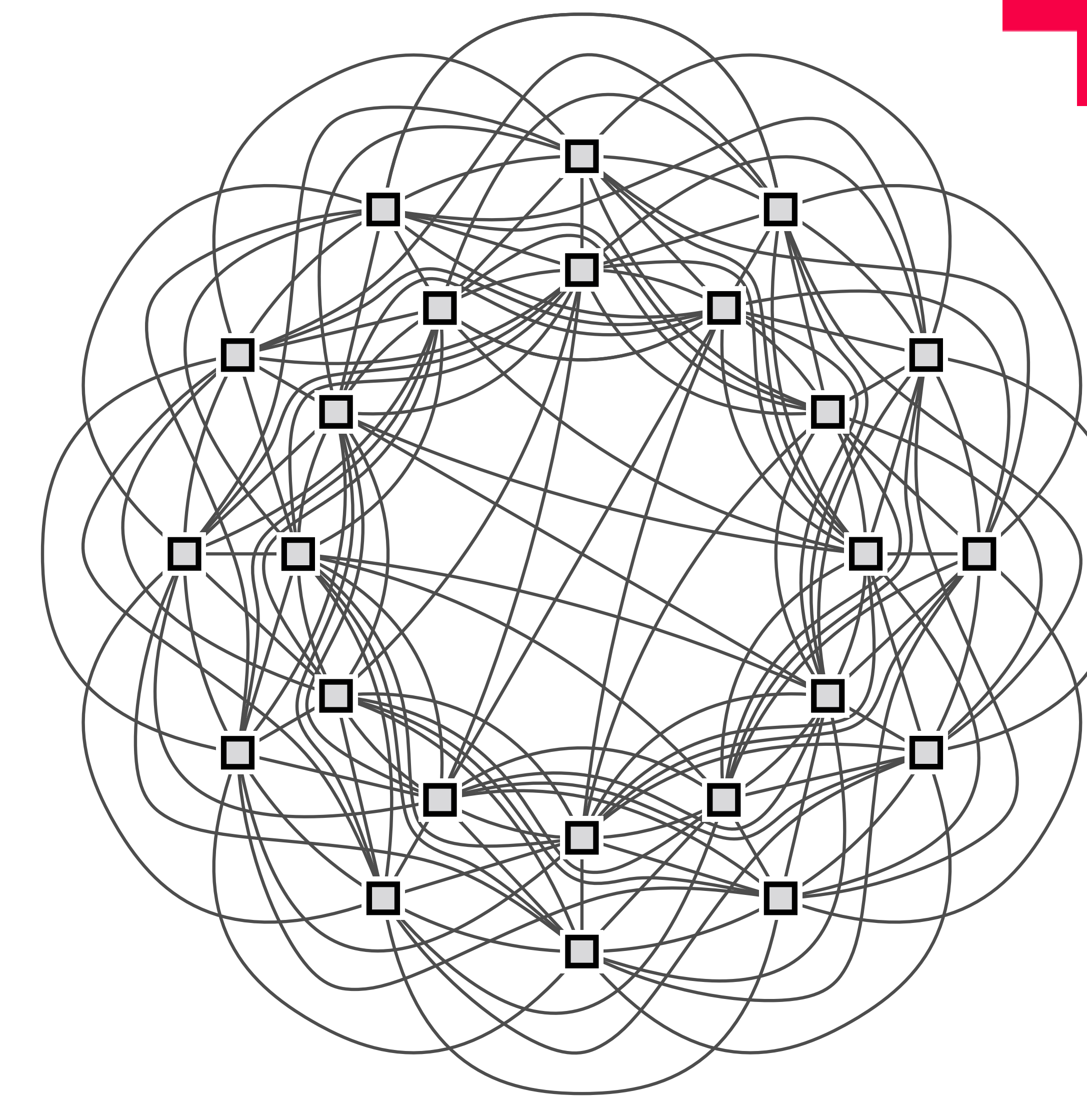
Happily ever after

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\mathbf{\Omega(n^2)}$* *non-homotopic	$\binom{n}{2}$

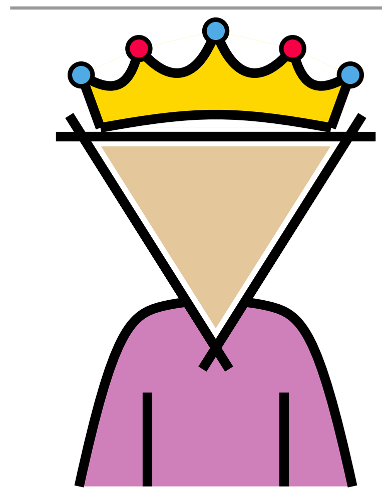
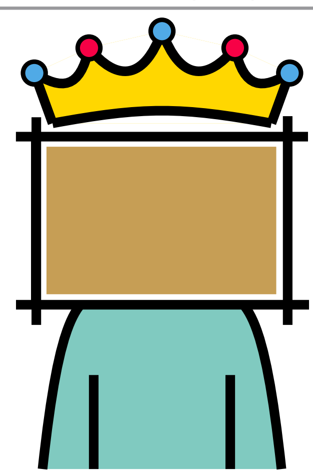
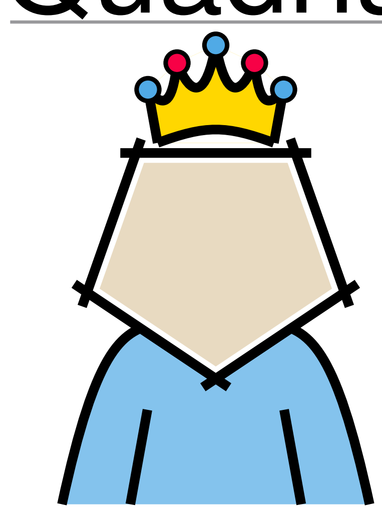
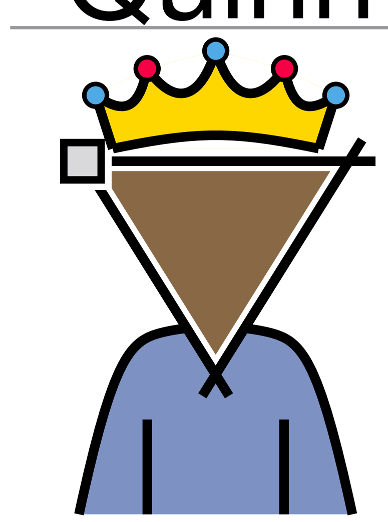


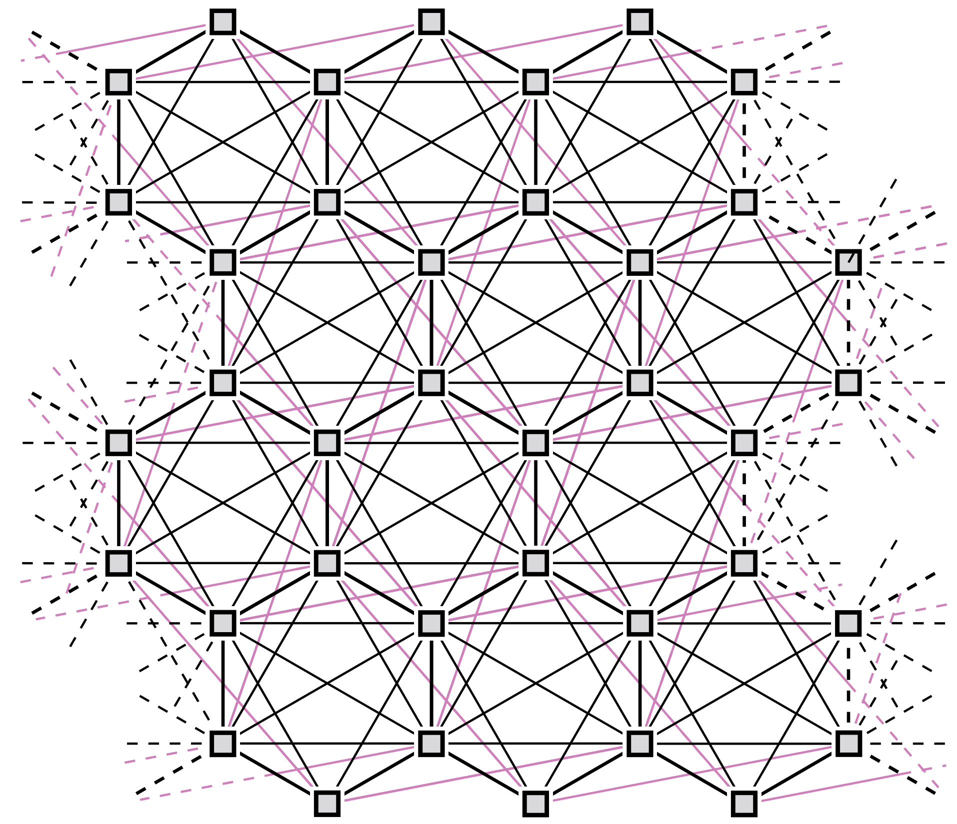


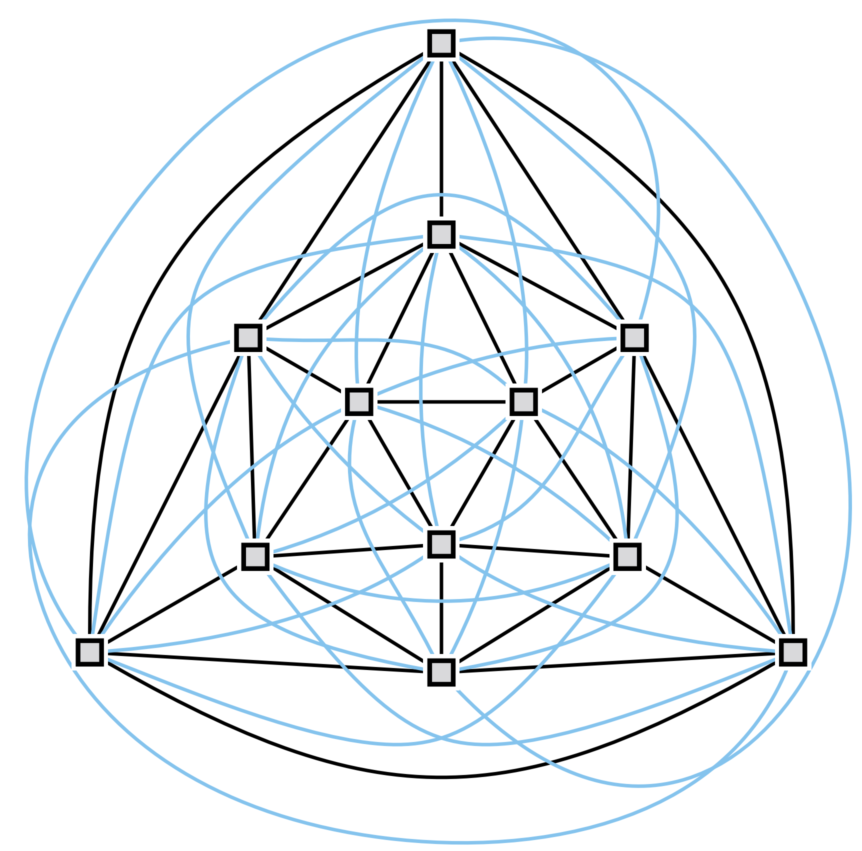


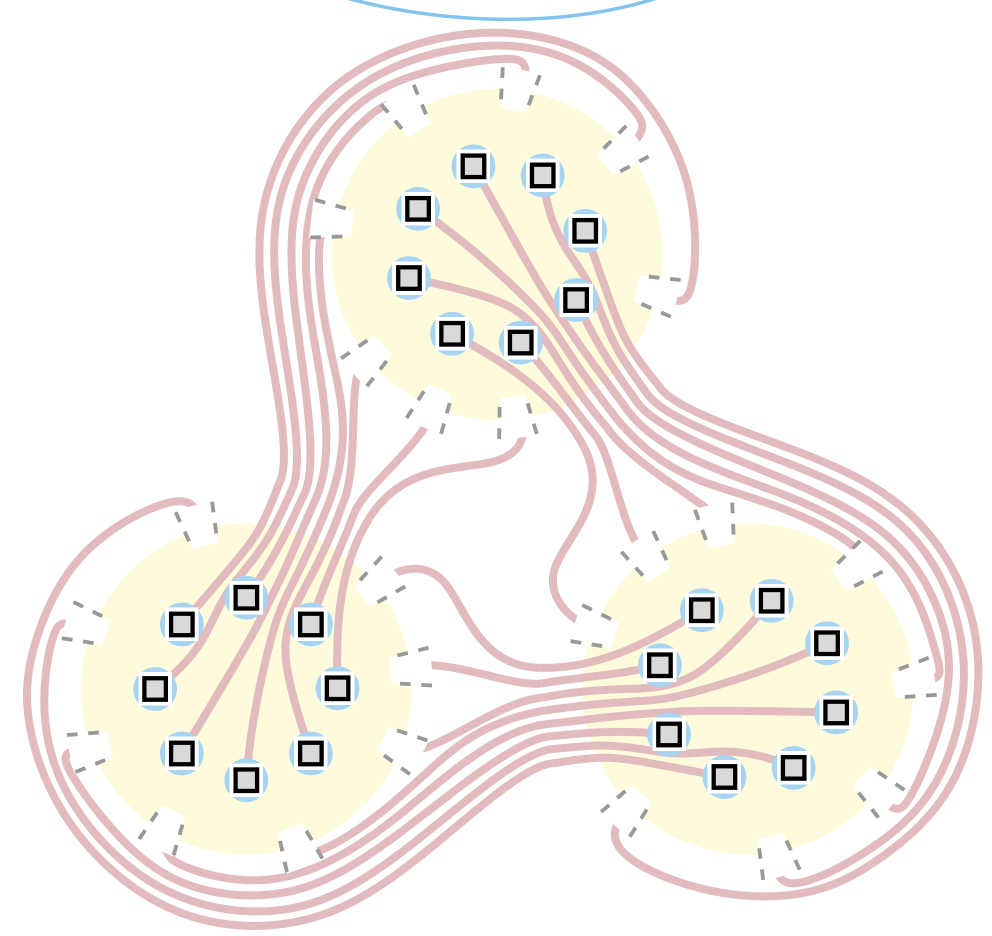


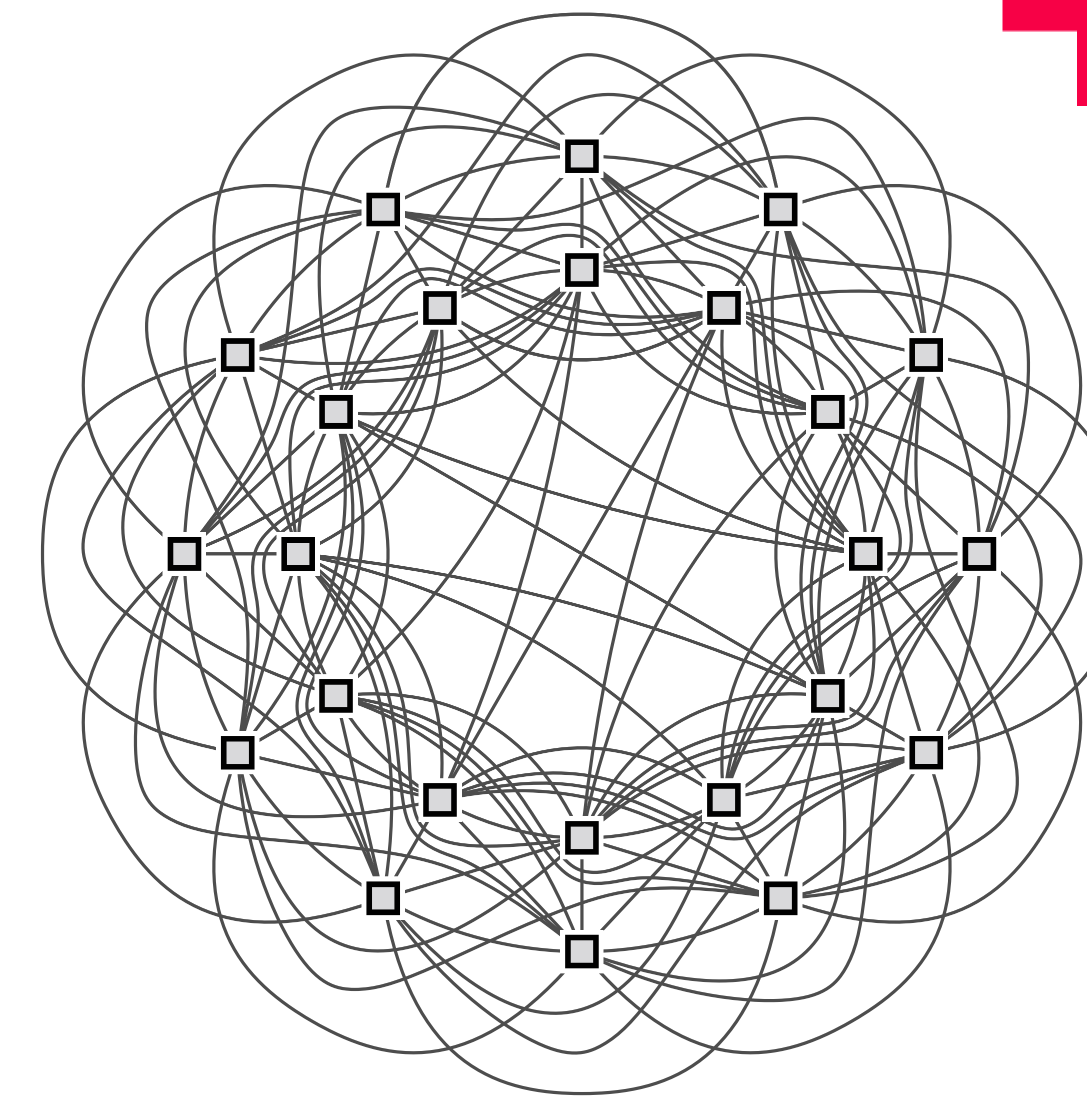
Happily ever after?

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7n-30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7n-20}$
 Quadria	$\mathbf{6n-12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6n-12}$	$\mathbf{6n-12}$
 Trimon	$\Omega(n^2)^*$ *non-homotopic	$\binom{n}{2}$





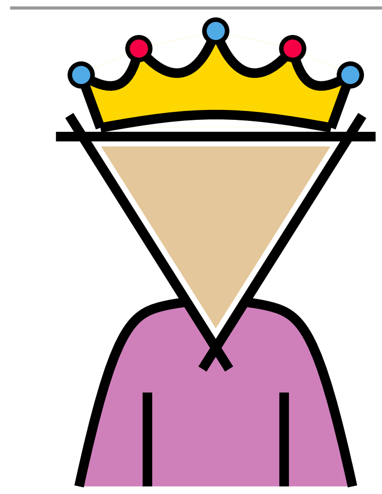
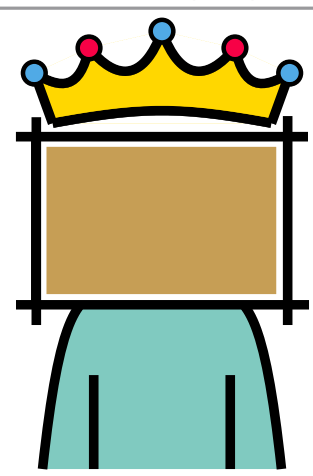
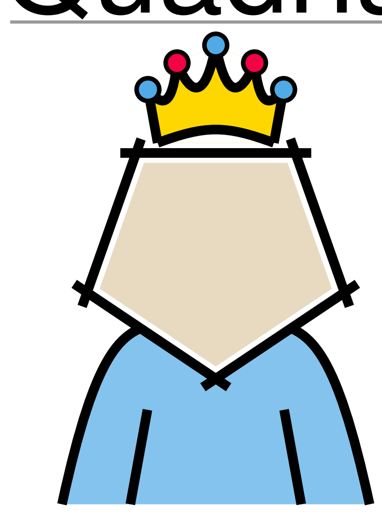
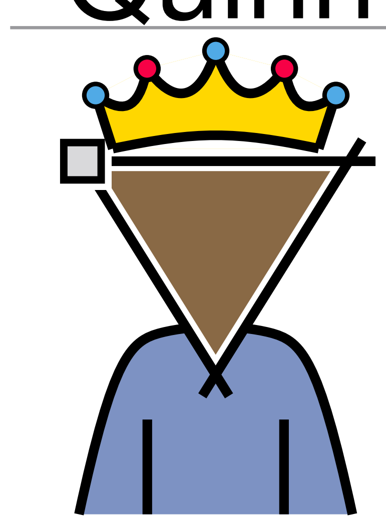


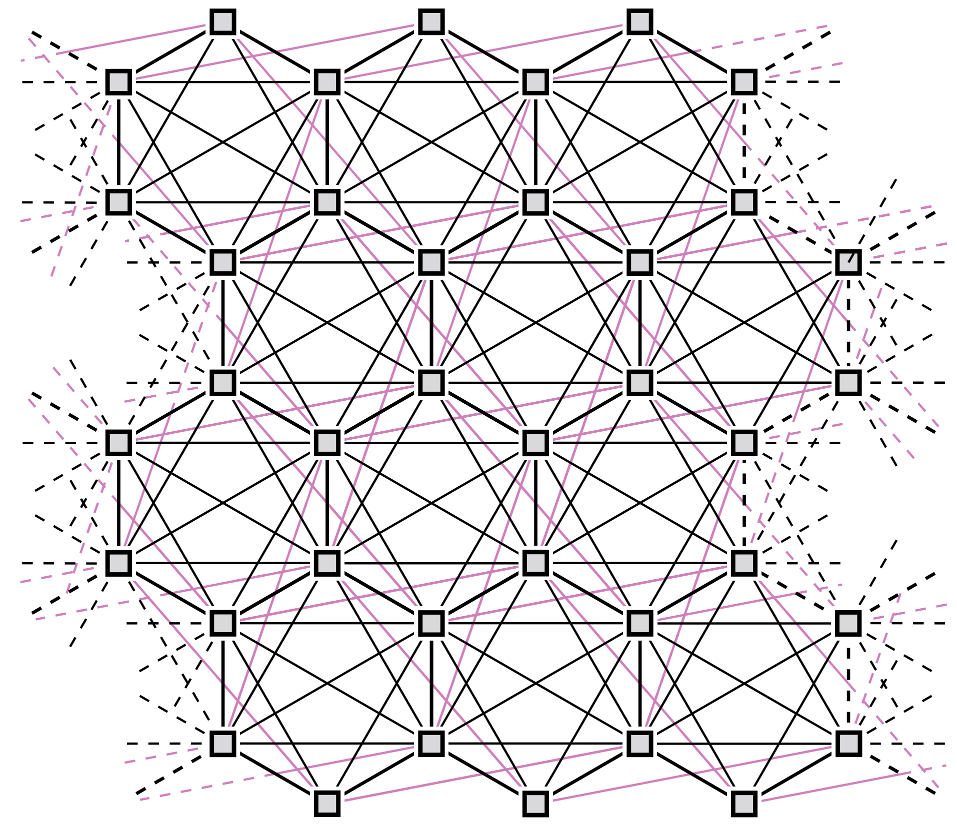


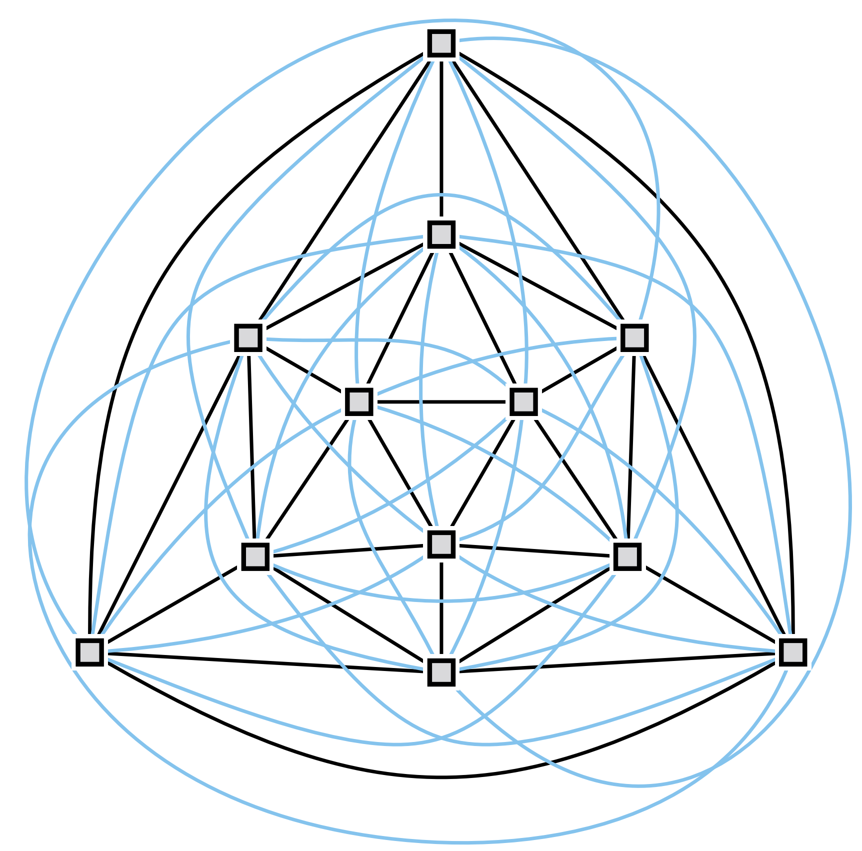
Questions:

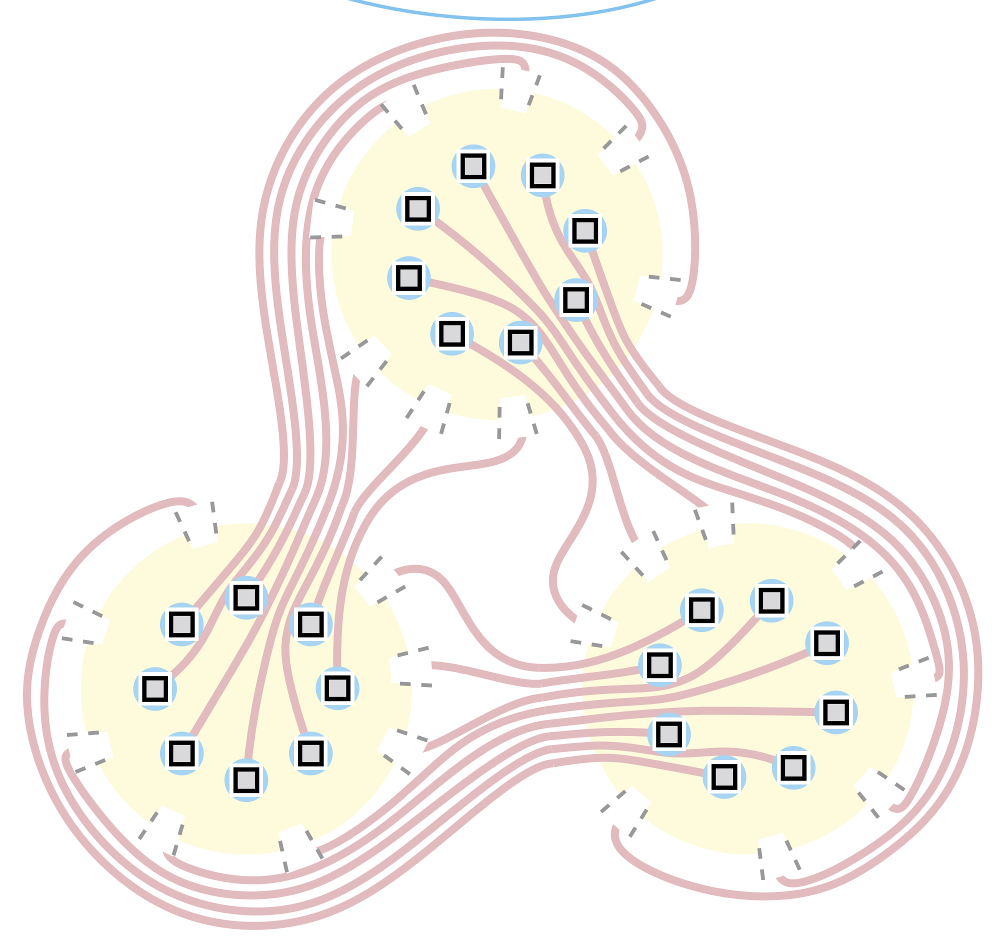
- What about Quadria?
- Can the principle of simplicity be upheld in a dense empire for Trimon?
- Is there a ruler for whom every empire can be realized?

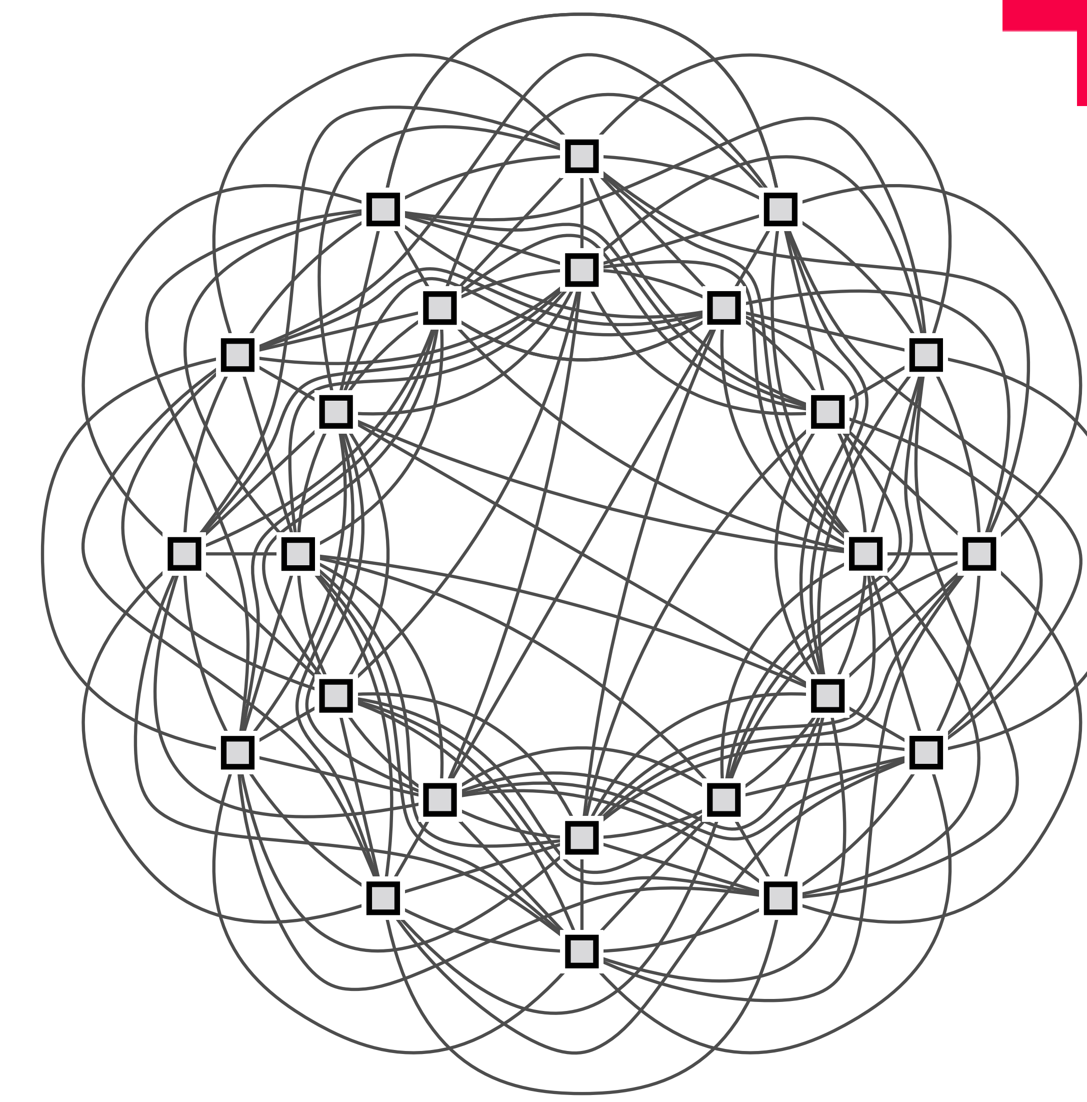
Happily ever after?

Forbidden cell	Lower Bound	Upper Bound
 Triss	$\mathbf{7}\boldsymbol{n}-\mathbf{30}$ based on [Ackerman, Tardos; '07]	$8n-20$ [Ackerman, Tardos; '07] $\mathbf{7}\boldsymbol{n}-\mathbf{20}$
 Quadria	$\mathbf{6}\boldsymbol{n}-\mathbf{12}$	$\binom{n}{2}$
 Quinn	$\mathbf{6}\boldsymbol{n}-\mathbf{12}$	$\mathbf{6}\boldsymbol{n}-\mathbf{12}$
 Trimon	$\boldsymbol{\Omega(n^2)}^*$ *non-homotopic	$\binom{n}{2}$









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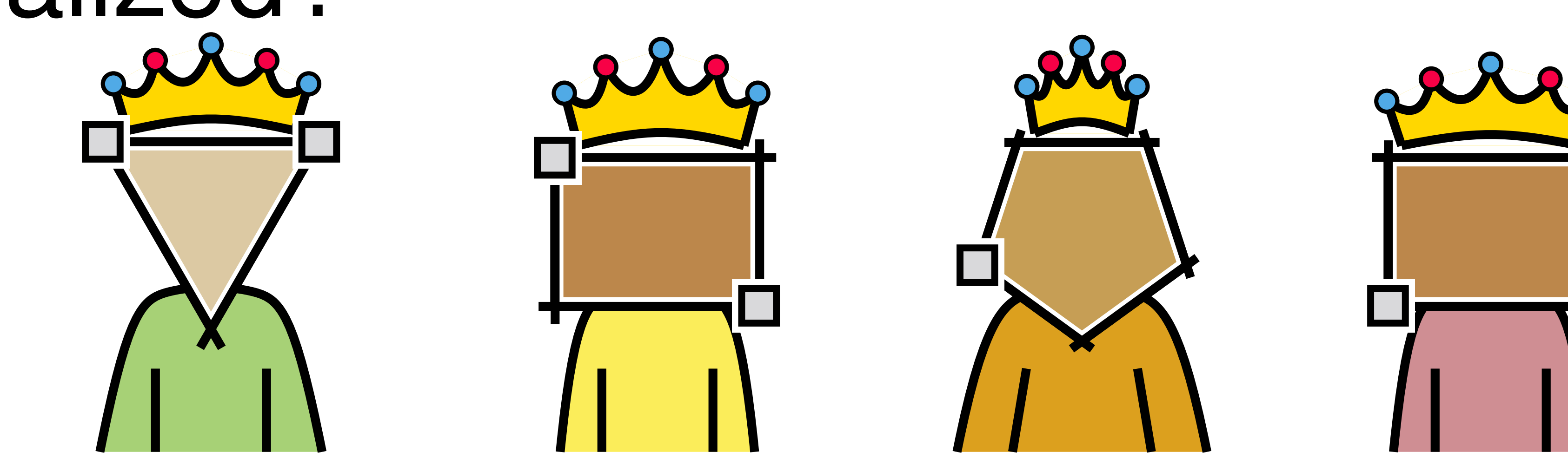
- Can the principle of simplicity be upheld in a dense empire for Trimon?
- Is there a ruler for whom every empire can be realized?



Thank you!